

Who Gets Screened Out? The Opioid Crisis and Employer Skill Requirements*

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Abstract

While growing evidence suggests that the opioid crisis has reduced employment levels, little is known about how the crisis has affected job skill requirements—tools that employers use to screen job candidates. Using data on the near universe of US job vacancies, this paper studies the impact of the opioid crisis on employers' job skill requirements. Specifically, we investigate the effect of the reformulation of OxyContin, which represents one of the most substantial reductions in the availability of abusable prescription opioids. Prior studies have documented that the reformulation resulted in a large transition from prescription opioids to more dangerous illicit opioids. Using a difference-in-differences event study design that exploits firm-level variation in exposure to reformulation, we show that this transition toward illicit opioids has reduced employment at the firm level. Furthermore, we find that firms have increased requirements for cognitive and computer skills in response to this crisis. Our findings emphasize the distributional consequences of this crisis: less-skilled workers may experience a disproportionate impact from the increased skill requirements, even among workers without a history of opioid use disorders.

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1 Introduction

The opioid epidemic continues to be a serious public health crisis in the United States. Between 1999 and 2019, opioid-involved overdose deaths increased by over fivefold, resulting in about half a million deaths from opioid overdose ([National Institute on Drug Abuse, 2023](#)). In 2019 alone, nearly 50,000 died from opioid overdose, surpassing the numbers resulting from motor vehicle accidents or breast cancer ([Centers for Disease Control and Prevention, 2021](#)).¹ Beyond the widely studied health and mortality consequences, recent research has investigated the far-reaching impacts of the opioid crisis on various outcomes, including crime ([Deiana and Giua, 2021](#); [Mallatt, 2022](#)), children well-being ([Ziedan and Kaestner, 2020](#); [Evans et al., 2022](#); [Buckles et al., 2023](#)), housing market ([Custodio et al., 2023](#); [D’Lima and Thibodeau, 2023](#)), and consumer and municipal finance ([Cornaggia et al., 2022](#); [Jansen, 2023](#)). Of particular interest to researchers and policymakers have been its implications for the labor market, given its potential to have substantial impacts on both workers and employers. Indeed, opioid use has broad workplace relevance, with 75% of employers reporting that their workplace had been directly affected by employee opioid use ([National Safety Council, 2019](#)).²

A growing literature has investigated the causal impact of the opioid crisis on the labor market, where most studies focus on the equilibrium employment effects and labor supply consequences ([Krueger, 2017](#); [Currie et al., 2019](#); [Savych et al., 2019](#); [Harris et al., 2020](#); [Cho et al., 2021](#); [Park and Powell, 2021](#); [Aliprantis et al., 2023](#); [Beheshti, 2023](#)). However, there remains a significant gap in understanding how this crisis affects employers in their recruitment and screening decisions. One important strategy employers may use to adapt to the challenges posed by the opioid crisis is to adjust job skill requirements—tools that can help screen job candidates. Recent survey evidence suggests that firms use skill requirements not only to describe job tasks but also as a screening device: skills-based hiring is widespread among employers, and two-thirds of those using it apply it at the initial screening stage ([National Association of Colleges and Employers, 2024](#)).³

In this paper, we study the impact of the opioid crisis on job skill requirements for new hires

¹The opioid crisis worsened during the COVID-19 pandemic and remained at historically high levels through 2023. CDC data show that the United States recorded over 100,000 drug overdose deaths in 2023, nearly 80,000 of which involved opioids ([Simha et al., 2023](#); [Centers for Disease Control and Prevention, 2025](#)).

²Of all the respondents, 38% have experienced absenteeism or impaired worker performance, and 31% have had an overdose, arrest, a near-miss, or an injury because of employee opioid use.

³A summary of the survey results is available at [the NACE website](#) (accessed March 2026).

using data on the near universe of US online job postings. To estimate the causal impact, we focus on an intervention that inadvertently shifted users from prescription opioids towards riskier and unregulated illicit opioids, including heroin. In 2010, Purdue Pharma introduced a reformulated version of OxyContin in an effort to increase its resistance to misuse. OxyContin was a widely misused prescription medication in the 2000s, and the reformulation represents one of the largest reductions in the availability of abusable prescription opioids. This reformulation provides a unique opportunity to study the effects of a shock to the drug market that was plausibly unrelated to direct changes in the labor market.

Previous studies have documented that this supply-side intervention induced opioid-dependent individuals to switch to illicit opioids with higher addictive potential, such as heroin and other substances. Prior evidence indicates that the reformulation of OxyContin was associated with the growth of heroin markets, increased rates of heroin-related crime, heroin-involved overdose deaths, and hepatitis B and C infections, highlighting both the transition toward illicit opioids and their greater health risks ([Alpert et al., 2018](#); [Beheshti, 2019](#); [Evans et al., 2019](#); [Powell et al., 2019](#); [Mallatt, 2022](#)). Importantly, evidence suggests that the reformulation expanded demand in illicit opioid markets and generated spillovers throughout those markets ([Powell and Pacula, 2021](#)). As these markets grew and suppliers adapted, heroin and other potent opioids likely became more widely available, reaching not only existing prescription opioid users but also people without prior access to prescription opioids. In line with this interpretation, [Powell and Pacula \(2021\)](#) find that areas more exposed to the reformulation experienced greater increases in overall opioid-related mortality.

The expansion of illicit opioid markets may therefore have had important labor market consequences. Because illicit opioid use is more strongly associated with health risks and legal consequences than prescription opioid use, employers may view workers at risk of misuse as more costly to hire. Even when only a small share of applicants actually use illicit drugs, employers may still respond to perceived opioid-related risks among a broader pool of workers because drug use is hard to observe and because of uncertainty about future drug use, risk aversion, limited understanding of substance use disorders, and liability concerns.

These considerations may increase employers' incentives to screen out applicants perceived to be at higher risk. One way employers may do so is to use applicants' job skill levels as a proxy

for opioid-related risk.⁴ This paper examines whether employers adjust skill requirements in job postings in response to changes in local opioid use, building on prior work showing that firms use skill requirements as an important hiring tool (Michaels et al., 2014; Modestino et al., 2016; Burstein and Vogel, 2017; Hershbein and Kahn, 2018; Carluccio et al., 2019; Modestino et al., 2020; Ballance et al., 2020; He et al., 2021; Clemens et al., 2021; Alekseeva et al., 2021).⁵

To formalize how employers respond to the transition toward illicit drug use and utilize skill requirements as a screening tool, we first present a conceptual framework by adopting a search model inspired by Morgan and Várdy (2009). In this model, an employer seeks to fill a job vacancy for a specific task. Employers can observe the candidate’s skill level but cannot directly observe whether a job candidate misuses opioids. Instead, employers know that less-skilled workers are more likely to be drug users than more-skilled workers. Combined with a noisy signal about the candidate’s productivity from the résumé, employers form posterior beliefs about the likelihood that the candidate is a drug user and make hiring decisions by comparing these beliefs against an optimal threshold. Employers only learn whether a candidate is a drug user after hiring them. In this context, the model predicts that an increase in the likelihood of opioid misuse among low-skilled candidates, as a result of the opioid crisis, leads employers to raise their hiring standards. We also show that this hiring strategy reduces the overall share of less-skilled workers, regardless of whether they misuse opioids.

To examine these predictions empirically, we construct unique firm-level data that measure how firms’ employment and skill requirements have changed following the transition towards illicit opioids. These firm-level data are constructed from two sources. First, we use online job posting data from Lightcast over the years 2007 to 2019.⁶ The data provide detailed information on job positions and specific skill requirements, covering both quantitative and qualitative skills. Second, we complement this dataset with firm-level employment data obtained from the Compustat North America Database. For our analysis, we focus on publicly traded companies, whose shares are available for anyone to buy and sell on the open market. Publicly traded companies account for

⁴This is consistent with evidence that opioid use and opioid-related fatalities disproportionately affect individuals with lower educational attainment (Case and Deaton, 2015; Altekruze et al., 2020), suggesting that skill levels—such as cognitive and computer skills, education, and experience—may serve as observable proxies for opioid-related risk.

⁵Section 2.5 discusses the limitations of drug testing as a screening tool and outlines additional channels, beyond screening, through which the opioid crisis may influence employers’ skill requirements.

⁶Note that the Lightcast job posting data are unavailable for 2008 and 2009.

one-third of U.S. employment in the non-farm business sector (Davis et al., 2006).

To estimate the causal effects of the reformulation on job skill requirements and employment, we use a difference-in-difference event study design that exploits geographic variation in local prescription opioid use prior to the reformulation, following an approach widely adopted in the literature (Alpert et al., 2018; Evans et al., 2022). The idea is to investigate whether firms in counties with higher initial prescription opioid use—which are likely more affected by the shift towards illegal substances resulting from reformulation—experienced larger changes in their job skill requirements. Specifically, we construct a firm-level exposure measure based on pre-reformulation local prescription opioid use by aggregating exposure across a firm’s establishments using a weighted average. The key identification assumption is that, for any exposure level, the observed outcome path of firms at that exposure level is the counterfactual path for firms at other exposure levels had they instead experienced that same exposure level, which Callaway et al. (2024) refer to as the *strong parallel trends assumption*. Because this assumption cannot be directly tested, we assess the plausibility of this assumption by examining pre-treatment trends and conducting robustness checks, including estimating an event study specification based on a binary treatment definition.

Our findings are as follows. First, we show that the transition toward illicit opioids reduces employment at the firm level, consistent with findings from prior studies looking at aggregate-level local employment (e.g., Park and Powell (2021)). Our estimates reveal that a one standard deviation increase in firm-level exposure to the reformulation (equivalent to an additional 0.22 per capita opioid prescriptions in the pre-reformulation period) resulted in a 5.1 percent decrease in employment at the firm level. Second, we then show that the transition toward illicit opioids has significant and long-lasting positive impacts on skill requirements. Our results indicate that a one standard deviation increase in firm-level exposure to the reformulation led to a 7.9 percent increase in the average number of cognitive skills, such as statistical analysis, mathematical capability, and industry knowledge, and a 5.4 percent increase in the average number of computer skills required in an online job posting. We find little evidence indicating that firms adjust their education and experience requirements following the reformulation, although our heterogeneity analyses reveal significant effects within specific subgroups.

An important question is whether the observed upskilling reflects compositional change, with

firms posting more high-skill jobs, or increases in skill requirements within existing job categories. To distinguish between these two channels, we construct a firm-occupation-year panel and estimate regressions with firm-occupation and occupation-year fixed effects. This specification isolates within-firm changes in skill requirements for the same occupation group (6-digit SOC) over time. We find evidence of both channels, but the increase in cognitive skill requirements is driven primarily by within-firm-occupation upskilling. This suggests that employers are raising skill requirements within the same occupation, not just changing the composition of jobs posted, which is consistent with the interpretation that screening is one important mechanism.

To understand how the effects of reformulation vary based on different characteristics of firms and local labor markets, we examine heterogeneity in relation to factors such as firm size, educational requirements, minimum wage levels, and local employment protection laws. We document the following key findings. First, we find that larger firms raise computer skill requirements while small and medium-sized firms increase cognitive skill requirements in response to the opioid crisis. A possible explanation is that cognitive skill requirements may be less costly to impose than technical skill requirements, given that higher computer skill requirements may require small and medium-sized firms to offer higher compensating wages. Second, we show that upskilling effects are more pronounced among firms with lower baseline education requirements. This finding aligns with evidence that opioid use and opioid-related mortality are more prevalent among individuals with lower educational attainment ([Case and Deaton, 2015](#); [Altekruse et al., 2020](#)). Finally, we find that upskilling effects are more pronounced among firms located in areas with higher minimum wages or stronger employment protection legislation. This suggests that where hiring and separation costs are higher, firms face stronger incentives to screen applicants carefully at the hiring stage.

One important identification concern is the presence of confounders correlated with baseline local opioid use rates that may independently influence skill requirements. Of particular concern is the 2008 Great Recession, which preceded the reformulation and represents a plausible source of such confounding. We address this in three ways. First, we directly control for the interaction between local Great Recession exposure and the full set of year dummies in all regressions throughout the paper. Even after accounting for this exposure, upskilling remains statistically

significant. Second, a firm’s exposure to the Great Recession and its exposure to the OxyContin reformulation are only weakly correlated ($r = 0.08$). Third, in our robustness analysis, we estimate an event study using half-yearly data, with the first half of 2010—one half-year before the reformulation—as the reference period. In this exercise, we find no evidence of pre-existing differential trends in outcomes across areas with varying reformulation exposure, alleviating concerns that our results are driven by the Great Recession or other pre-reformulation shocks correlated with baseline opioid use.⁷

Another concern is that our measure of firm-level exposure to the reformulation may be confounded by other concurrent firm-specific factors. To address this, we conduct a firm-by-state level analysis, aggregating establishments owned by the same firm within a state into a single entity. This analysis includes entity fixed effects as well as firm-by-year fixed effects to absorb firm-specific time-varying shocks. We find strong evidence of upskilling even when comparing entities of the same firm operating in states with different levels of reformulation exposure.

Our paper contributes to several strands of literature. First, we add to the literature studying the causal impact of opioid use on the labor market. Prior work uses geographic variation in opioid prescribing and documents that a higher local prescription rate is associated with lower labor market participation rates (Krueger, 2017; Harris et al., 2020; Aliprantis et al., 2023).⁸ Other work exploits variation generated by the OxyContin reformulation or policies aimed at reducing misuse of prescription opioids and documents that the opioid crisis is associated with lower employment and lower labor force participation rates (Cho et al., 2021; Park and Powell, 2021; Beheshti, 2023).⁹ Recent work shows that beyond its effects on labor supply along the extensive margin, opioid use is associated with higher employee absenteeism (Armando et al., 2019) and reduced on-the-job productivity within the military (Alpert et al., 2022). While

⁷While earlier work attributed increases in “deaths of despair”—including suicide and poisoning by alcohol and drugs—to socioeconomic despair (e.g., Case and Deaton, 2015), more recent work shows that short-term changes in local economic conditions explain only a small fraction of such increases (Ruhm, 2019; Deaton and Case, 2020; Currie and Schwandt, 2021). This further mitigates concerns about confounding from local economic conditions in our setting.

⁸In contrast, Currie et al. (2019) find a small and positive impact of opioid prescribing on women’s employment with no effect on men’s employment. They use the prescribing rates for adults aged 65 and older in a county as an instrument for the prescribing rates among younger adults.

⁹Park and Powell (2021) and Cho et al. (2021) show that the transition from prescription opioids to illicit opioids triggered by the OxyContin reformulation led to declines in employment, labor force participation rates, hours worked, and earnings. Beheshti (2023) documents that the rescheduling of hydrocodone, which reduced access to it, resulted in higher labor force participation rates.

previous work has primarily focused on the equilibrium employment effects and labor supply outcomes, we provide new evidence on how employers adjust job skill requirements in response to the opioid crisis. To the best of our knowledge, this is the first paper to examine the impact of the opioid crisis on skill requirements using the U.S. job posting data.

Second, our study contributes to the extensive literature examining the causes and consequences of the opioid crisis. This literature has studied the roles of opioid policies, physicians, manufacturers, insurers, and pharmacists in contributing to the opioid crisis, as thoroughly reviewed in [Maclean et al. \(2020\)](#). A small but growing literature has looked at the effect of the crisis on firms and their responses. [Ouimet et al. \(2025\)](#) find that increased opioid prescriptions are associated with reduced employment and firm value, and firms substitute relatively scarce labor with capital, particularly when they face fewer financial constraints. [Chen et al. \(2021\)](#) document that the opioid epidemic adversely affects local firms' innovation. We complement this literature by documenting a new margin of firm adjustment: skill requirements in job postings.

Third, our paper also contributes to the literature investigating the factors that influence firms' skill requirements. Prior research suggests that firms' skill requirements are influenced by the business cycle ([Hershbein and Kahn, 2018](#)), the pool of labor supply ([Modestino et al., 2016, 2020](#)), technological progress ([Michaels et al., 2014](#); [Alekseeva et al., 2021](#)), international trade ([Burstein and Vogel, 2017](#); [Carluccio et al., 2019](#); [He et al., 2021](#)), and labor market institutions ([Ballance et al., 2020](#); [Clemens et al., 2021](#)). We add to this literature by examining this question in the unique context of the opioid crisis, which may reduce the quality of labor supply. Our findings suggest that firms respond to this decline in labor quality by tightening their hiring standards.

Lastly, our study is closely related to the broad literature on statistical discrimination in labor markets. [Becker et al. \(2019\)](#) provide evidence of hiring discrimination against women based on potential and realized fertility, highlighting how employers may discriminate under uncertainty about applicants' future productivity. [Doleac and Hansen \(2020\)](#) show that policies that prevent employers from asking questions regarding job applicants' criminal records during the initial application process reduce employment among young Black men without a college degree by 5 percent. Relatedly, [Bartik and Nelson \(2024\)](#) find that banning employers from accessing credit

histories of job applicants reduces job-finding rates for Black job-seekers by 3 percentage points. Another work by [Cortés et al. \(2022\)](#) documents that bans on the use of credit reports in hiring decisions lead to reductions in job vacancies, especially for positions involving routine tasks. This body of evidence raises the possibility that our upskilling results partly reflect statistical discrimination, as employers may use skill requirements as a proxy for opioid-related risk among applicants.

Our findings have important policy implications. First, they highlight the *distributional consequences* of the opioid crisis: by raising skill requirements for new hires, employers disproportionately disadvantage less-skilled workers, including those with no history of opioid use disorder. Second, our results suggest that addressing the opioid crisis calls for diversifying policy responses beyond prevention and treatment, such as occupational training programs that help less-skilled workers adapt to higher skill requirements.

The remainder of the paper proceeds as follows. Section 2 provides background on the OxyContin reformulation and its potential impacts on hiring decisions. Section 3 presents a conceptual framework that links changes in opioid-related risk in the local labor market to employers' screening strategies. Section 4 describes the data and key variables. Section 5 outlines the empirical strategy. Section 6 presents the main results, heterogeneity analyses, and robustness checks. Section 7 discusses policy implications and concludes.

2 Background

2.1 The Three Waves of the Opioid Crisis

Opioids are a class of drugs that includes both natural and synthetic substances used to reduce pain. Opioids range from common prescription medications such as oxycodone and morphine to illicit substances such as heroin and fentanyl. While opioids can effectively manage pain when used properly, they can be dangerous when used inappropriately due to their potential for recreational and harmful use, along with associated risks of morbidity and mortality. Moreover, opioids are often used in combination with alcohol or other depressant medications like benzodiazepines or tranquilizers, which can greatly increase the risk of overdose.

The United States has been experiencing a devastating opioid crisis, unfolding across three waves of overdose deaths. As illustrated in the overdose trends in Figure 1, these three waves are linked to different types of opioids: (1) prescription opioid pills (labeled as “Commonly prescribed opioids” in Figure 1), (2) heroin, and (3) synthetic opioids excluding methadone (labeled as “Other synthetic opioids” in Figure 1). The first wave, beginning around the year 2000, witnessed deaths from prescription opioids steadily increasing until 2011 and remaining relatively stable thereafter. The second wave was characterized by a sharp increase in heroin-related deaths beginning around 2010. This surge coincided with an intervention in 2010 when Purdue Pharma released a reformulated, abuse-deterrent version of OxyContin. The transition from the second wave to the third, since 2013, has been marked by a surge in fatalities associated with illicitly-made fentanyl. Overall, Figure 1 suggests that the shift from prescription to illicit opioids, which became especially pronounced after the reformulation of OxyContin, was followed by a large and persistent rise in overdose deaths involving illicit opioids.

2.2 The Reformulation of OxyContin

OxyContin, introduced by Purdue Pharma in 1996, is a brand-name version of the extended-release form of oxycodone that acts for 12 hours. Purdue Pharma aggressively marketed OxyContin to primary care providers for the treatment of non-cancer chronic pain, actively encouraging more lenient prescribing standards (Van Zee, 2009). This marketing strategy led to OxyContin being prescribed to a broader population, and as a result, OxyContin’s sales skyrocketed from \$48 million in 1996 to nearly \$1.1 billion by 2000 (Van Zee, 2009). The widespread availability of OxyContin was associated with a rise in its misuse, diversion, and addiction rates, making it one of the most widely misused drugs in the U.S. by 2004 (Cicero et al., 2005). Recent studies suggest that its introduction and aggressive promotional targeting account for a substantial share of the increase in prescription opioid supply and overdose deaths since 1996 (Alpert et al., 2022; Arteaga and Barone, 2022).

In response to growing misuse rates, Purdue Pharma introduced an abuse-deterrent formulation of OxyContin tablets in April 2010. The abuse-deterrent formulation was designed to make the pill difficult to break, crush, or dissolve. Reformulated OxyContin became commercially available

in August 2010, with the distribution of the original formulation ending within the same month. By December of 2010 and 2011, the reformulated OxyContin constituted 90% and 99% of all OxyContin prescriptions dispensed, respectively (Beachler et al., 2022).

Appendix Figure A1 presents national trends in the legal distribution of OxyContin and oxycodone from January 2008 to December 2012, using data from the Drug Enforcement Administration's (DEA) Automated Reports and Consolidated Ordering System (ARCOS).¹⁰ In both panels, blue triangles represent the per capita Morphine Equivalent Dose (MED) of OxyContin, and black hollow circles indicate the per capita MED of oxycodone. Panel (a) focuses on high-dosage OxyContin, defined as 80 mg OxyContin, while panel (b) presents trends in the total distribution of OxyContin.

As both panels illustrate, the legal distribution of OxyContin fell sharply in both levels and trends immediately following the release of the new formula in August 2010. As panel (a) shows, this decline was driven largely by reductions in high-dosage OxyContin, the formulation most susceptible to misuse (Janssen and Zhang, 2023). In both panels, total oxycodone—a broader category that includes OxyContin and other oxycodone products—shows an immediate negative trend break following the reformulation, followed by a gradual and persistent downward trend.

2.3 Transition Towards Illicit Opioids

Consequences of the OxyContin reformulation. The reformulation of OxyContin reduced the availability of abusable prescription opioids. However, this reformulation had unintended consequences, leading to a large shift from prescription opioids to illicit opioids. A study by Cicero and Ellis (2015) surveyed 153 recreational OxyContin users, finding that 33% of them switched to other substances due to the reformulation, with 70% of this group moving to heroin. In line with these survey results, subsequent research suggests that while the reformulation reduced OxyContin misuse, it inadvertently led to an increase in fatal overdoses involving heroin and other illicit drugs (Alpert et al., 2018; Evans et al., 2019; Powell and Pacula, 2021).

Beyond increasing illicit drug use among existing users, the shift toward illicit opioids may

¹⁰ARCOS records certain controlled substances from their point of manufacture through commercial distribution channels to the point of sale or distribution at the dispensing/retail level. This includes hospitals, retail pharmacies, practitioners, mid-level practitioners, and teaching institutions.

have broader effects on both existing users and the broader community in several ways. First, increased heroin use raises the risk of blood-borne infections, as heroin users often inject drugs and share needles and syringes. Prior studies have reported increases in hepatitis B and C infections in regions with a history of higher OxyContin use following its reformulation (Beheshti, 2019; Powell et al., 2019). Second, this shift contributed to the expansion of the illicit drug market, potentially broadening access to potent illicit opioids among individuals who could not previously obtain opioids through legal channels. This suggests that the effects of illicit market expansion may reach beyond existing opioid users. Consistent with this view, Powell and Pacula (2021) show that areas more exposed to the reformulation experienced disproportionate growth in fatal overdoses involving synthetic opioids such as illicit fentanyl (as well as other substances such as cocaine) and an increase in overall opioid-related mortality. Third, the expansion of the illicit drug market may increase the criminal activity associated with it. Mallatt (2022) documents increases in heroin possession and in the number of heroin dealers in areas where OxyContin prescriptions were more widespread.

Evidence from Google Trends. Figure 2 provides complementary evidence of a sharp transition from prescription opioids to illicit opioids following the reformulation. Using Google Trends, we plot search intensity for "OxyContin," "Heroin," and "Fentanyl" from 2004m1–2019m12, normalized relative to January 2008. The monthly frequency of this series allows us to track changes in opioid-related attention around the reformulation across opioid categories. The figure shows an immediate decline in searches for "OxyContin" following the reformulation. This decline coincides with a sharp rise in searches for "Heroin," which reach a peak in 2017. Searches for "Fentanyl" begin to increase around 2013 and continue to rise thereafter, while heroin searches decline after their 2017 peak. Overall, these patterns indicate an immediate and sharp post-reformulation shift from OxyContin to heroin, followed by a growing role for illicit fentanyl in later years. For comparison, panel (b) plots search intensity for "Marijuana," which remains relatively stable throughout this period.

2.4 Employer Perspectives on Opioid Use: Concerns and Workplace Responses

Employer concerns around employee opioid use. According to the National Safety Council’s (NSC) 2019 National Employer Survey, 75% of employers report that their workplace has been directly affected by employee opioid use ([National Safety Council, 2019](#)).¹¹ Figure 3 presents a subset of NSC survey results summarizing employer concerns related to employee opioid use and their perceptions of its impact on job performance. Panel (a) displays the share of respondents who view each workplace issue as a “major” or “minor” concern, with the combined percentage shown above each bar. Light blue shading highlights three opioid-related issues. Panel (b) shows employer perceptions of the potential impact of prescribed opioid use on job performance. Respondents were asked to rate their agreement with the statement, “Opioids taken as prescribed can impair job performance.” Response options are ordered from strongest to weakest agreement, with the corresponding share of respondents shown in each bar.

Panel (a) of Figure 3 shows that the most commonly cited concern among employers is the ability to hire qualified workers, with 90% reporting it as either a major or minor concern. Other top concerns include employee benefits costs (86%) and worker compensation costs (86%). Opioid-related issues also rank prominently: 83% of employers expressed concern about illicit opioid use, and 79% and 75% were concerned about legal prescription opioid use and the illicit use or sale of opioids, respectively. These results suggest that while traditional workforce and cost issues remain top of mind, opioid misuse is also perceived as a major workplace challenge.

Panel (b) of Figure 3 highlights employers’ perceptions of the impact of prescription opioids on job performance. A combined 86% of respondents either strongly agree (43%) or somewhat agree (43%) that opioids taken as prescribed can impair job performance. In contrast, only a small fraction expressed neutrality (9%) or disagreement (5% in total). These results indicate a broad consensus among employers that even medically appropriate opioid use may negatively affect workplace functioning.

¹¹The survey was based on responses from 526 employer decision-makers at U.S. organizations with 50 or more employees.

Responses to employee substance use by substance type. The NSC survey also indicates that employers respond more severely to illicit opioid use than to the misuse of legal substances such as prescription medications or alcohol. Appendix Table A1 summarizes how employers respond to different types of substance misuse in the workplace. Responses are shown by substance type, including prescription opioids, prescription stimulants, benzodiazepines, alcohol, legal marijuana, illicit marijuana, illicit opioids, heroin/fentanyl, and other illicit drugs.¹²

The table suggests that employers are generally more lenient toward prescription opioid misuse than illicit drug use. For instance, 10% of employers report ignoring prescription opioid misuse, and 41% say they would return the employee to their position after treatment. In contrast, only 2–3% of employers say they would ignore heroin or fentanyl use, and just 25% report a return-to-position approach. Instead, a majority of employers indicate they would dismiss employees for heroin (54%), fentanyl (55%), or other illicit opioid use (49%).

2.5 Potential Impacts of Reformulation on Firm Skill Requirements

Trends and limitations of drug testing as a hiring screen. In response to rising concerns about illicit opioid use, employers may seek to screen out potential users through pre-employment drug testing. Figure 4 provides suggestive evidence that employer attention to drug screening increased over time. Panel (a) plots Google Trends search intensity for "drug test" and "drug screen," normalized to 2008m1, over the period 2004m1–2019m12. Panel (b) plots two series we construct from our Lightcast job posting data: the quarterly share of online job ads mentioning "drug screen" and the quarterly share mentioning "drug test," covering 2010q1–2019q4.¹³ The August 2010 reformulation is marked by a vertical dashed line in both panels.

In panel (a), search intensity for both terms increased by roughly 150 to 250 percent relative to the 2008m1 baseline by the late 2010s. In panel (b), the share of job ads mentioning "drug screen" rose from about 2.1 percent in 2010q1 to about 3.4 percent by 2019q4, while the share mentioning "drug test" increased from about 1.7 percent to about 2.4 percent over the same period. These

¹²Respondents were asked: "Which of the following would you say best reflects your organization's approach to an employee who is found to be misusing...?" Employers selected one of the following options: ignore the problem, return the employee to their position after treatment, ensure the employee is carefully monitored for the remainder of their employment, relocate the employee to a position of lesser responsibility, or dismiss them.

¹³These data are available only from 2010 onward.

trends suggest that drug screening became an increasingly salient feature of hiring practices during the period of rising illicit opioid use.

Despite this increasing employer attention to drug screening, pre-employment drug testing has important limitations that undermine its effectiveness as a screening tool. First, testing is typically conducted only after a conditional offer is made, limiting its deterrent power. Legal restrictions also vary across states—for example, Iowa requires employers to notify applicants of testing in job postings and to list the substances tested. Besides, drug tests often cannot distinguish between illicit and prescribed opioids, making it difficult to identify heroin use specifically. Moreover, substances like heroin are detectable for only 2 to 8 hours after use, reducing the likelihood of detection. Finally, the rise in drug test cheating further undermines the reliability of these tests in identifying actual substance misuse.¹⁴

Skills-based hiring as a screening tool. These limitations of drug testing suggest that employers may also rely on other screening tools. One important screening device is skill requirements. The National Association of Colleges and Employers (NACE) reports in its Job Outlook 2025 survey that 64.8% of employers use skills-based hiring for new entry-level hires, and more than half do so always or most of the time ([National Association of Colleges and Employers, 2024](#)). Among employers that use skills-based hiring, two-thirds apply it at the screening stage and 90% at the interview stage. Firms implement skills-based hiring through competency-based job descriptions, skills-based interview rubrics, and candidate assessment tools. Although this evidence is not specific to opioid exposure, it is consistent with our interpretation that skill requirements may help screen applicants when direct screens—such as drug testing—are imperfect, costly, or legally constrained.

Potential impacts on firm skill requirements. The shift from prescription to illicit opioids may have increased employer concern about opioid use among job applicants, as illicit opioid use is more strongly associated with criminal justice involvement and health risks, including illicit drug

¹⁴It is also challenging to identify potential legal opioid users once they are employed. According to the NSC 2020 survey, 34% of employers report that employee responsibility to disclose prescription opioid use is not covered in their organization's written policies.

overdoses and blood-borne diseases.¹⁵ In turn, this shift could affect job skill requirements through several channels.

First, employers may use skill requirements as a screening tool, adjusting hiring criteria to avoid applicants perceived to be at greater risk of opioid use. Previous research suggests that opioid use and fatalities are more prevalent among individuals with lower levels of education and those from lower socioeconomic backgrounds (Case and Deaton, 2015; Altekruse et al., 2020). Employers may view skill levels as an indicator of potential risks associated with opioid use and overdoses.

Second, employers may adjust skill requirements in response to changes in applicant quality by changing the composition of the workers they seek to hire. For instance, employers may expand postings for higher-skilled jobs or replace postings for lower-skilled jobs with those for higher-skilled jobs in order to maintain the expected productivity of new hires.

Third, the reformulation may reduce labor supply and limit the pool of job applicants. How firms adapt to this labor shortage is ex ante ambiguous. Firms may choose to relax hiring standards to fill positions more quickly. On the other hand, they may raise skill requirements to maximize productivity with fewer new hires.

Importantly, even when only a small subset of job candidates are at high risk of illicit drug use, firms may respond disproportionately to the perceived risk of opioid use among applicants. This response may be attributed to factors such as uncertainty about applicants' future substance use, low tolerance for risk related to employee drug use, lack of understanding about substance use disorders, and concerns about legal liability. Furthermore, if the reformulation made opioid-related risk more salient to employers and increased concern about opioid use more broadly, the overall response in skill requirements could be larger, as employers may adjust hiring criteria in response to opioid-related risk more generally, not only to the risk associated with illicit opioid use.

Beyond these channels, the reformulation may also affect skill requirements through a selection channel: firms that employ more lower-skilled workers may be more likely to exit after the reformulation, which would mechanically raise average skill requirements among surviving firms. In this paper, however, we focus on publicly traded firms, for which sample attrition is

¹⁵Increased illicit opioid use among current employees may also shape expectations for job candidates. For instance, employers could extrapolate the changes in productivity observed among current workers onto job seekers, thereby adjusting their expectations for candidates' future productivity.

relatively limited.¹⁶ Our discussion therefore focuses on changes in skill requirements within firms following the reformulation, rather than on compositional changes driven by differential firm exit.

3 Conceptual Framework

As discussed in the previous section, one key potential mechanism is screening: employers may use skill requirements to screen out higher-risk applicants in response to perceived opioid-related risk. In this section, we formalize this mechanism by presenting a theoretical framework for how employers adjust their hiring standards in response to the opioid crisis. We focus on the screening mechanism because it is most consistent with our empirical evidence and, relative to other channels, is less immediately intuitive. A simple theoretical framework helps clarify how changes in perceived opioid-related risk can translate into adjustments in hiring standards.

We adapt the model structure from [Morgan and Várdy \(2009\)](#) to the context of job posting requirements and résumé screening. In our model, an employer seeks to fill a vacancy for a specific occupation. The employer reviews résumés from a pool of job applicants for this position. Each résumé review incurs a fixed cost k , representing the time and resources spent evaluating candidates.

We assume each candidate has two characteristics: skill level (γ) and productivity type (θ). Skill level can be either more-skilled or less-skilled, and is observable to the employer through credentials, education, and experience listed on résumés. In contrast, productivity type—which depends on whether the candidate is a drug user or not—is unobservable to the employer. There are two productivity types: productive workers (i.e., non-drug users) and non-productive workers (i.e., drug users). Non-drug users have high productivity and are able to perform the job effectively, and $\theta = 1$. Conversely, drug users have low productivity and cannot perform the job adequately, so $\theta = 0$. Since this type is unobservable, employers cannot distinguish between drug users and non-users based on observable résumé characteristics.

Let m_H and m_L denote the probabilities that a candidate applying for this specific occupation is more-skilled or less-skilled, respectively. For simplicity, we assume $m_H = m_L$. A key assumption of the model is that both more-skilled and less-skilled workers are capable of

¹⁶We discuss sample attrition in more detail in Section 4.

performing the job once hired; however, their actual productivity depends solely on their drug use status. This simplifying assumption allows us to isolate how increases in local drug use affect employers' hiring decisions when they must rely on imperfect signals to screen candidates. Among more-skilled workers, the probability of being highly productive (i.e., a non-drug user) is p_H , while for less-skilled workers, this probability is p_L . We assume more-skilled candidates are more likely to be non-drug users (i.e., $p_H > p_L$) and that employers know these probabilities.¹⁷

The timing of the model is illustrated in Figure 5. In the first period, the employer reviews a résumé from a randomly selected applicant for the specific job opening, observing their skill level, γ (more-skilled or less-skilled, based on certifications, education, and experience listed). The employer also observes other information in the résumé that serves as signals sent by the candidate about their underlying productivity type θ , represented by $S_\gamma = \theta + \varepsilon_\gamma$. The noise term ε_γ arises because these observable characteristics are imperfect predictors of actual drug use and job performance. The noise term ε_γ is assumed to follow a normal distribution: $\varepsilon_\gamma \sim \mathcal{N}(0, \sigma_\gamma^2)$. We assume that the signal noise is identical for both more-skilled and less-skilled applicants for this occupation, i.e., $\sigma_H = \sigma_L = \sigma$.

After reviewing the résumé and evaluating the signal from the applicant's qualifications, the employer forms a posterior belief about the likelihood that the candidate with skill level γ can perform the specific job effectively. Based on this belief, the employer decides whether to hire the candidate for the position.¹⁸

In period 2, if the candidate is not hired, the employer continues reviewing résumés for the same position with a new draw from the pool of workers. If the candidate is hired, their true productivity type θ is revealed through their work performance. If the candidate has high productivity (i.e., is a non-drug user, $\theta = 1$), the employer receives a net present value payoff of $\nu > 0$, and the worker is retained in the position. If the candidate has low productivity (i.e., is a drug user, $\theta = 0$) and is

¹⁷In practice, these probabilities reflect employers' perceptions of drug use prevalence among each skill group, which may differ from the actual shares. For instance, employers may overestimate the prevalence of drug use among less-skilled workers, or they may form beliefs based on the broader share of workers at high risk of drug use rather than actual users—either of which would result in perceived probabilities exceeding the true population shares. Our model does not explicitly distinguish between employers' perceived probabilities and the actual population shares.

¹⁸As discussed in Section 2, pre-employment drug testing has several limitations: it can typically only be conducted after a conditional job offer is made, has legal restrictions that vary by state, often cannot distinguish between illicit and prescribed opioids, and substances like heroin are detectable for only 2–8 hours after use. Given these constraints, we do not incorporate drug testing into our model and instead focus on how employers use skill requirements as a screening mechanism.

therefore unable to perform the job, the employer dismisses the worker and incurs a cost of c .

3.1 Posterior Beliefs

We first derive the employer's posterior belief that the worker with skill level γ can perform the job productively after receiving signal S . Using Bayes' formula, we have the equation below.

$$q_\gamma(S) = \frac{\phi\left(\frac{S-1}{\sigma}\right) \cdot p_\gamma}{(1-p_\gamma) \cdot \phi\left(\frac{S}{\sigma}\right) + p_\gamma \cdot \phi\left(\frac{S-1}{\sigma}\right)}, \quad (1)$$

where the $\phi(\cdot)$ denotes the probability density function of a standard Normal distribution. Equation (1) shows that after receiving a signal S , the employer can form a belief about the probability that a worker of kind γ (more-skilled or less-skilled) is capable of performing the job. Conversely, by using the inverse function, we can determine the signal S that corresponds to a given posterior belief q about the probability that the worker can perform the job. Importantly, because the mapping between signals and posterior beliefs is one-to-one, each signal uniquely identifies a posterior belief. The inverse function of equation (1) can be expressed as follows:

$$S_\gamma(q) = \frac{1}{2} - \sigma^2 \cdot \log \left[\frac{1-q}{q} \cdot \frac{p_\gamma}{1-p_\gamma} \right]. \quad (2)$$

Note that even though the employer knows the candidate's kind γ —that is, whether the worker is a more-skilled or less-skilled worker—the success probability $Q_\gamma = q_\gamma(S_\gamma)$ remains a random variable prior to the realization of the signal S_γ . Therefore, we can define the cumulative distribution function (CDF) of the success probability $G_\gamma(q)$ and the probability density function (PDF) as below.

$$G_\gamma(q) = p_\gamma \cdot \Phi\left(\frac{S_\gamma(q)-1}{\sigma}\right) + (1-p_\gamma) \cdot \Phi\left(\frac{S_\gamma(q)}{\sigma}\right) \quad (3)$$

$$g_\gamma(q) = \frac{\sigma}{q(1-q)} \left[p_\gamma \cdot \phi\left(\frac{S_\gamma(q)-1}{\sigma}\right) + (1-p_\gamma) \cdot \phi\left(\frac{S_\gamma(q)}{\sigma}\right) \right], \quad (4)$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ are the CDF and PDF of the standard normal distribution, respectively.

Before the employer observes the applicant's type γ , the CDF of the success probability, $G(q)$,

is the weighted average of the type-specific distributions:

$$G(q) = m_H G_H(q) + m_L G_L(q). \quad (5)$$

3.2 Value Function and Optimal Hiring Decisions

We next derive the firm's value function and characterize the optimal hiring rule. As explained by [Morgan and Várdy \(2009\)](#), the firm's optimal strategy in this setting takes the form of a threshold rule. Specifically, the firm chooses an optimal threshold $\underline{q}^*$ and hires a job candidate only if the posterior belief about the success probability based on the observed signal exceeds this threshold, regardless of the skill level of the job applicant.¹⁹

Let V^* denote the employer's expected payoff under an optimal threshold search strategy, and let $H(q, V^*)$ denote the expected payoff from hiring when the employer believes that the applicant performs the job successfully with probability q . Then the employer hires the applicant if and only if the expected payoff from hiring exceeds the continuation value of rejecting the applicant and proceeding to the next period. Hence, we can express the value function V^* as:

$$\begin{aligned} V^* &= \delta \int_0^1 \max \{H(q, V^*), V^*\} dG(q) - k \\ &= \delta \int_0^1 \max \{qv + (1 - q)(V^* - c), V^*\} dG(q) - k. \end{aligned} \quad (6)$$

Under a generic threshold rule $\underline{q}$, where the firm hires an applicant only when the posterior belief about success exceeds $\underline{q}$, the firm's expected payoff can be expressed as:

$$V(\underline{q}) = \delta \int_0^1 \max \{qv + (1 - q)(V(\underline{q}) - c), V(\underline{q})\} dG(q) - k. \quad (7)$$

Alternatively, using the equation above, the firm's expected payoff $V(\underline{q})$ can be expressed as:

$$V(\underline{q}) = \frac{\delta \int_{\underline{q}}^1 (vq + (1 - q)(-c)) dG(q) - k}{1 - \delta \left(1 - \int_{\underline{q}}^1 q dG(q)\right)}. \quad (8)$$

¹⁹The firm uses the same hiring threshold for both kinds of candidates (i.e., more-skilled and less-skilled candidates) because, conditional on q , the candidate's kind γ is irrelevant to the hiring decision ([Morgan and Várdy, 2009](#)).

Hence, the firm's problem is to choose the optimal $\underline{q}$, which can maximize $V(\underline{q})$ in equation (8). The optimal solution to equation (8) is characterized by Proposition 1.²⁰

PROPOSITION 1: *The optimal hiring strategy, $\underline{q}^*$, can be implicitly defined as*

$$\begin{aligned} \underline{q}^* = & \frac{c}{c + (1 - \delta)v + k} - \frac{c\delta}{c + (1 - \delta)v + k} (1 - m_H p_H - m_L p_L) \\ & + \frac{c\delta}{c + (1 - \delta)v + k} \int_0^{\underline{q}^*} G(q) dq. \end{aligned} \quad (9)$$

The opioid crisis likely reduces the share of productive workers in both skill groups, i.e., both p_H and p_L decrease. However, since opioid use and related fatalities are disproportionately concentrated among less-skilled workers, we assume that p_L decreases by a larger magnitude. For tractability, we focus on the case where only p_L decreases. The following policy implication is derived from this assumption and Proposition 1, with the proof provided in Appendix B.

POLICY IMPLICATION 1: *When the proportion of low-productivity workers (drug users) increases in the population of less-skilled applicants for this occupation (i.e. p_L decreases), the optimal hiring threshold, $\underline{q}^*$, increases.*

Intuitively, this implies that if the success CDF of low-skilled workers, $G_L(q)$, does not shift too dramatically in response to a decline in the proportion of non-drug users, p_L , firms respond by raising their screening threshold $\underline{q}^*$. In our context, we interpret this as when the opioid crisis increases the share of drug users, particularly among less-skilled workers, firms respond by raising their hiring standards to better screen out drug users. In Appendix B, we show that this condition can be supported under reasonable parameter values.

3.3 Permanent Composition

Finally, we characterize the steady-state composition of worker types in the economy. Let r_γ denote the probability that a vacancy is permanently filled by a job applicant of kind γ , given that the firm follows an optimal screening strategy with threshold $\underline{q}$. The value of r_γ can be expressed recursively

²⁰The proof of Proposition 1 is provided in Appendix B.

as:

$$r_\gamma = m_\gamma \left\{ P_r \left[\left(1 - G_\gamma(\underline{q} \mid \theta_\gamma = 1) \right) + G_\gamma(\underline{q} \mid \theta_\gamma = 1) r_\gamma \right] + (1 - P_\gamma) r_\gamma \right\} \\ + (1 - m_\gamma) \left\{ r_\gamma \left[1 - P_{-\gamma} \left(1 - G_{-\gamma}(\underline{q} \mid \theta_\gamma = 1) \right) \right] \right\}. \quad (10)$$

Let's further define:

$$G_{\gamma\theta} \equiv G_\gamma(\underline{q} \mid \theta_\gamma = \theta) = \begin{cases} \Phi\left(\frac{S_\gamma(\underline{q})-1}{\sigma}\right) & \text{when } \theta = 1 \\ \Phi\left(\frac{S_\gamma(\underline{q})}{\sigma}\right) & \text{when } \theta = 0. \end{cases} \quad (11)$$

Then equation (10) can be expressed as:

$$r_\gamma = \frac{m_\gamma P_\gamma (1 - G_{\gamma 1})}{m_\gamma P_\gamma (1 - G_{\gamma 1}) + (1 - m_\gamma) P_\gamma (1 - G_{-\gamma 1})}. \quad (12)$$

From the expression for r_γ in equation (12), which gives the probability that a vacancy is permanently filled by a job applicant of a given type, we derive Policy Implications 2 and 3.²¹

POLICY IMPLICATION 2: *As the proportion of drug users among less-skilled workers increases (i.e., as P_L decreases), the permanent composition of less-skilled workers declines, given a recruiting threshold $\underline{q}$ (i.e. $\frac{\partial \gamma_L}{\partial P_L} > 0$).*

POLICY IMPLICATION 3: *As the recruiting threshold $\underline{q}$ increases, the permanent composition of less-skilled workers declines.*

Policy Implication 2 indicates that, holding the recruiting threshold fixed, an increase in the proportion of drug users reduces the permanent share of less-skilled workers in the population.²²

Policy Implication 3 further implies that an increase in the recruiting threshold also lowers the permanent share of less-skilled workers in the population.

²¹Proofs of the policy implications are provided in Appendix B.

²²Intuitively, a decline in p_L reduces the permanent share of less-skilled workers through two reinforcing channels. First, since the prior probability of being productive is lower, the employer's posterior belief is lower for any given signal S by Bayes' rule, meaning that less-skilled workers need to send a stronger signal to clear the same hiring threshold. Second, as the number of non-drug users in the less-skilled population declines, the distribution of signals sent by less-skilled workers shifts down, since each signal combines true productivity with noise. Both channels make it harder for less-skilled workers to meet the employer's hiring threshold, reducing their permanent share in the workforce.

Policy Implication 2 suggests that the opioid crisis reduces the share of less-skilled workers in the population. Policy Implications 1 and 3 together imply that the opioid crisis raises the recruiting threshold, which amplifies this effect. Both channels therefore point to a decline in the share of less-skilled workers. In our context, this provides an important insight: the opioid crisis raises hiring standards and increases the likelihood that less-skilled workers are screened out. In Section 6.1, we present empirical evidence consistent with this insight.

4 Data

This section outlines the datasets we use in our analysis. First, we describe county-level data on prescription opioid use. Second, we describe our firm-level data sets—Compustat North America Database and Lightcast Online Job Posting Data—and explain how we link these datasets at the firm level.

Prescription opioid use. Our data on county-level prescription opioid use are from the Centers for Disease Control (CDC). This dataset comprises an 85 percent sample of retail pharmacy providers, excluding hospitals. We explain in Section 5 how we derive firm-level exposure to the reformulation using these data on county-level prescription opioid use in the pre-reformulation period.

Skill requirements. Our skill demand measures are from the online job posting data provided by Lightcast, an employment analytics firm (Lightcast, 2010–2019). Lightcast collects job ads from about 40,000 online job boards and uses its own machine-learning algorithm to unify duplicate job ads and parse the postings into a systematic form. Lightcast claims that its database includes the near-universe of online job ads. The database has been widely used in academia and industry. It provides detailed information on each job posting, including job title, standard occupation classification (SOC), employer name, location, and employer industry. More importantly, Lightcast collects detailed job requirements of each job ad, such as education, work experience, and a list of skill requirements. There are more than 10,000 unique skill keywords in the skill requirements. The list of skills includes general skills (such as communication skills, teamwork, critical thinking, quality control, etc.), specific skills (such as foreign languages, legal

compliance, computer numerical control, revenue projections, etc.), and specific software names (such as SAP, Python, Java, SQL, Tensor Flow, ND4J, etc.).

It is worth noting that there are some limitations to our job posting data. Job vacancies in Lightcast may not accurately represent the overall employment distribution. Prior studies report that job vacancies in Lightcast are skewed toward certain areas of the economy, although this limitation also applies to other widely used job vacancy databases, such as the Job Openings and Labor Turnover Survey (JOLTS) (Lazear and Spletzer, 2012; Davis et al., 2013). Despite this limitation in representing the employment distribution, Lightcast is known to be consistent with overall labor market trends. For example, Hershbein and Kahn (2018) show that the national and industry trends in the number of online job postings from Lightcast closely track those of employment from the Current Population Survey (CPS), Occupational Employment and Wage Statistics (OEWS), and job vacancies from JOLTS.

Lightcast uses its algorithm to develop a robust skills taxonomy. We follow Deming and Kahn (2018), who create categories of skill requirements based on the skills taxonomy in a way that the categories are useful for economic research.²³ Appendix Table A2 lists the skill categories and the corresponding keywords or phrases that belong to each category. For instance, a cognitive skill should include keywords or phrases such as "problem-solving," "research," and "statistics." These keywords are chosen by Deming and Kahn (2018) to match the "non-routine analytical" job tasks that are classified by Autor et al. (2003) based on the O*NET database. A computer skill should include keywords such as "computer," "spreadsheets," or specialized software such as "Java," "SQL," and "Python."²⁴ We quantify a firm's demand for a particular skill by calculating the average count of skills within that skill category required in a job posting.²⁵

Employment. We complement job posting data with employment measures constructed from the Compustat North America Database (Compustat), which collects standardized financial and

²³Deming and Kahn (2018) also demonstrates that the skills in each category are prevalent in the job ads of relevant occupations.

²⁴Our computer skill category combines general computer skills and specific software skills, categorized separately in Deming and Kahn (2018). Computer skills are basic software, such as Microsoft Excel and PowerPoint, while specific software skills include names of specialized software.

²⁵If a job posting does not require such a skill, it is coded as zero. Therefore, our outcomes capture both the extensive margin (whether to require at least one skill within that category in a posting) and the intensive margin (how many skills to include for that skill category in a posting) responses.

employment information for all publicly traded U.S. firms. We exclude funds, trusts, and other financial vehicles (NAICS 525) from the sample.

Linkage process. Our analyses focus on publicly traded companies.²⁶ To identify publicly traded companies within our job posting data, we link the Compustat and Lightcast databases at the firm level. Unfortunately, there is no simple way to link these two datasets because there is no common firm identifier between them. Moreover, many firm names in the job posting data contain abbreviations or misspellings. For instance, a firm name "Micron Technology" can be expressed as "Micron," "Micron Tech," "Micron Incorporation," or "Micron Technology, Inc." in the job posting data. We use a fuzzy matching algorithm and employ another proprietary database called the Computer Intelligence Technology Database (CiTDB) to overcome this problem. CiTDB has covered 3.2 million establishments since 2010 and provides a firm structure and address for each establishment. Specifically, there is a unique firm ID, and one can identify a firm's headquarters, branches, and addresses. We first standardize company names using the algorithm provided by [Wasi and Flaaen \(2015\)](#). As an additional input for matching, we also construct a measure of industry linkage based on the input-output table. Then, we link Compustat and Lightcast using a fuzzy match based on the standardized names and industry linkage. Lastly, we compare the name and address of a matched establishment from Lightcast and that from CiTDB to ensure that the matched establishment is a branch of the matched firm from Compustat.

Using the linked dataset, we construct a firm-year panel on the following outcomes: (1) the average number of specific skills required in a job posting, including cognitive and computer skills; (2) the average years of schooling required in a job posting; (3) the average years of experience required in a job posting; and (4) employment level. Our sample covers 2007 and 2010–2019 for job posting outcomes,²⁷ and 2005–2019 for firm outcomes from Compustat. Our final firm-level

²⁶Limiting our focus to publicly traded companies offers two advantages. First, since Compustat data only cover publicly traded companies, focusing on these companies allows us to zoom in on those included in both Compustat and Lightcast datasets. Therefore, we can explore the impact of reformulation on a wide range of firm outcomes within the same group. Second, since job posting data lack a time-invariant company identifier, focusing on publicly traded firms facilitates better identification of the same firm. In the job posting data, there are instances where observations for the same company have varying names across years. However, publicly traded firms, on average, are typically much larger in size and have standardized names, making it easier to merge them into a single entity. As previously noted, publicly traded firms constitute one-third of U.S. employment in the non-farm business sector ([Davis et al., 2006](#)).

²⁷The Lightcast job posting data are unavailable for 2008 and 2009.

analysis sample comprises 2,104 publicly traded companies included in both databases.^{28,29} Table 1 reports summary statistics for the firm-level exposure measure and the main outcome variables used in the analysis.

5 Empirical Strategy

To explore the causal impact of the OxyContin reformulation on skill requirement responses, we employ difference-in-difference and event study designs that exploit variation in firm-level exposure to pre-reformulation local prescription opioid use. This section explains how we construct the measure of exposure to the OxyContin reformulation and describes our empirical models.

5.1 Measuring Exposure to the OxyContin Reformulation

County-level exposure. To quantify the causal impact of the OxyContin reformulation on skill demand, we leverage geographic variations in pre-intervention exposure to prescription opioids, the approach that has been suggested by [Alpert et al. \(2018\)](#) and [Evans et al. \(2019\)](#), and widely adopted in the literature. [Alpert et al. \(2018\)](#) use state-level variation by constructing a pre-intervention exposure measure based on the population-weighted rate of OxyContin misuse at the state level from 2004 to 2009, calculated using data from the National Survey on Drug Use and Health (NSDUH). Since this measure is only available at the state level, we follow the approach suggested by [Evans et al. \(2022\)](#) to construct a pre-intervention exposure measure at the county level. Specifically, we use the population-weighted mean number of all Schedule II opioid prescriptions per capita in each county for the years 2006 to 2009, obtained from CDC data. Unlike [Alpert et al.](#)'s measure, which is specific to OxyContin misuse, our county-level CDC measure captures local prescribing

²⁸Since our main focus is on the labor market, we exclude firm-by-year observations with missing employment information from the Compustat database. As a result, 6.3% of the total observations are dropped.

²⁹We do not impose any sample restrictions based on the number of years a firm is observed in the job posting data. The only explicit restriction is that we drop 60 firms (2.8% of the sample) observed only once, as such firms do not contribute to identification in specifications with firm fixed effects. Note that concerns about firm attrition and sample selection are limited in our setting for two reasons. First, we focus on publicly traded firms, for which sample attrition is relatively low. Only 7.76% of firms in our sample are observed for fewer than four years, and 74.3% are observed in all years or are missing in only one year over the sample period. Second, because our empirical model includes firm fixed effects, our estimates are identified from within-firm changes over time rather than differences across firms, which helps address concerns that the results are driven by compositional changes in the sample, for example, due to firm exit.

intensity for all Schedule II prescription opioids. It therefore reflects exposure to the legal supply of prescription opioids and can be viewed as a proxy for local availability of prescription opioids prior to the reformulation.

While this broader county-level measure offers a more precise representation of local variation in pre-intervention exposure to prescription opioids than state-level aggregates (Evans et al., 2022), we acknowledge important limitations. The CDC measure captures all Schedule II opioid prescriptions rather than OxyContin specifically, and reflects legal distribution rather than misuse. Because the reformulation targeted the misuse of OxyContin, our exposure variable is therefore subject to measurement error to the extent that opioid prescribing does not strongly correlate with actual OxyContin misuse. We nevertheless use the CDC prescribing measure because it provides the county-level geographic variation essential for constructing our firm-level Bartik-style exposure across establishments, an advantage unavailable with state-level misuse data alone.

Despite these limitations of our exposure measure, multiple pieces of evidence validate our approach. Appendix Figure A2 shows that when aggregated to the state level, our county-level CDC exposure measure correlates positively ($r = 0.53$) with Alpert et al.'s measure of OxyContin misuse rates from NSDUH data.³⁰ More importantly, Appendix Figure A3 demonstrates that higher CDC exposure predicts larger subsequent reductions in OxyContin misuse rates from 2008 to 2012 and greater increases in heroin mortality rates from 2008 to 2016,³¹ confirming that this measure captures areas more affected by the transition to illicit opioids following reformulation.³² In Section 6.3, we further assess robustness using the state-level sample with Alpert et al.'s direct measure of OxyContin misuse. Figure 6 displays geographic variation in pre-intervention per capita opioid prescriptions across counties using the CDC exposure measure.

Firm-level exposure. To investigate the impact of the intervention on firm-level outcomes, we create a firm-level exposure measure based on our county-level exposure measure. One challenge with this approach is that a single firm often has multiple establishments dispersed across different

³⁰Data on the NSDUH measure of OxyContin misuse are obtained from Alpert et al. (2018).

³¹The NSDUH data on OxyContin misuse are obtained from Alpert et al. (2018), and the data on heroin mortality are from the National Vital Statistics System (NVSS).

³²Similarly, Evans et al. (2022) validates the county-level CDC exposure measure by showing that counties with higher CDC opioid exposure experienced larger reductions in OxyContin misuse between 2008 and 2012, based on Alpert et al.'s state-level measure.

counties. To address this, we construct a firm-level exposure that combines exposure across the various establishments owned by the same firm. This Bartik-style measure is constructed by calculating a weighted average of the county-level opioid exposure measure across a firm's establishments based on each establishment's physical location. It is often challenging to obtain suitable weights for aggregating establishment-level outcomes into the firm level. To address this, we create weights based on the pre-intervention period number of job ads at the establishment level. One implicit assumption here is that the number of job postings represents the size of establishments, which is widely accepted in the literature. Appendix Figure A4 displays the histogram of the distribution of the firm-level exposure measure in our sample. Summary statistics for this exposure measure are reported in Panel A of Table 1. On average, during the pre-intervention period from 2006 to 2009, a firm is exposed to 0.7 Schedule II opioid prescriptions per capita, with a standard deviation of 0.22.³³

A firm-level analysis leveraging a firm-level exposure measure offers several advantages over local-level or establishment-level analyses. First, our firm-level approach allows us to distinguish between changes within firms and compositional changes across different firms.³⁴ This distinction is particularly important for identifying the mechanisms behind potential changes in the composition of the aggregate labor force. For example, if we observe that the shift towards illicit opioid use has changed the labor force composition at aggregate local levels, investigating whether this is due to within-firm adjustments (such as intentional changes in labor inputs) or shifts in the composition of firms with heterogeneous labor inputs can provide important evidence for understanding the underlying mechanisms.

Second, our firm-level exposure measure is less prone to unobserved local- or industry-level heterogeneity, thus implying fewer confounding effects related to the opioid shock. While the geographic variation in the opioid shock is concentrated in some local areas, which could also be subject to unobserved regional economic shocks, our firm-level shock mitigates this concern by smoothing the exposure to these areas. Note that we also net out industry-specific concurrent shocks by directly controlling for industry-by-time fixed effects in our empirical models, as

³³In addition, Appendix Figure A5 presents the average exposure by industry groups. Note that our analysis focuses on within-industry across-firm comparison rather than across-industry comparison.

³⁴Many prior studies have attempted to separately identify within-firm effects and compositional changes across firms as potential mechanisms in various contexts (Bloom et al., 2016; Hershbein and Kahn, 2018).

explained in Section 5.2.

Finally, our firm-level approach captures aggregate decisions related to hiring strategies. Even when a firm operates multiple establishments across regions, it typically makes critical decisions such as hiring collectively (Hazell et al., 2022). Examining changes in employment and skill requirements at the firm level offers insights into possible strategic responses by firms to the opioid crisis.

5.2 Empirical Specifications

In our firm-level analysis, we estimate the following event study specification:

$$\ln(y_{fgt}) = \sum_{t=2005, t \neq 2009}^{2019} \delta_t \text{Exposure_Pre}_f + \alpha_f + \gamma_{gt} + X'_{ft}\beta + \varepsilon_{fgt}, \quad (13)$$

where y_{fgt} denotes the outcome for firm f in industry g during year t . All outcomes are log-transformed. Exposure_Pre_f indicates the standardized pre-intervention exposure to local prescription opioid use at the firm level. The coefficients of interest are δ 's, presenting how the effect of the reformulation on our firm-level outcomes changes over time. For the firm employment outcome from the Compustat database, we use 2009, the year before the reformulation, as the reference period, so the coefficient for that year is normalized to zero. Because the Lightcast job posting data are unavailable for 2008 and 2009, we use 2007 as the reference period. The industry classification is based on the three-digit NAICS code. γ_{gt} denotes industry-by-year fixed effects and controls for industry-specific time-varying confounding factors. X_{ft} denotes time-varying firm-level characteristics, including two control variables: the log of the investment tax credit obtained from Compustat, and the interaction of firm-level exposure to the 2008 Great Recession shock with the full set of year dummy variables.³⁵ Standard errors are clustered at the firm level.

³⁵Following Hershbein and Kahn (2018), we use unemployment rates during the Great Recession (2007–2009) as a measure of the recession shock. We construct the weighted average of the local unemployment rate for each firm in a manner similar to how we construct firm-level opioid shock, which involves taking the weighted average of the county-level exposure across establishments. This control mitigates the concern that the recession influences the outcomes through the increased use of illicit opioids as a scarring effect.

Identification assumption. The key identification assumption is that, for any exposure level, the outcome path of firms at that exposure level is the same as the path firms at other exposure levels would have followed had they instead experienced that same exposure level—what [Callaway et al. \(2024\)](#) refer to as the *strong parallel trends assumption*. This differs from the standard parallel trends assumption in a binary treatment framework, which requires parallel trends in untreated potential outcomes. Importantly, the strong parallel trends assumption is not strictly stronger than the binary treatment assumption; for example, it does not require parallel trends in untreated potential outcomes for all exposure groups.

Because counterfactual outcome paths are unobserved, this assumption cannot be tested directly. We therefore assess its plausibility by examining pre-treatment trends. In addition, we test the robustness of our results by estimating a difference-in-differences event study specification using a binary treatment definition, classifying higher-exposure firms as treated and lower-exposure firms as the comparison group (see Section 6.3). The results are consistent with our baseline continuous treatment specification, suggesting that our findings hold under the less restrictive standard parallel trends assumption and are robust to the choice of treatment definition.

To summarize the average effect in the post-period, we estimate the following difference-in-differences specification:

$$\ln(y_{fgt}) = \lambda Exposure_Pre_f \times Post_t + \rho_f + \tau_{gt} + X'_{ft}\sigma + u_{fgt}, \quad (14)$$

where $Post_t$ is an indicator denoting the post-reformulation period covering years from 2010 to 2019. The coefficient of interest is λ , representing the effect of the reformulation on our firm- or county-level outcomes. All the other variables are defined as in equation (14). Standard errors are clustered at the firm level.

Finally, we use a difference-in-differences specification that separately estimates the short- and medium-run effects, which takes the following form:

$$\begin{aligned} \ln(y_{fgt}) = & \lambda_1 Exposure_Pre_f \times SR\ Post_t + \lambda_2 Exposure_Pre_f \times MR\ Post_t \\ & + \rho_f + \tau_{gt} + X'_{ft}\sigma + u_{fgt}, \end{aligned} \quad (15)$$

where $SR Post_t$ is an indicator for the first five years following reformulation, 2010 to 2014. $MR Post_t$ is an indicator for the subsequent years, 2015 to 2019. All the other variables are defined as in equation (14). Standard errors are clustered at the firm level.

6 Results

6.1 Effects on Firm Employment and Skill Requirements

We examine the impacts of the OxyContin reformulation on various labor market outcomes, including firm employment and firms' requirements for cognitive skills, computer skills, education, and work experience in job postings. Throughout our analyses, we report the impact of a one standard deviation increase in firm-level exposure to the reformulation, which corresponds to an additional 0.22 per capita opioid prescriptions in the pre-reformulation period.

Effect on firms' employment levels. In Figure 7, we present the results for firm-level employment. Specifically, we show the coefficients and their associated 95 percent confidence intervals derived from the estimation of equation (13). Our analysis indicates no statistically significant evidence of pre-existing trends prior to the reformulation, which supports the parallel trends assumption. Beginning in 2010, the year of the OxyContin reformulation, firm employment began to decline. This negative effect continued to grow through 2015 and has remained relatively stable since then.

Column 1 of Table 2 reports the corresponding difference-in-difference estimates. Panel A presents estimates summarizing the overall effect from 2010 to 2019 from the estimation of equation (14). Panel B reports estimates separately for the short-run post-reformulation period (2010–2014) and the medium-run period (2015–2019) based on equation (15). Over the post-reformulation period from 2010 to 2019, a one standard deviation increase in exposure to the reformulation is linked to a 5.1 percent reduction in employment at the firm level (p -value=0.002). Splitting the post-reformulation period, the estimated effect is a 4.2 percent reduction in 2010–2014 (p -value=0.002) and a 6.2 percent reduction in 2015–2019 (p -value=0.003).

To put our estimates in context, we compare them with those reported in a closely related study

by [Park and Powell \(2021\)](#), which finds that a one standard deviation increase in exposure to the reformulation resulted in a 1.2 percent reduction in state-level per capita employment five years after reformulation.³⁶ The larger magnitude of our estimates is consistent with the fact that our results for publicly traded firms do not capture the potential reallocation of reduced employment toward self-employed workers and other small firms outside our sample.

Effect on firms’ skill requirements. Next, we examine the impact of reformulation on skill requirements using our job posting data. Figure 8 illustrates the dynamic effects of the OxyContin reformulation on firm-level skill requirements, as estimated using equation (13). The outcome variables include the average number of cognitive skills (panel (a)), the average number of computer skills (panel (b)), the average years of schooling (panel (c)), and the average years of work experience (panel (d)) required in a job posting at the firm level, where all outcomes are log-transformed.

The estimates in panels (a) and (b) indicate that the requirements for cognitive and computer skills increased following the reformulation, with effect sizes growing until 2014–2016 and remaining relatively stable through 2019. This increase in effect sizes over time is consistent with prior evidence that the reformulation contributed to the gradual expansion of illicit drug markets in areas with higher pre-reformulation rates of prescription opioid use ([Alpert et al., 2018](#); [Park and Powell, 2021](#); [Evans et al., 2022](#)). Panels (c) and (d) provide little evidence of changes in years of schooling or work experience requirements following the reformulation, although we do find statistically significant effects for certain subgroups, as we discuss in Section 6.2.³⁷

Columns 2–5 of Table 2 provide the corresponding difference-in-difference estimates for these outcomes. In Panel A, we observe that a one standard deviation increase in firm-level exposure to reformulation resulted in a 7.9 percent increase in the average number of cognitive skills (column 2; p -value=0.027) and a 5.4 percent increase in the average number of computer skills (column 3;

³⁶A key difference from [Park and Powell \(2021\)](#) is that their estimates capture overall employment changes in the broader state labor market, whereas ours capture within-firm changes in employment among publicly traded firms. Note that a one standard deviation difference in their state-level exposure to reformulation, measured by the pre-reformulation rate of OxyContin misuse, is 0.23. By comparison, a one standard deviation difference in our firm-level exposure to reformulation, calculated by pre-reformulation per-capita opioid prescriptions in the counties where a firm is located, is equal to 0.22.

³⁷In Appendix Figure A6, we also present our event study estimates for the following skills: social skills (panel (a)), service skills (panel (b)), management skills (panel (c)), and writing skills (panel (d)). Our analysis reveals little evidence suggesting that the reformulation is linked to an increase in requirements for these skills.

p -value=0.095) required in an online job posting. In Panel B, our estimates reveal that both the magnitude and statistical significance of these effects increased over time. In the medium term, from 2015 to 2019, cognitive and computer skills increased by 8.3 percent (p -value=0.021) and 7.8 percent (p -value=0.021), respectively. In the last two columns of both panels, the estimates for education and experience requirements are statistically insignificant.

Overall, our main analysis finds evidence of upskilling in cognitive and computer skill requirements at the firm level, but finds little evidence of changes in education or experience requirements. One plausible explanation is that cognitive and computer skills serve as effective yet relatively low-cost screening margins, whereas raising education or experience requirements is more costly because it may require offering higher wages and narrows the pool of eligible applicants.³⁸

Within-firm-occupation analysis. So far, we have shown that the transition toward illicit opioid use led to an increase in skill requirements at the firm level. A key question is whether this upskilling effect is driven by an increase in skill requirements within an occupation (i.e., within-occupation change) or by a shift in the composition of job vacancies, with a greater share of high-skilled positions relative to low-skilled ones (i.e., compositional change).

An increase in skill requirements within occupations in response to the opioid crisis would provide strong evidence that employers are strategically adjusting job requirements to screen out potential opioid misusers. Importantly, this analysis helps address the concern that the observed upskilling simply reflects changes in labor supply rather than changes in employers' job requirements. If upskilling were driven solely by shifts in workforce composition induced by the opioid crisis, we would be unlikely to observe increases in skill requirements within the same occupation.

To explore this question, we examine the dynamic effect of the OxyContin reformulation on the log of skill requirements at the firm-by-occupation level. To do this, we create a yearly panel

³⁸Cognitive skills may serve as a job-relevant proxy for opioid-related risk for several reasons. Employers report concerns about absenteeism, impaired worker performance, and workplace injuries or incidents associated with employee opioid use ([National Safety Council, 2019](#)). This may increase the value of screening for attention, reliability, trainability, and follow-through. In addition, because cognitive skills such as problem-solving are already common in job postings and skills-based hiring is widely used, raising cognitive requirements may be a relatively easy and low-cost way to screen applicants.

dataset of skill requirements for each firm and 6-digit SOC (Standard Occupational Classification) occupation. We then estimate a version of our event study model (equation (13)) in which we replace firm fixed effects with firm-by-occupation fixed effects and occupation-by-year fixed effects. All other controls remain unchanged, including industry-by-year fixed effects and time-varying firm-level characteristics. This approach allows us to capture the within-firm-occupation upskilling effects.³⁹

Panel (a) of Figure 9 shows that the within-firm-occupation effects on cognitive skills are comparable in magnitude to the baseline firm-level estimates presented in panel (a) of Figure 8, indicating that within-occupation upskilling accounts for much of our baseline result for cognitive skills. This suggests that employers raise skill requirements for the same occupations, not just by adjusting the composition of job postings toward higher-skilled roles. This evidence of within-occupation upskilling provides strong support for the screening mechanism outlined in Section 3.

Panel (b) shows that the within-occupation effect on the number of computer skills is small and statistically insignificant, in contrast to the larger firm-level effect presented in panel (b) of Figure 8. It is worth noting that this pattern is not inconsistent with our interpretation, because we do not claim that screening is the only mechanism driving the results. Rather, it suggests that compositional changes across occupations also play a role in the firm-level effect.⁴⁰

Panel (c) presents statistically significant within-occupation effects on schooling requirements: a one standard deviation increase in opioid exposure is associated with an approximately 7% increase in required years of schooling.⁴¹ The estimates for experience requirements shown in

³⁹This analysis includes firm-by-occupation fixed effects and therefore restricts the sample to firm-occupation pairs observed in more than one year during the sample period (number of firms= 1,961). This restriction is more stringent than in the baseline analysis, which requires firms to be observed in more than one year regardless of occupation ($N = 2,104$). To address concerns that changes in firm composition may affect the results, we estimate the baseline firm-level event study model (equation (13)) using both the full baseline sample and a subsample consisting of firms included in the firm-occupation analysis. Both samples in these regressions are based on firm-level panel data. As shown in Appendix Figure A7, the results are robust across samples, indicating that compositional changes do not drive the findings in the firm-occupation analysis.

⁴⁰Such compositional changes would be expected if firms respond to the opioid crisis by expanding hiring in occupations that require more computer use and are likely to involve work that is easier to monitor or coordinate through formalized, software-based tasks. This may reflect employers' efforts to maintain the overall productivity of new hires as the quality of the applicant pool deteriorates, consistent with the second mechanism discussed in Section 2.5.

⁴¹These findings, which contrast with the firm-level results depicted in Figure 8, suggest that changes in the composition of job postings across occupations with different schooling and experience requirements partly offset the within-occupation upskilling effects. As discussed above, one possible explanation is that increasing educational

panel (d) provide no evidence of changes in skill requirements following reformulation, either at the firm level or the firm-occupation level.

6.2 Heterogeneity Analysis

Having shown that the opioid crisis led to upskilling at the firm level, we examine heterogeneity in effects of the reformulation on skill requirements across firm and labor market characteristics, including firm size, educational requirements, and local minimum wage levels.

First, we examine heterogeneity by firm size by estimating equation (14) separately for small, medium, and large firms.⁴² Given that the costs of monitoring and hiring workers who misuse opioids increase with firm size (Garen, 1985; Oi and Idson, 1999), larger firms may have a greater incentive to screen out these individuals during the hiring process. We present the coefficients and associated 95% confidence intervals for the difference-in-differences effects of the OxyContin reformulation on cognitive and computer skill requirements, as well as on years of schooling and experience, for each subgroup in Figure 10. The figure shows that larger firms raise computer skill requirements, while small and medium-sized firms raise cognitive skill requirements, in response to the opioid crisis. One possibility is that smaller firms may increase cognitive skill requirements because doing so is potentially less costly than raising computer skill requirements, which may be associated with higher compensating wages. Larger firms, by contrast, may be better equipped to meet such wage demands while using higher computer skill requirements to screen out higher-risk applicants or to improve the overall skill composition of their new hires.

Second, we explore heterogeneity by firm education requirements. We categorize firms within each sector into two groups based on whether the percentage of job postings requiring a four-year college degree is above or below the median.⁴³ In Figure 11, we present coefficients and

requirements can be costly for firms and may not serve purely as a screening tool. As a result, employers might compensate by hiring more in occupations that require less formal education. Consistent with this interpretation, our heterogeneity analysis shows that the offsetting effect is more pronounced in states with lower minimum wages (see Figure 12), where the costs of hiring and separation are relatively low. In such environments, firms may screen within certain occupations while simultaneously increasing hiring in lower-skilled roles, potentially with higher turnover.

⁴²We categorize firms into these three groups within each sector (utility, manufacturing, service, IT, finance, and professional services) based on the average total assets measured over the last three years prior to the reformulation.

⁴³Due to missing job posting data for some firms in 2007, we use the 2010 data for consistent baseline education requirements. In 2010, the average percentage of job postings requiring a four-year degree was 14.5% for the low-requirement group and 59.6% for the high-requirement group. These figures are weighted according to the number of job postings for each firm in 2010. Notably, we observed similar results when calculating baseline education

corresponding 95% confidence intervals for the difference-in-differences effects of the OxyContin reformulation, derived from equation (14). Our findings suggest that upskilling in computer skill requirements is more pronounced among firms with a lower percentage of job postings requiring a four-year college degree at baseline. This aligns with previous research indicating that fatal opioid overdoses are more common among less-educated individuals (Case and Deaton, 2015; Altekruse et al., 2020), suggesting that firms depending more heavily on less-educated labor may face greater opioid-related risks among their employees and job candidates, and thus have stronger incentives to tighten hiring standards.

Third, we explore heterogeneity in effects across firms in areas with low, medium, and high minimum wage levels.⁴⁴ Prior work shows that higher minimum wages lead employers to set more demanding hiring standards, as the higher cost of labor requires a correspondingly higher level of applicant productivity to remain profitable (Butschek, 2022). By the same logic, a higher minimum wage amplifies the cost of hiring less productive workers at risk of opioid misuse, giving employers stronger incentives to raise skill requirements in response to the opioid crisis. Figure 12 presents coefficients and 95% confidence intervals from the estimation of equation (14) for each group. The upskilling effects are more pronounced for firms exposed to higher minimum wage levels, consistent with the prediction that higher labor costs strengthen employers' incentives to screen applicants more carefully and to attract a more productive workforce. Importantly, firms in the high minimum wage group show statistically significant increases not only in cognitive and computer skill requirements but also in years of schooling, whereas these effects are small and statistically insignificant in the low minimum wage group.

Similarly, we analyze heterogeneity in effects on skill requirements across firms in areas with low, medium, and high employment protection levels in Appendix Figure A8.⁴⁵ The results are

requirements using averages from both 2007 and 2010 rather than just 2010.

⁴⁴We use state-level minimum wages provided by the US Department of Labor. Using the minimum wages averaged over the last three years before the reformulation, we construct the firm-level average minimum wage based on the geographic variation of a firm's establishments.

⁴⁵We use a state-level employment protection score constructed by Oxfarm, a non-governmental organization. The data are available at: <https://www.oxfamamerica.org/explore/countries/united-states/poverty-in-the-us/best-states-to-work/> (accessed May 2024). The index is based on state policies related to protections around paid sick leave, advance notice, flexible scheduling, sexual harassment, equal pay, etc. We construct a firm-level exposure to employment protection by taking the weighted average of state-level scores based on the geographic distribution of a firm's establishments. We then classify firms into three groups based on this firm-level score.

broadly consistent with those from the minimum wage analysis. We find that upskilling effects are more pronounced for firms operating under stronger employment protection regimes, in line with the idea that higher hiring and separation costs give firms stronger incentives to screen applicants more carefully at the hiring stage.

6.3 Robustness Analysis

We test the robustness of our results by considering potential alternative explanations, using a different policy exposure measure, and constructing alternative samples.

Binary treatment. Our baseline specification treats exposure to the OxyContin reformulation as continuous. To assess the sensitivity of our results to this choice, we estimate a version of our event study specification (equation (13)) using a binary treatment definition that classifies firms with above-median pre-reformulation exposure as treated and those below the median as the comparison group. Appendix Figure A9 compares the event study coefficients to those from this baseline continuous treatment specification. The two sets of estimates show similar dynamic patterns and lead to similar qualitative conclusions. As discussed in Section 5, the similarity of results across the two specifications indicates that our findings hold under the less restrictive standard parallel trends assumption, alleviating concerns about sensitivity to the stronger parallel trends assumption required by the continuous treatment framework. It also suggests that our findings are not sensitive to the choice of treatment definition.

Half-year level sample. A key concern in our study is an omitted factor that affects both initial opioid use and skill requirements. This concern is especially relevant because data for 2008 and 2009 are unavailable in the Lightcast job postings data, and one may therefore worry that our upskilling findings reflect a differential increase in skill requirements in high-exposure areas between 2007 and 2010, rather than a sudden change following the 2010 reformulation. To mitigate this concern, we conduct our regressions using a more granular, half-yearly panel dataset instead of our yearly panel dataset. Estimates from yearly and half-yearly samples may not be directly comparable because some firms' hiring decisions may be made on an annual basis, making it challenging to compare skill requirements between the first and second halves.

Nevertheless, this analysis is useful as it allows for a more rigorous assessment of whether there is any pre-existing trend in our outcomes prior to the reformulation.

Figure 13 presents coefficients and 95% confidence intervals from estimation of a version of equation (13) in which the time unit is a half-year. The first half of 2010, the last pre-reformulation period, is used as the reference period. Relative to the first half of 2010, we find that the estimated effects on cognitive skill requirements remain statistically significant. Although the estimated effects for computer skills are smaller both in size and statistical significance when using half-yearly data, we still observe a comparable pattern of results.⁴⁶ Most importantly, using this more granular dataset, we find no evidence of a pre-existing trend across the observed pre-reformulation half-years for any of our skill outcomes. The effect on skill requirements begins to emerge only in the latter half of 2010 or later. This reassures us that our results are unlikely to be influenced by other confounding factors that might have impacted skill demand before the reformulation of OxyContin.

Alternative explanation: Great Recession. The 2008 economic downturn, known as the Great Recession, is a key potential confounding factor due to its timing and profound economic consequences. We address this concern in three ways. First, as described in Section 5, we directly control for the dynamic effect of the Great Recession in our regressions. Second, we observe a small correlation of 0.083 between the recession shock and the reformulation shock, mitigating the concerns regarding this confounder. Lastly, our half-year level analysis indicates no pre-existing trends in the pre-period, suggesting that the Great Recession is unlikely to be the primary driver of our results.

Firm-by-state level sample. A potential concern is that our firm-level exposure measure may be correlated with other firm-specific time-varying factors that independently influence skill requirements. To address this concern, we conduct a firm-by-state level analysis in which we aggregate establishments owned by the same firm within a given state into a single entity.⁴⁷ For

⁴⁶In contrast, our findings regarding education and experience requirements are less stable across analyses. While our baseline analysis reveals no evidence of increases in requirements for education and experience, we find suggestive evidence of an increase in experience requirements in our half-year analysis. Furthermore, our firm-by-state level analysis indicates evidence of increases in requirements for both education and experience, as discussed below.

⁴⁷In our baseline analysis, we aggregate all establishments operated by the same firm across multiple states into a single firm entity.

this analysis, we control for firm-by-year fixed effects to account for any firm-specific time-varying shocks. The idea is to investigate whether entities located in states with higher exposure to reformulation experienced greater changes in skill requirements compared to other entities under the same firm but located in states with lower exposure, even after accounting for firm-specific time-varying factors.⁴⁸ Specifically, we run a version of equation (13) that replaces firm fixed effects with firm-by-state fixed effects (i.e., entity fixed effects) and additionally controls for firm-by-year fixed effects. For the opioid exposure measure, we follow [Alpert et al. \(2018\)](#) and use state-level OxyContin misuse rates from the pre-reformulation period, as further described in Section 6.3. Each entity-year observation is weighted by the entity’s share of the firm’s total job postings in that year.

Figure 14 presents the event study results from our baseline analysis in dark blue circles and the results from this entity-level analysis in light blue hollow circles. Although the point estimates are slightly smaller in magnitude compared to our baseline estimates, we find statistically significant evidence that reformulation is associated with increases in cognitive and computer skill requirements at the entity level, even after accounting for any potential firm-level concurrent shocks. In contrast to our baseline findings indicating little effect on education and experience requirements at the firm level, this analysis suggests that entities in high-exposure states increase their education and experience requirements following reformulation.

Alternative policy exposure measure. For the robustness analysis using the firm-by-state sample discussed above, we use an alternative measure of exposure—state-level misuse of OxyContin from [Alpert et al. \(2018\)](#). [Alpert et al.](#)’s measure captures misuse of OxyContin in the state during the pre-reformulation period and may provide a more accurate state-level exposure measure for OxyContin misuse. As mentioned above, we observe statistically significant increases in cognitive and computer skill requirements using this alternative measure, as shown in Figure 14.

Dropping Appalachian states. Another concern might be that shocks associated with the decline of the coal mining industry may confound our estimates. As shown in Figure 6, opioid prescription rates are particularly high in the Appalachian area, comprising 423 counties across 13

⁴⁸To reduce noise, we aggregate job postings at the firm-state-year level rather than using firm-county-year observations following [Giroud and Mueller \(2019\)](#).

states spanning from southern New York to northern Mississippi. Moreover, coal mining began to decline in these areas in 2008, with coal production falling by 50% between 2008 and 2019.⁴⁹

This decline in coal production may have contributed to increases in both opioid use and unemployment. We explore how results change when we exclude these regions from our analysis. Appendix Figure A10 demonstrates that our results from the state-level analysis, as described above, remain relatively stable even after excluding the 13 Appalachian states. This suggests that our results cannot be solely explained by the decline of the mining industry in Appalachian states.

Using a 4-digit industry code. We test the robustness of our upskilling results to replacing 3-digit industry-by-year fixed effects with 4-digit industry-by-year fixed effects. In Appendix Figure A11, we present event study estimates from equation (13) using two specifications: our baseline specification with industry-by-year fixed effects based on 3-digit industry code (dark blue circles), and an alternative specification with industry-by-year fixed effects based on 4-digit industry code (hollow circles in light blue). Overall, our upskilling effects are much larger and stronger when using the 4-digit industry code, suggesting that our estimates provide a more conservative measure of the effect on job skill requirements.

Controlling for labor supply measures. Another concern is that our results may solely reflect changes in the local labor supply in response to the OxyContin reformulation. For instance, the supply of less-skilled or young workers, who may have been more affected by the transition to illicit opioids, may have declined disproportionately. As a result, firms may have reduced job postings for less-skilled positions, leading to higher equilibrium skill requirements in the average job posting.

While there is no way to test this directly, we conduct an exercise that explicitly controls for labor supply measures for worker subgroups, which may reflect labor supply responses to the reformulation. The idea is that, if our results are entirely driven by factors that predict labor supply responses, our estimates should be sensitive to the inclusion of these controls. We use state-year labor force participation rates and average wages by gender, education level, and race/ethnicity from the National Historical Geographic Information System (NHGIS), and construct corresponding firm-level labor supply measures based on the geographic distribution of

⁴⁹Data are available at: <https://www.eia.gov/outlooks/steo/data/browser/#?v=18> (accessed April 2024).

each firm’s establishments across states.

In Table 3 (Appendix Table A3), we report the sensitivity of our results when controlling for labor force participation rates (wage levels) for the subgroup of workers. In both tables, we reproduce our baseline estimates from the estimation of equation (14) reported in Table 2, separately for the average post-reformulation effect over 2010–2019 (Panel A) and the medium-run effect over 2015–2019 (Panel B). Each panel reports results for employment (sub-panel 1), cognitive skill requirements (sub-panel 2), and computer skill requirements (sub-panel 3). In column 2, we add female and male workers’ labor force participation rates (wage levels). In column 3, we add measures for education subgroups—college graduates and non-college graduates. Finally, in column 4, we add measures for race/ethnicity subgroups—non-Hispanic White, non-Hispanic Black, and Hispanic.

The employment impact of the reformulation remains robust to the inclusion of these measures in both tables. Importantly, the medium-run effects for both cognitive and computer skill requirements remain statistically significant and similar in magnitude to the baseline across all specifications in both tables, suggesting that our upskilling results are unlikely to be entirely driven by labor supply responses to the opioid crisis.

Alternative explanation: Retiring baby boomers. Another possible alternative explanation is the potential confounding impact of the change in the age structure. Around 2011, there was a sharp increase in the growth rate of the share of the population aged over 65 due to the retirement of the large Baby Boom birth cohorts (Maestas et al., 2023).⁵⁰ This change in the age structure may have wide-reaching impacts on firm performance and the labor market. For instance, this increase in the retiring population might have reduced local labor supply and employment levels. The reduction in elderly labor supply and the change in the age structure of the labor force could eventually impact companies’ hiring strategies.

To address this concern, we investigate the correlation between exposure to the reformulation and two measures: the elderly population share in the pre-reformulation period and changes in the elderly population share between 2010 and 2015. Panel (a) of Appendix Figure A12 presents a scatter plot in which each observation represents the population-weighted average opioid

⁵⁰Maestas et al. (2023) document an increase in the older population share by more than 20% between 2010 and 2020 and that this was associated with lower capita GDP.

prescriptions per capita in a given county between 2006 and 2009—our measure for county-level exposure to the reformulation—on the x-axis, and the population-weighted average share of the population over age 65 in that county during the same period on the y-axis. We do not observe any notable correlation between these two variables.

In panel (b) of Appendix Figure [A12](#), we further examine the correlation between exposure to the OxyContin reformulation and the elderly share growth rate from 2010 to 2015. As emphasized in [Maestas et al. \(2023\)](#), we observe that the majority of counties experienced an increase in the elderly population during this period, as most observations lie above the horizontal dashed line representing zero change. Once again, we do not witness any meaningful correlation between elderly share growth and exposure to the OxyContin reformulation, assuring that our results are not likely to be driven by the potential confounding effect of the age structure change.

7 Conclusion and Policy Implications

The opioid overdose epidemic is a crisis of both health and economic dimensions in the United States. To measure the economic consequences of the opioid crisis, it is crucial to understand the economic impacts of the opioid crisis on employers and their responses in the labor market. This is particularly important in understanding who bears the costs of the crisis, quantifying the magnitude of the burden, and designing policy interventions to tackle these challenges.

In this study, we examine the impact of a large transition toward illicit opioids following the OxyContin reformulation on employers' skill requirements for new hires. Using comprehensive firm-level longitudinal data, we find that increased illicit opioid use led to increased requirements for cognitive and computer skills.

Our findings have several policy implications. First, it underscores the distributional effects of the opioid crisis on workers. Our findings reveal that employers increase their skill requirements for new hires in response to the crisis, disproportionately affecting less-skilled workers. Previous research suggests that the toll of the opioid crisis is particularly heavy on individuals with lower socioeconomic status and those with lower levels of educational attainment ([Case and Deaton, 2015](#); [Altekruse et al., 2020](#)). Our study highlights that even less-skilled workers without a history of opioid use disorders can be affected by the crisis due to these increased skill requirements.

Therefore, interventions aimed at addressing the adverse impact of the opioid crisis, such as those designed to improve employment outcomes, should not be limited to individuals with opioid use disorders.

Second, our study highlights the need for diverse types of resources tailored to targeted populations. Policy discussions surrounding the opioid crisis have largely concentrated on health outcomes and resources for the prevention and treatment of opioid use disorders. However, our results suggest that providing occupational training programs to enhance the skills of less-skilled workers could be a meaningful approach to mitigating the adverse impact of the opioid crisis on this group.

Third, our study implies that employers may have strong incentives to prevent and address opioid use disorders not only among their employees but also within their communities. Our findings suggest that employers may be adversely affected by the opioid crisis not just in terms of employee productivity but also through local opioid use, as an increase in local drug misuse can result in reductions in the number of qualified job candidates. Firms might benefit themselves in the long term by contributing to efforts to combat the opioid crisis. In fact, employers are uniquely positioned to play a pivotal role in preventing and treating opioid use disorder. About 60 percent of adults who report past-year opioid misuse are currently employed ([SAMHSA, 2021](#)), and the workplace is a significant part of employees' daily lives. Policymakers may consider that employers can play a critical role in preventing and addressing the opioid crisis.

Lastly, our study focuses on the transition from prescription to illicit opioids, which carries important implications beyond our sample period. Since this transition began around 2010, illicit opioids have been a major driver of escalating overdose mortality rates, with illicitly-made fentanyl playing an increasingly prominent role from 2013 onward. As the illicit opioid epidemic continues to take a heavy toll on communities across the United States, our findings provide important evidence for understanding and addressing its ongoing labor market consequences.

References

- Alekseeva, Liudmila, José Azar, Mireia Giné, Sampsa Samila, and Bledi Taska**, “The Demand for AI Skills in the Labor Market,” *Labour Economics*, 2021, 71, 102002.
- Aliprantis, Dionissi, Kyle Fee, and Mark E Schweitzer**, “Opioids and the labor market,” *Labour Economics*, 2023, p. 102446.
- Alpert, Abby, David Powell, and Rosalie Liccardo Pacula**, “Supply-side drug policy in the presence of substitutes: Evidence from the introduction of abuse-deterrent opioids,” *American Economic Journal: Economic Policy*, 2018, 10 (4), 1–35.
- Alpert, Abby E, Steve Schwab, and Benjamin D Ukert**, “Opioid Use and Employment Outcomes: Evidence from the US Military,” Technical Report, National Bureau of Economic Research 2022.
- Altekruse, Sean F, Candace M Cosgrove, William C Altekruse, Richard A Jenkins, and Carlos Blanco**, “Socioeconomic risk factors for fatal opioid overdoses in the United States: Findings from the Mortality Disparities in American Communities Study (MDAC),” *PloS one*, 2020, 15 (1), e0227966.
- Armando, Franco, Wagner Zachary, and M Whaley Christopher**, “Doped up or dropped out: the effects of opioid use on workplace absences,” *Working Paper*, 2019.
- Arteaga, Carolina and Victoria Barone**, “A Manufactured Tragedy: The Origins and Deep Ripples of the Opioid Epidemic,” Technical Report, Working paper. [http://www. carolina. com/s/Opioids_ArteagaBarone_Jan2022. pdf](http://www.carolina.com/s/Opioids_ArteagaBarone_Jan2022.pdf) 2022.
- Autor, David H, Frank Levy, and Richard J Murnane**, “The Skill Content of Recent Technological Change: An Empirical Exploration,” *The Quarterly Journal of Economics*, 2003, 118 (4), 55.
- Ballance, Joshua, Robert Clifford, and Daniel Shoag**, ““No More Credit Score”: Employer Credit Check Bans and Signal Substitution,” *Labour Economics*, 2020, 63, 101769.
- Bartik, Alexander W. and Scott T. Nelson**, “Deleting a Signal: Evidence from Pre-Employment Credit Checks,” *The Review of Economics and Statistics*, 02 2024, pp. 1–47.
- Beachler, Daniel C, Kelsey Hall, Renu Garg, Geetanjoli Banerjee, Ling Li, Luke Boulanger, Huseyin Yuce, and Alexander M Walker**, “An evaluation of the effect of the oxycontin reformulation on unintentional fatal and nonfatal overdose,” *The Clinical Journal of Pain*, 2022, 38 (6), 396.
- Becker, Sascha O, Ana Fernandes, and Doris Weichselbaumer**, “Discrimination in hiring based on potential and realized fertility: Evidence from a large-scale field experiment,” *Labour Economics*, 2019, 59, 139–152.
- Beheshti, David**, “Adverse health effects of abuse-deterrent opioids: Evidence from the reformulation of OxyContin,” *Health economics*, 2019, 28 (12), 1449–1461.

- , “The impact of opioids on the labor market: Evidence from drug rescheduling,” *Journal of Human Resources*, 2023, 58 (6), 2001–2041.
- Bloom, Nicholas, Mirko Draca, and John Van Reenen**, “Trade induced technical change? The impact of Chinese imports on innovation, IT and productivity,” *The review of economic studies*, 2016, 83 (1), 87–117.
- Buckles, Kasey, William N Evans, and Ethan MJ Lieber**, “The drug crisis and the living arrangements of children,” *Journal of Health Economics*, 2023, 87, 102723.
- Burstein, Ariel and Jonathan Vogel**, “International Trade, Technology, and the Skill Premium,” *Journal of Political Economy*, 2017, 125 (5), 1356–1412.
- Butschek, Sebastian**, “Raising the Bar: Minimum Wages and Employersâ Hiring Standards,” *American Economic Journal: Economic Policy*, 2022, 14 (2), 91–124.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro HC Sant’Anna**, “Difference-in-differences with a continuous treatment,” Technical Report, National Bureau of Economic Research 2024.
- Carluccio, Juan, Alejandro Cuñat, Harald Fadinger, and Christian Fons-Rosen**, “Offshoring and Skill-Upgrading in French Manufacturing,” *Journal of International Economics*, 2019, 118, 138–159.
- Case, Anne and Angus Deaton**, “Rising morbidity and mortality in midlife among white non-Hispanic Americans in the 21st century,” *Proceedings of the National Academy of Sciences*, 2015, 112 (49), 15078–15083.
- Centers for Disease Control and Prevention**, “Understanding the Opioid Overdose Epidemic,” 2025. <https://www.cdc.gov/overdose-prevention/about/understanding-the-opioid-overdose-epidemic.html>. Accessed: March 2026.
- Centers for Disease Control and Prevention, National Center for Health Statistics.**, “National Vital Statistics System, Mortality 1999-2020 on CDC WONDER Online Database,” 2021. <http://wonder.cdc.gov/mcd-icd10.html>. Accessed: May 2024.
- Chen, Chong, Qianqian Huang, Chang Shi, and Tao Yuan**, “Opioid Epidemic and Corporate Innovation,” *Available at SSRN 4229228*, 2021.
- Cho, David, Daniel Garcia, Joshua Montes, and Alison Weingarden**, “Labor Market Effects of the Oxycodone-Heroin Epidemic,” 2021.
- Cicero, Theodore J. and Matthew S. Ellis**, “Abuse-Deterrent Formulations and the Prescription Opioid Abuse Epidemic in the United States: Lessons Learned From OxyContin,” 2015, 72 (5), 424–430.
- Cicero, Theodore J, James A Inciardi, and Alvaro Muñoz**, “Trends in abuse of OxyContin® and other opioid analgesics in the United States: 2002-2004,” *The Journal of Pain*, 2005, 6 (10), 662–672.

- Clemens, Jeffrey, Lisa B. Kahn, and Jonathan Meer**, “Dropouts Need Not Apply? The Minimum Wage and Skill Upgrading,” *Journal of Labor Economics*, 2021, 39 (S1), S107–S149.
- Cornaggia, Kimberly, John Hund, Giang Nguyen, and Zihan Ye**, “Opioid crisis effects on municipal finance,” *The Review of Financial Studies*, 2022, 35 (4), 2019–2066.
- Cortés, Kristle R., Andrew Glover, and Murat Tasci**, “The Unintended Consequences of Employer Credit Check Bans for Labor Markets,” *The Review of Economics and Statistics*, 2022, 104 (5), 997–1009.
- Currie, Janet and Hannes Schwandt**, “The opioid epidemic was not caused by economic distress but by factors that could be more rapidly addressed,” *The ANNALS of the American Academy of Political and Social Science*, 2021, 695 (1), 276–291.
- , **Jonas Jin, and Molly Schnell**, “US Employment and Opioids: Is There a Connection?â,” in “Health and labor markets,” Emerald Publishing Limited, 2019, pp. 253–280.
- Custodio, Claudia, Dragana Cvijanovic, and Moritz Wiedemann**, “Opioid crisis and real estate prices,” *Available at SSRN 3712600*, 2023.
- Davis, Steven J, John Haltiwanger, Ron Jarmin, Javier Miranda, Christopher Foote, and Eva Nagypal**, “Volatility and dispersion in business growth rates: Publicly traded versus privately held firms [with comments and discussion],” *NBER macroeconomics annual*, 2006, 21, 107–179.
- Davis, Steven J., R. Jason Faberman, and John C. Haltiwanger**, “The Establishment-Level Behavior of Vacancies and Hiring,” *Quarterly Journal of Economics*, 2013, 128 (2), 581–622.
- Deaton, Angus and Anne Case**, “Deaths of Despair and the Future of Capitalism,” 2020.
- Deiana, Claudio and Ludovica Giua**, “The intended and unintended effects of opioid policies on prescription opioids and crime,” *The BE Journal of Economic Analysis & Policy*, 2021, 21 (2), 751–792.
- Deming, David and Lisa B. Kahn**, “Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals,” 2018, 36, S337–S369.
- D’Lima, Walter and Mark Thibodeau**, “Health crisis and housing market effects-evidence from the US opioid epidemic,” *The Journal of Real Estate Finance and Economics*, 2023, 67 (4), 735–752.
- Doleac, Jennifer L. and Benjamin Hansen**, “The Unintended Consequences of âBan the Boxâ: Statistical Discrimination and Employment Outcomes When Criminal Histories Are Hidden,” *Journal of Labor Economics*, 2020, 38 (2), 321–374.
- Evans, Mary F, Matthew C Harris, and Lawrence M Kessler**, “The hazards of unwinding the prescription opioid epidemic: Implications for child maltreatment,” *American Economic Journal: Economic Policy*, 2022, 14 (4), 192–231.

- Evans, William N, Ethan MJ Lieber, and Patrick Power**, “How the reformulation of OxyContin ignited the heroin epidemic,” *Review of Economics and Statistics*, 2019, 101 (1), 1–15.
- Garen, John E.**, “Worker Heterogeneity, Job Screening, and Firm Size,” *Journal of Political Economy*, 1985, 93 (4), 715–739. Publisher: The University of Chicago Press.
- Giroud, Xavier and Holger M. Mueller**, “Firmsâ Internal Networks and Local Economic Shocks,” *American Economic Review*, 2019, 109 (10), 3617–3649.
- Harris, Matthew C, Lawrence M Kessler, Matthew N Murray, and Beth Glenn**, “Prescription opioids and labor market pains the effect of schedule ii opioids on labor force participation and unemployment,” *Journal of Human Resources*, 2020, 55 (4), 1319–1364.
- Hazell, Jonathon, Juan Herreno, Emi Nakamura, and Jón Steinsson**, “The slope of the Phillips Curve: evidence from US states,” *The Quarterly Journal of Economics*, 2022, 137 (3), 1299–1344.
- He, Chuan, Karsten Mau, and Mingzhi Xu**, “Trade Shocks and Firms Hiring Decisions: Evidence from Vacancy Postings of Chinese Firms in the Trade War,” *Labour Economics*, 2021, 71, 102021.
- Hershbein, Brad and Lisa B Kahn**, “Do recessions accelerate routine-biased technological change? Evidence from vacancy postings,” *American Economic Review*, 2018, 108 (7), 1737–1772.
- Jansen, Mark**, “Spillover effects of the opioid epidemic on consumer finance,” *Journal of Financial and Quantitative Analysis*, 2023, 58 (6), 2365–2386.
- Janssen, Aljoscha and Xuan Zhang**, “Retail pharmacies and drug diversion during the opioid epidemic,” *American Economic Review*, 2023, 113 (1), 1–33.
- Krueger, Alan B**, “Where have all the workers gone? An inquiry into the decline of the US labor force participation rate,” *Brookings papers on economic activity*, 2017, 2017 (2), 1.
- Lazear, Edward P. and James R. Spletzer**, “Hiring, Churn, and the Business Cycle,” *American Economic Review*, 2012, 102 (3), 575–579.
- Lightcast**, “Online Job Postings Data,” Lightcast, Moscow, ID 2010–2019. <https://lightcast.io/products/data/overview>.
- Maclean, Johanna Catherine, Justine Mallatt, Christopher J Ruhm, and Kosali Simon**, “Economic studies on the opioid crisis: A review,” 2020.
- Maestas, Nicole, Kathleen J Mullen, and David Powell**, “The effect of population aging on economic growth, the labor force, and productivity,” *American Economic Journal: Macroeconomics*, 2023, 15 (2), 306–332.
- Mallatt, Justine**, “Policy-induced substitution to illicit drugs and implications for law enforcement activity,” *American Journal of Health Economics*, 2022, 8 (1), 30–64.

- Michaels, Guy, Ashwini Natraj, and John Van Reenen**, “Has ICT Polarized Skill Demand? Evidence from Eleven Countries over Twenty-Five Years,” *The Review of Economics and Statistics*, 2014, 96 (1), 60–77.
- Modestino, Alicia Sasser, Daniel Shoag, and Joshua Ballance**, “Downskilling: Changes in Employer Skill Requirements over the Business Cycle,” *Labour Economics*, 2016, 41, 333–347.
- , —, and —, “Upskilling: Do employers demand greater skill when workers are plentiful?,” *Review of Economics and Statistics*, 2020, 102 (4), 793–805.
- Morgan, John and Felix Várdy**, “Diversity in the Workplace,” *American Economic Review*, 2009, 99 (1), 472–485.
- National Association of Colleges and Employers**, “Nearly Two-Thirds of Employers Use Skills-based Hiring Practices for New Entry-level Hires,” November 2024. Accessed: March 2026.
- National Institute on Drug Abuse**, “Drug Overdose Death Rates,” 2023. <https://nida.nih.gov/research-topics/trends-statistics/overdose-death-rates>. Accessed: May 2024.
- National Safety Council**, “National Employer Survey 2019: Opioid Usage in the Workplace,” Research report, National Safety Council February 2019.
- Oi, Walter Y. and Todd L. Idson**, “Chapter 33 Firm size and wages,” in “Handbook of Labor Economics,” Vol. 3, Elsevier, 1999, pp. 2165–2214.
- Ouimet, Paige, Elena Simintzi, and Kailei Ye**, “The impact of the opioid crisis on firm value and investment,” *The Review of Financial Studies*, 2025, 38 (5), 1291–1332.
- Park, Sujeong and David Powell**, “Is the rise in illicit opioids affecting labor supply and disability claiming rates?,” *Journal of Health Economics*, 2021, 76, 102430.
- Powell, David, Abby Alpert, and Rosalie L Pacula**, “A transitioning epidemic: how the opioid crisis is driving the rise in hepatitis C,” *Health Affairs*, 2019, 38 (2), 287–294.
- and **Rosalie L Riccardo Pacula**, “The evolving consequences of oxycontin reformulation on drug overdoses,” *American Journal of Health Economics*, 2021, 7 (1), 41–67.
- Ruhm, Christopher J**, “Drivers of the fatal drug epidemic,” *Journal of health economics*, 2019, 64, 25–42.
- Savych, Bogdan, David Neumark, and Randall Lea**, “Do opioids help injured workers recover and get back to work? The impact of opioid prescriptions on duration of temporary disability,” *Industrial Relations: A Journal of Economy and Society*, 2019, 58 (4), 549–590.
- Simha, Siddhartha, Yusuf Ahmed, Chad M Brummett, Jennifer F Waljee, Michael J Englesbe, and Mark C Bicket**, “Impact of the COVID-19 pandemic on opioid overdose and other adverse events in the USA and Canada: a systematic review,” *Regional Anesthesia & Pain Medicine*, 2023, 48 (1), 37–43.

Substance Abuse and Mental Health Services Administration, “Key substance use and mental health indicators in the United States: Results from the 2020 National Survey on Drug Use and Health,” Technical Report HHS Publication No. PEP21-07-01-003, NSDUH Series H-56, Center for Behavioral Health Statistics and Quality, Substance Abuse and Mental Health Services Administration, Rockville, MD 2021.

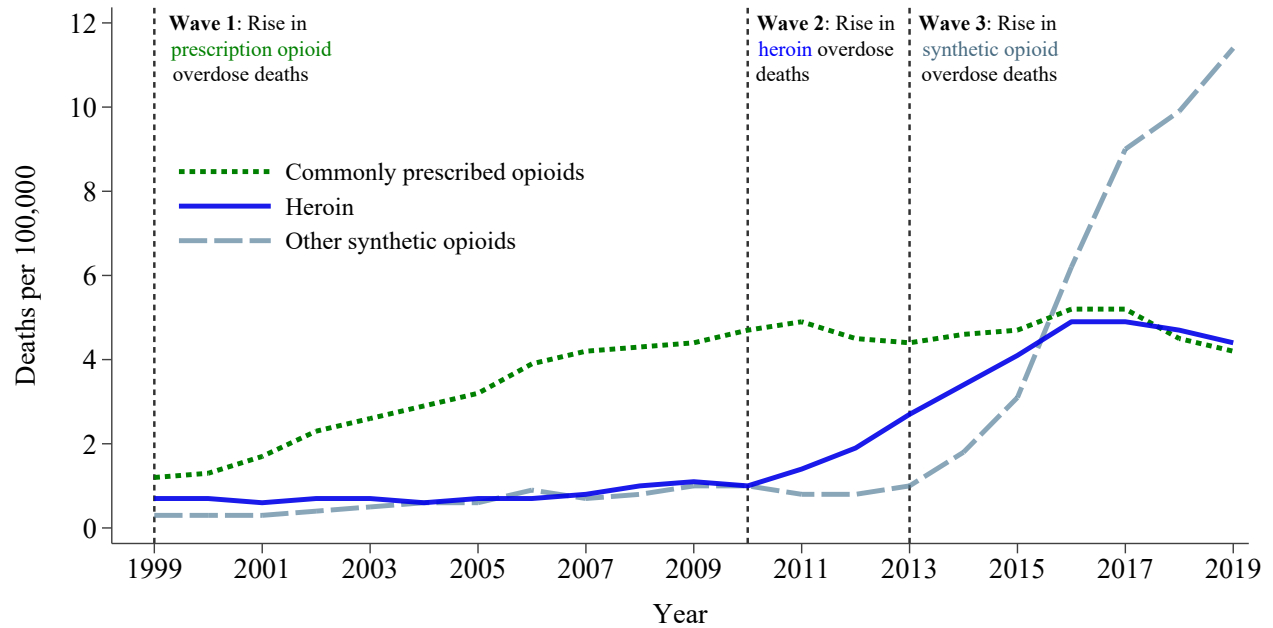
Wasi, Nada and Aaron Flaaen, “Record Linkage Using Stata: Preprocessing, Linking, and Reviewing Utilities,” *The Stata Journal*, 2015, 15 (3), 672–697.

Zee, Art Van, “The promotion and marketing of oxycontin: commercial triumph, public health tragedy,” *American journal of public health*, 2009, 99 (2), 221–227.

Ziedan, Engy and Robert Kaestner, “Effect of prescription opioids and prescription opioid control policies on infant health,” Technical Report, National Bureau of Economic Research 2020.

8 Figures and Tables

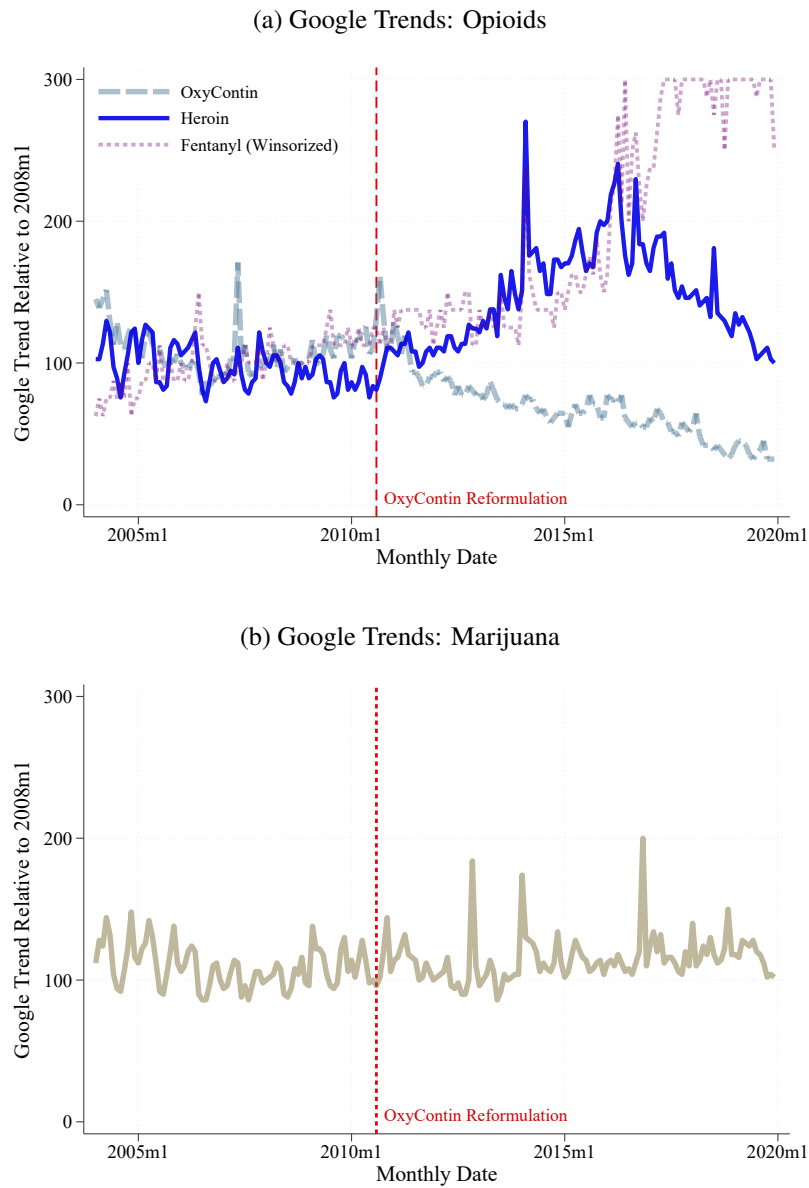
Figure 1: National Trends in Opioid Mortality



Source: National Vital Statistics, Mortality File

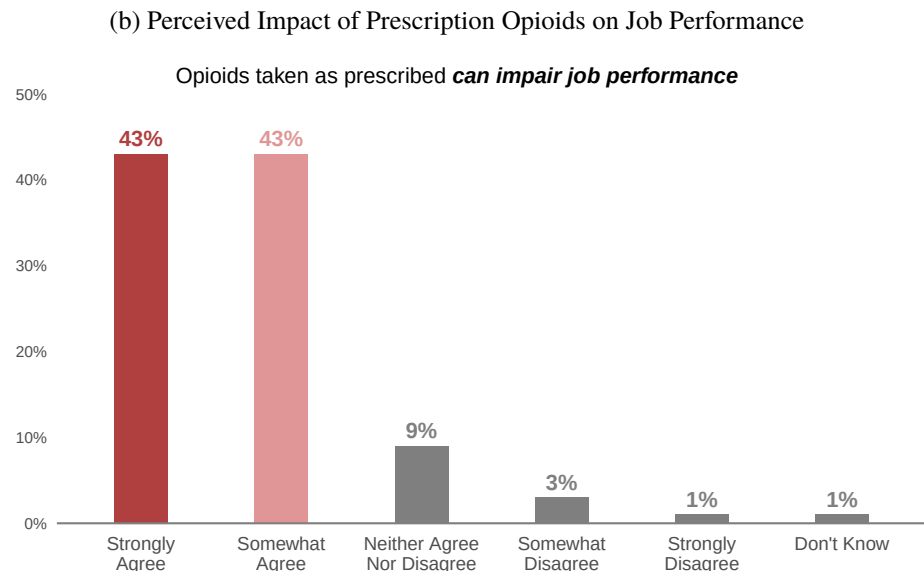
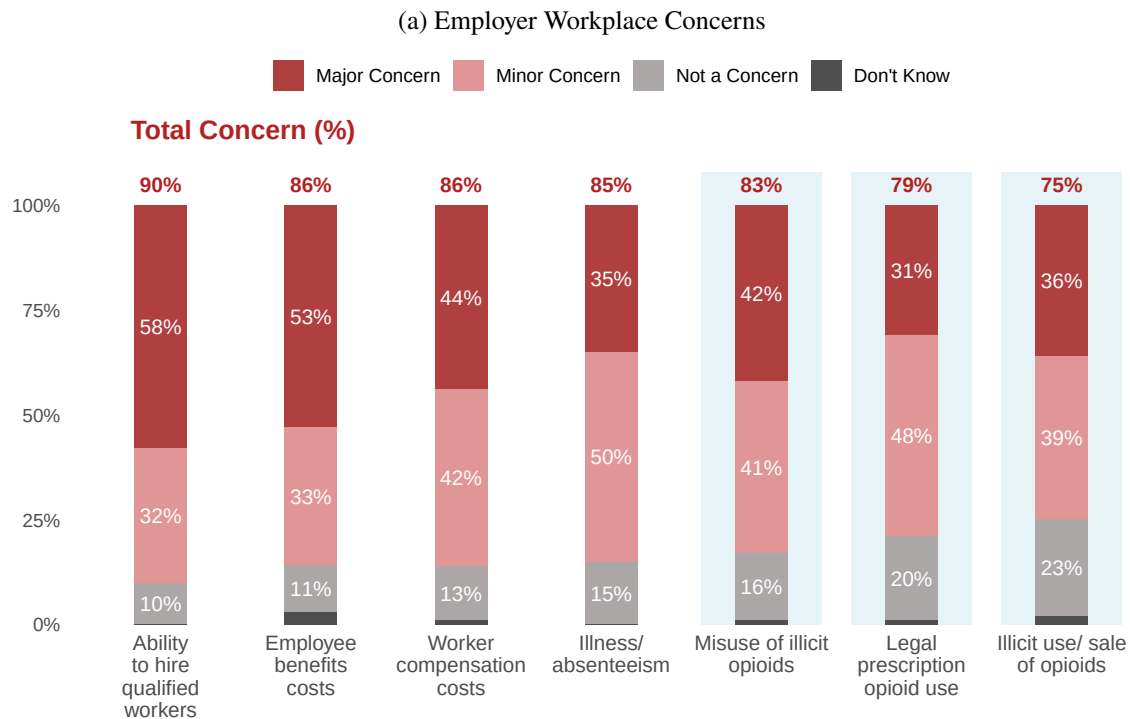
Notes: This figure plots national opioid overdose mortality rates (deaths per 100,000 population) from 1999–2019, separately for overdoses involving commonly prescribed opioids, heroin, and other synthetic opioids. The annotations highlight the three widely discussed “waves” of the opioid epidemic: the rise in prescription-opioid overdose deaths (Wave 1), the rise in heroin overdose deaths (Wave 2), and the rise in synthetic-opioid overdose deaths (Wave 3). Data are from the CDC National Vital Statistics System (NVSS) Mortality File (Multiple Cause of Death).

Figure 2: Trends in Public Search Interest in Opioids and Marijuana



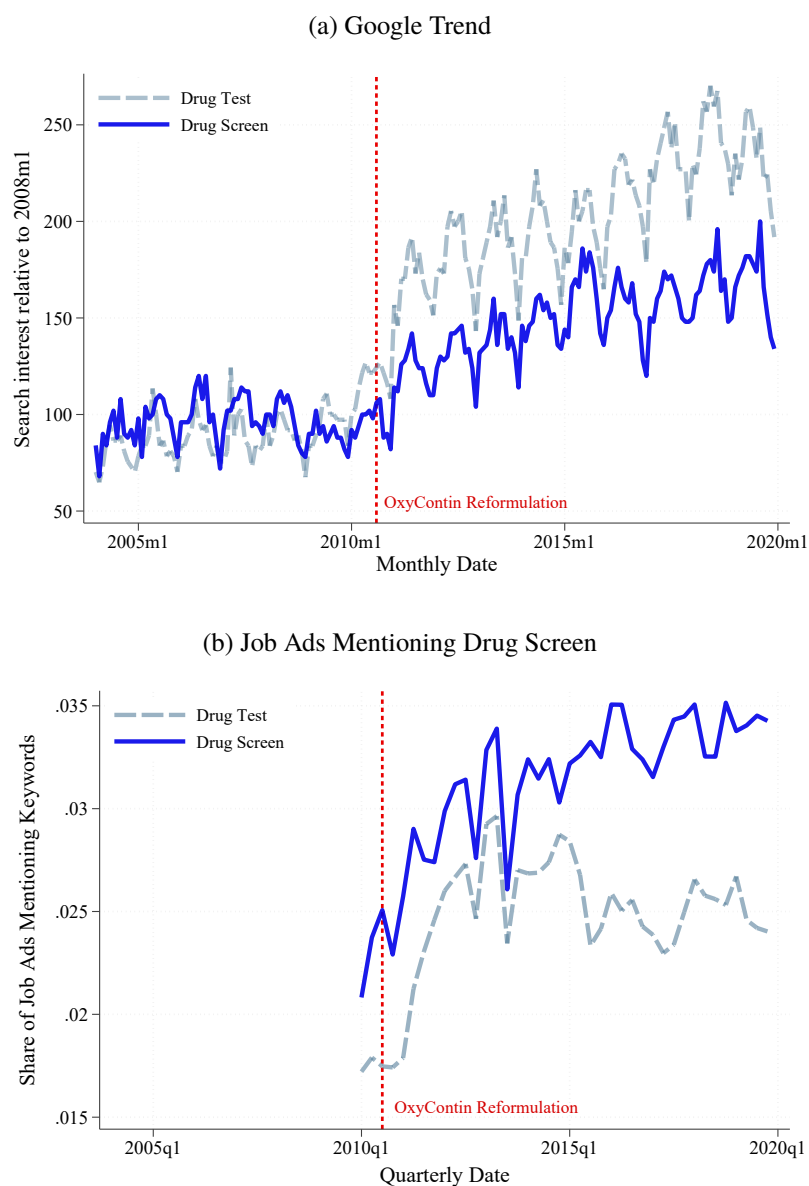
Notes: Panel (a) plots Google Trends search intensity for opioid-related keywords (“OxyContin,” “Heroin,” and “Fentanyl”; fentanyl is winsorized for readability) from 2004m1–2019m12. Panel (b) plots Google Trends search intensity for “Marijuana” over the same period. Each series is normalized to its value in 2008m1 (i.e., reported relative to 2008m1). The vertical dashed line marks the 2010 OxyContin reformulation.

Figure 3: Employer Workplace Concerns (National Employer Survey)



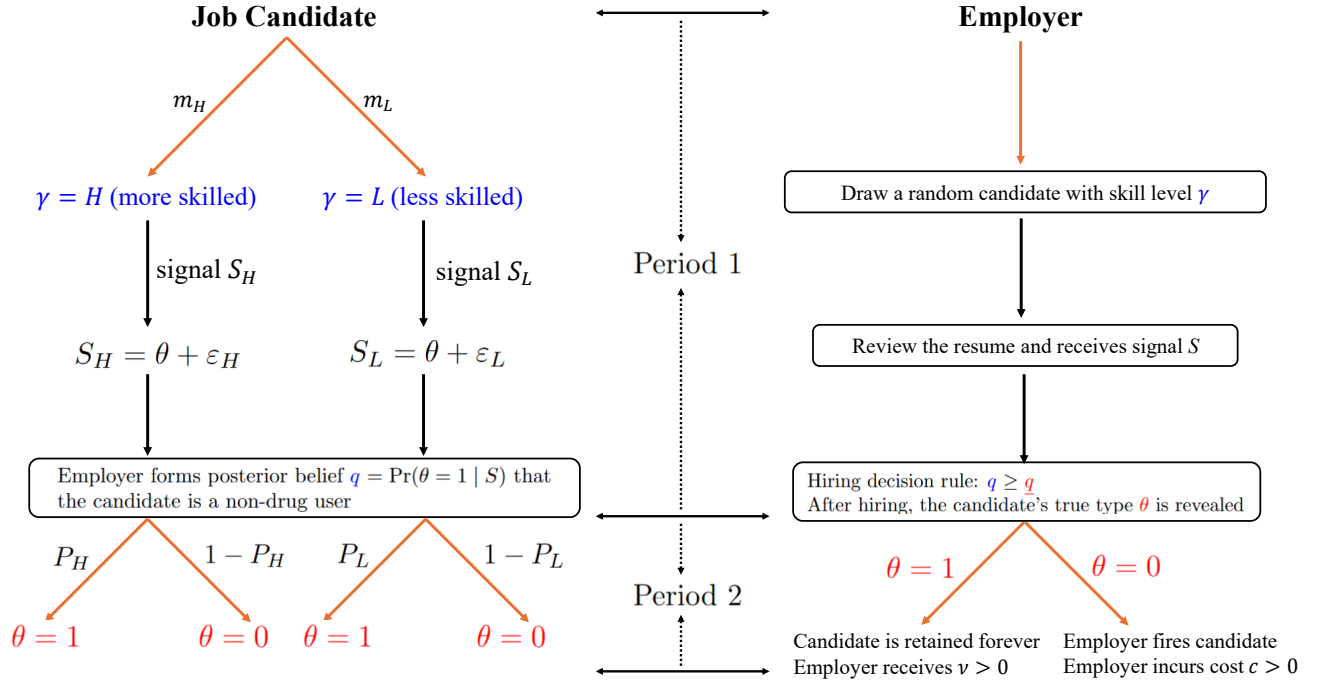
Notes: This figure is based on the National Safety Council’s (NSC) 2019 *National Employer Survey: Opioid Usage in the Workplace*, which includes responses from 526 U.S. employer decision-makers working in organizations with 50 or more employees. Panel (a) displays the share of respondents who view each workplace issue as a “major” or “minor” concern, with the total percentage of concern indicated at the top of each bar. Light blue shading highlights three opioid-related issues. Panel (b) reports the perceived impact of prescribed opioid use on job performance. Respondents were asked to rate their level of agreement with the statement “Opioids taken as prescribed can impair job performance.” Response options are shown in descending order of agreement, with bars representing the share of respondents selecting each option. Strong agreement responses are highlighted in dark red.

Figure 4: Trends in Drug Testing: Public Search Interest and Employer Job Postings



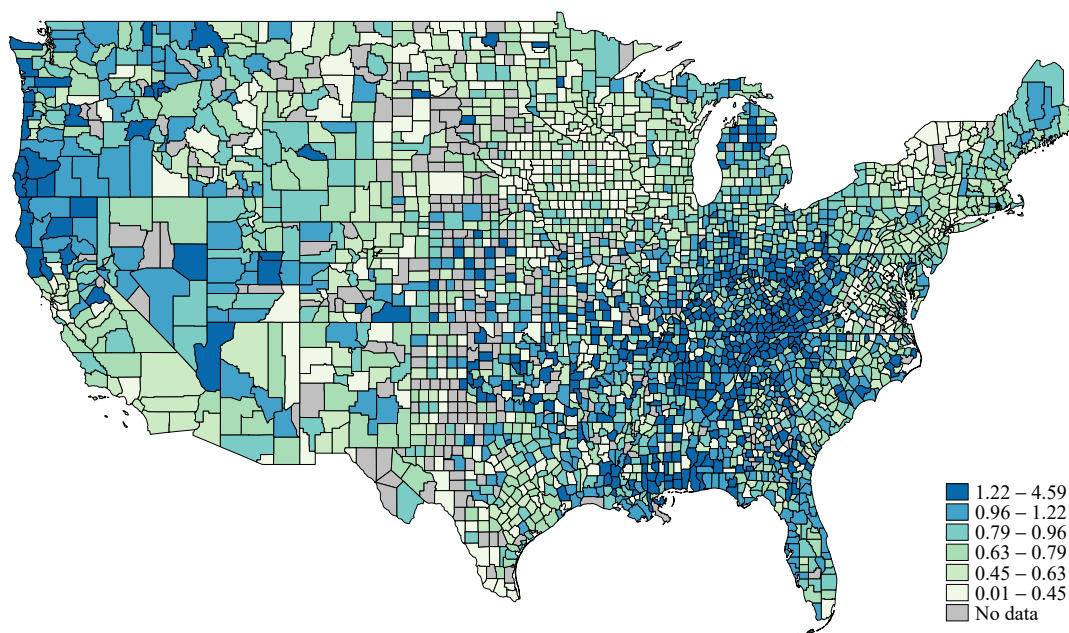
Notes: Panel (a) plots Google Trends search intensity for "Drug Test" and "Drug Screen," normalized to 2008m1, from 2005m1–2020m1. Panel (b) plots the quarterly share of online job postings that mention "drug test" or "drug screen." The vertical dashed line marks the August 2010 OxyContin reformulation.

Figure 5: Illustration of a Two-Period Hiring Model



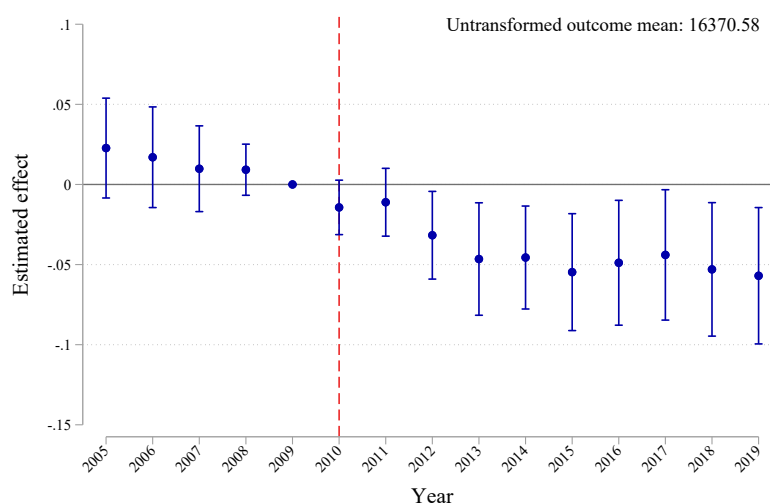
Notes: This figure illustrates the two-period hiring model for a specific occupation. In Period 1, the employer draws a random candidate from the applicant pool and observes their skill level γ (more-skilled H or less-skilled L) from their résumé. The employer then reviews the résumé and receives a noisy signal S about the candidate's productivity type θ , where $\theta = 1$ indicates a high-productivity worker (non-drug user) and $\theta = 0$ indicates a low-productivity worker (drug user). Based on the posterior belief q that the candidate is productive, the employer hires a job candidate only if the posterior belief about the success probability based on the observed signal exceeds the predetermined threshold $\bar{q}$. In Period 2, if hired, the candidate's true productivity type is revealed through work performance. High-productivity workers ($\theta = 1$) are retained and generate payoff $v > 0$, while low-productivity workers ($\theta = 0$) are dismissed, incurring cost c . The values P_H and P_L capture the probability that more-skilled and less-skilled candidates are high-productivity workers. In our model, we assume that P_H is greater than P_L , indicating that more-skilled candidates are more likely to be high-productivity workers.

Figure 6: Geographic Variation in Exposure to Pre-Intervention Prescription Opioids Across Counties



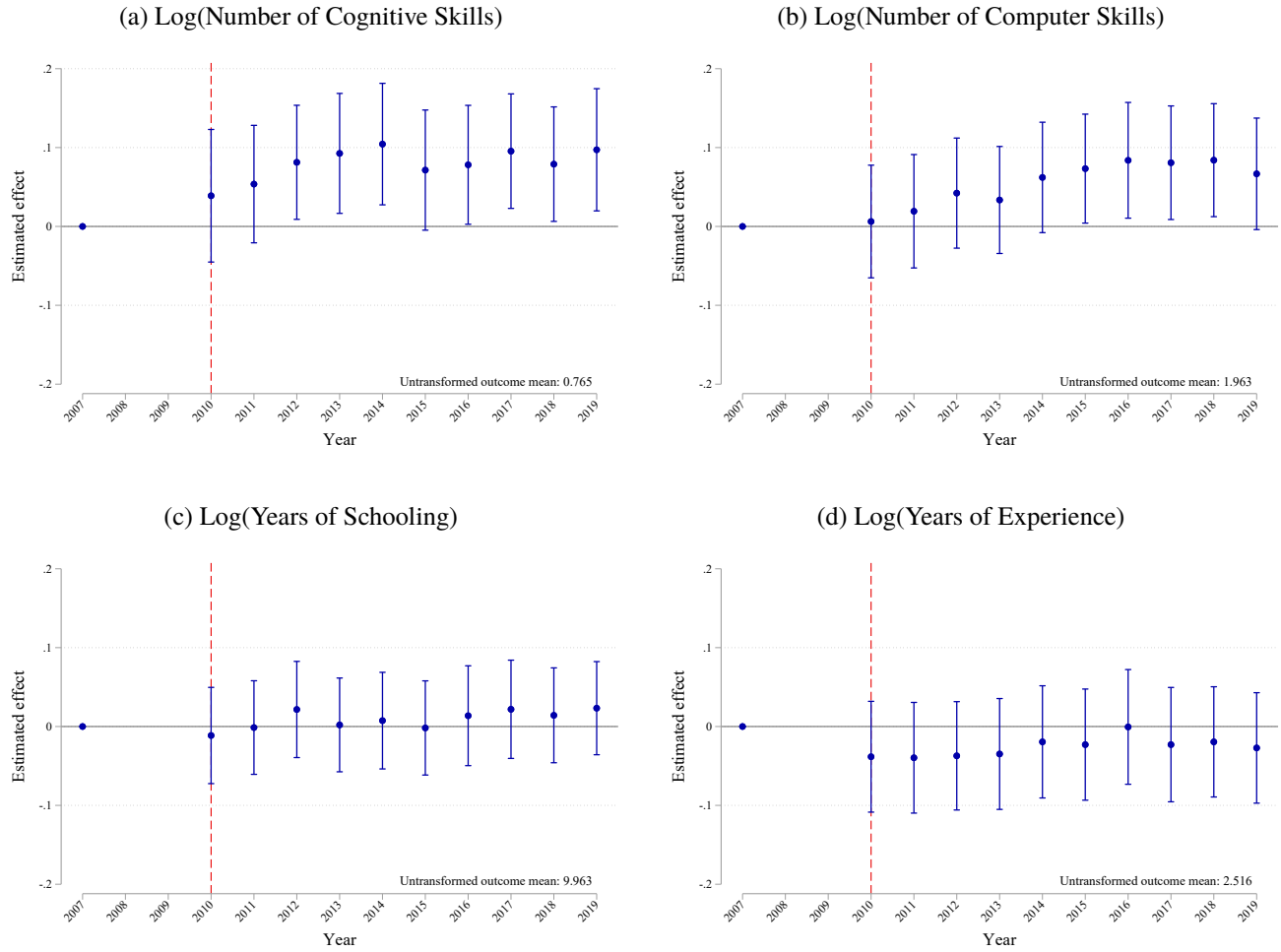
Notes: This figure presents the geographic distribution of pre-intervention per capita opioid prescriptions across US counties. We calculate the population-weighted mean number of all Schedule II opioid prescriptions per capita in each county for the years 2006 to 2009, obtained from the Centers for Disease Control (CDC).

Figure 7: Effects of the OxyContin Reformulation on Firm Employment



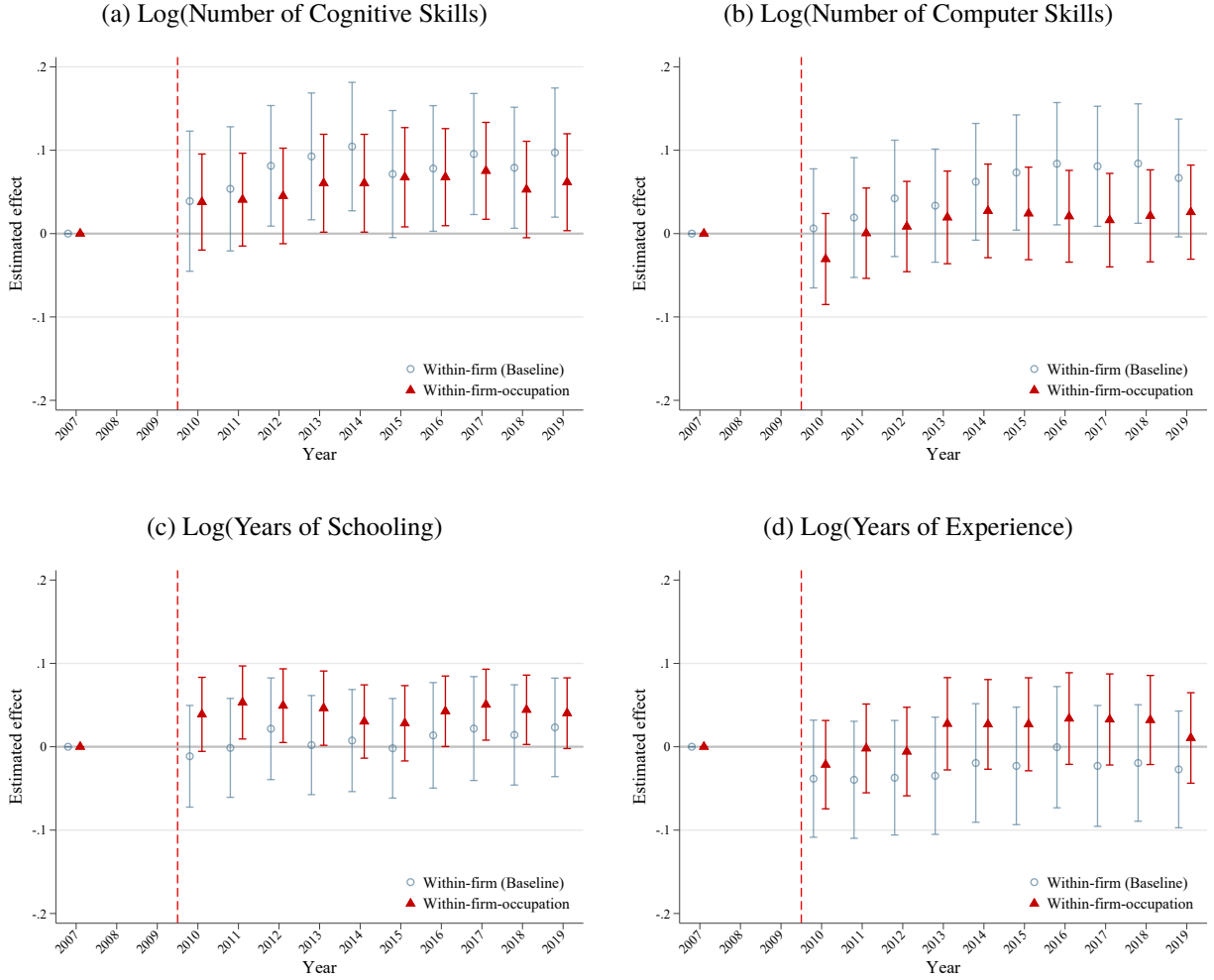
Notes: The figure shows the effect of firm-level exposure to the OxyContin reformulation on the log of employment. The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (13). The year 2009, which is one year prior to the OxyContin reformulation, is set as the reference point and normalized to zero. Standard errors are clustered at the firm level. The Untransformed outcome mean is calculated based on the pre-reformulation period.

Figure 8: Effects of the OxyContin Reformulation on Firm Skill Requirements



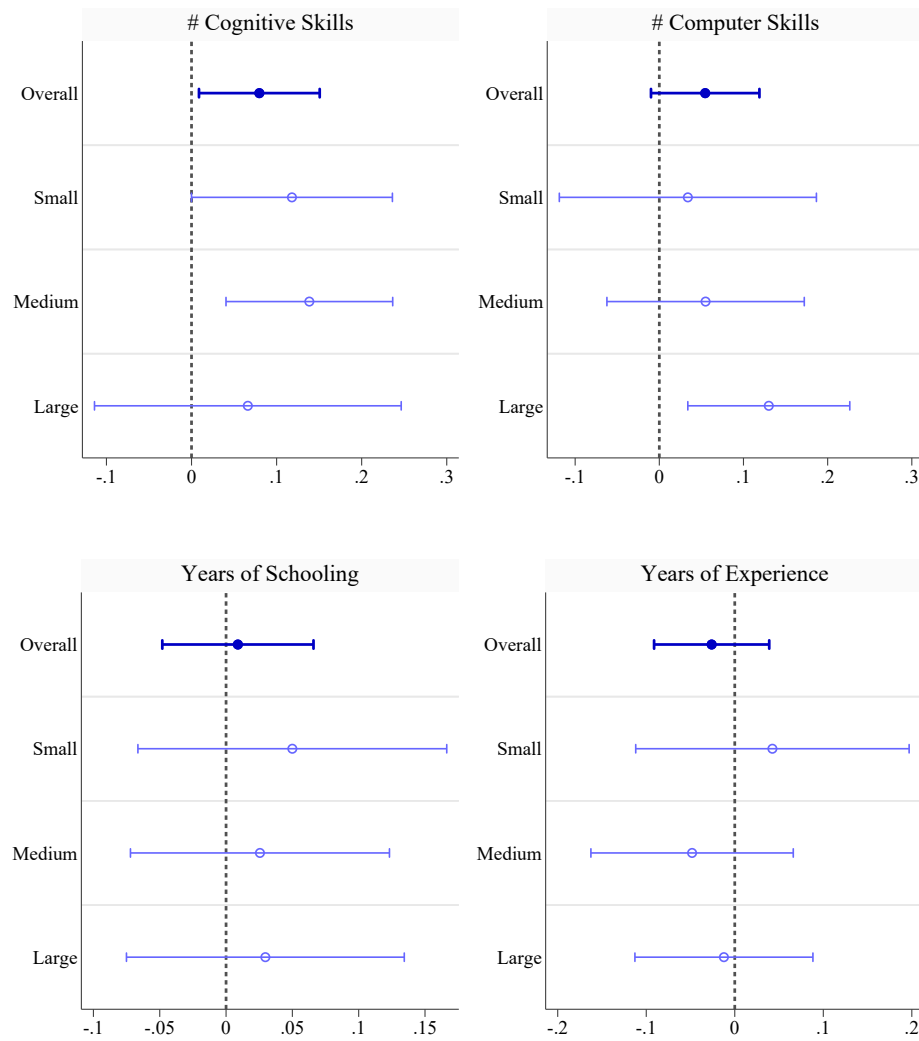
Notes: The figure shows the impact of firm-level exposure to the OxyContin reformulation on the following outcomes: average number of cognitive skills (panel (a)), average number of computer skills (panel (b)), average years of schooling (panel (c)), and average years of experience (panel (d)) required in a job posting. All outcomes are log-transformed. The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (13). The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level. Untransformed outcome means are calculated based on the pre-reformulation period.

Figure 9: Effects on Within-Firm-Occupation Skill Requirements



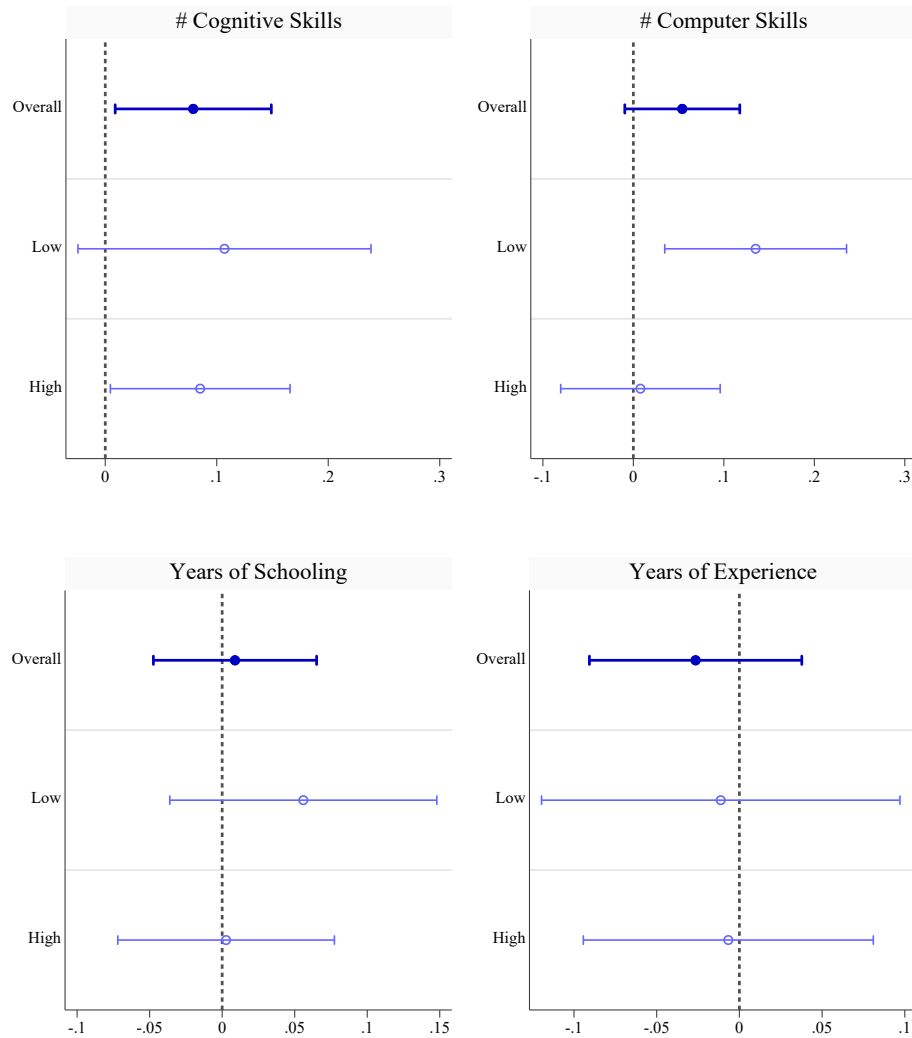
Notes: This figure compares event study estimates of the impact of firm-level exposure to the OxyContin reformulation on job skill requirements, using (i) our baseline firm-level specification (“Within-firm (Baseline)”) and (ii) a within-firm-by-occupation specification (“Within-firm-occupation”) constructed from a yearly firm $\times$ 6-digit SOC occupation panel. Outcomes are log-transformed measures of skill requirements in job postings: average number of cognitive skills (panel (a)), average number of computer skills (panel (b)), average years of schooling (panel (c)), and average years of experience (panel (d)). The figure plots the coefficients and corresponding 95% confidence intervals on the interaction terms between exposure and the full set of year dummies from an event study specification (analogous to equation (13)); year 2007 is the reference period and is normalized to zero. Standard errors are clustered at the firm level. The within-firm-occupation estimates additionally include firm-by-occupation fixed effects and occupation-by-year fixed effects, absorbing occupation-specific time trends.

Figure 10: Effects of Reformulation on Skill Requirements: Heterogeneity by Firm Size



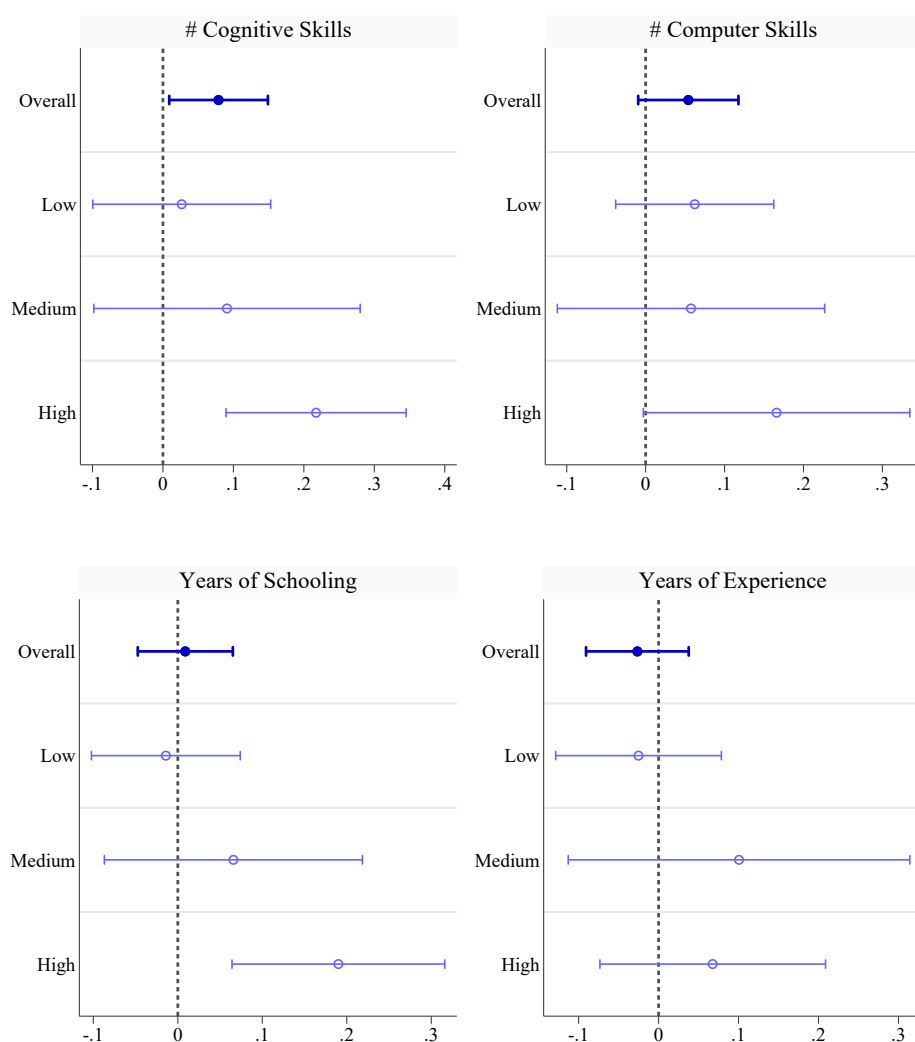
Notes: The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (14) separately for the sub-group denoted on the y-axis. Our baseline estimates are displayed at the top of each panel. All outcomes are log-transformed. Standard errors are clustered at the firm level.

Figure 11: Effects of Reformulation on Skill Requirements: Heterogeneity by Baseline Firm Education Requirement Levels



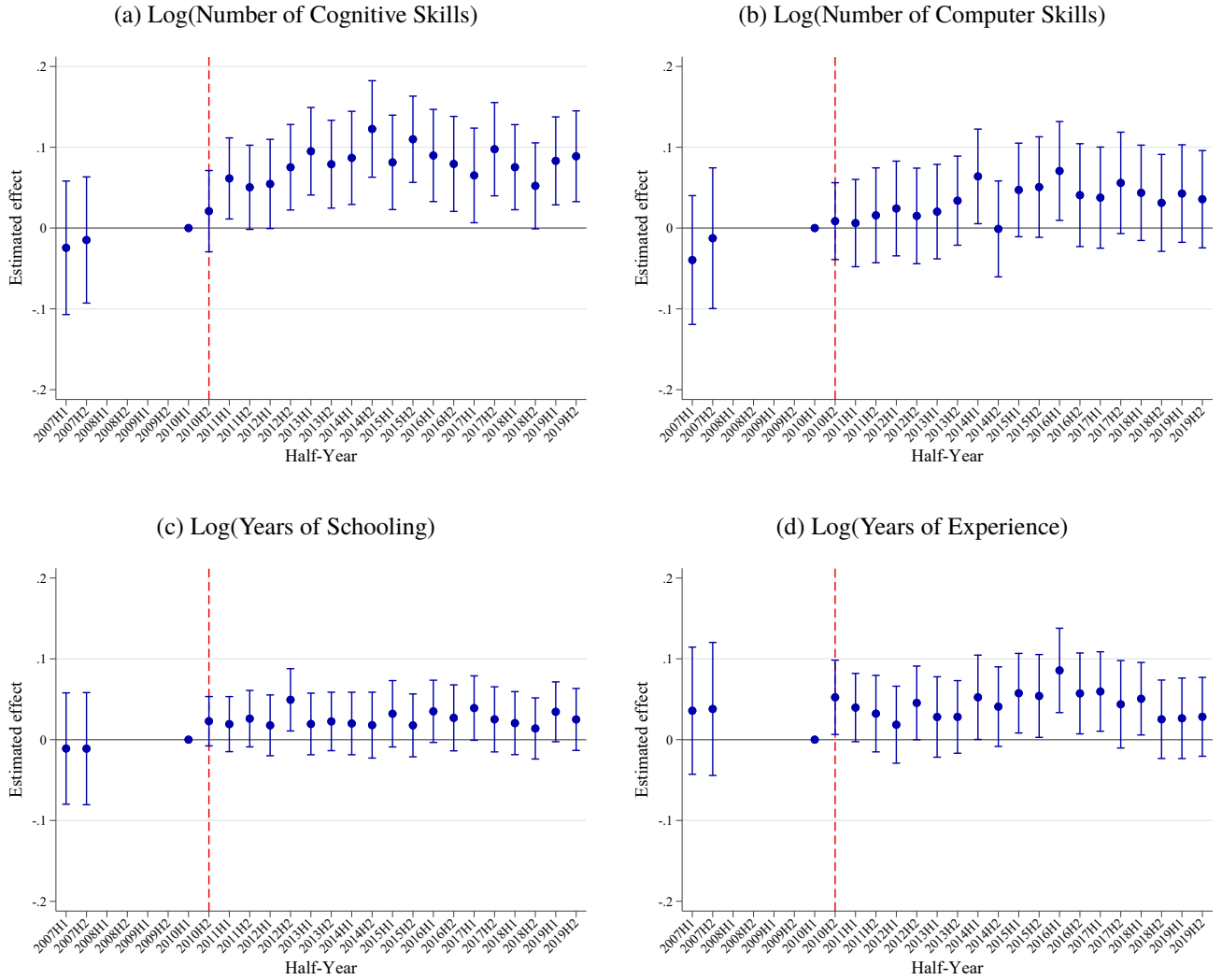
Notes: The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (14) separately for the sub-group denoted on the y-axis. Our baseline estimates are displayed at the top of each panel. All outcomes are log-transformed. We divide firms into two groups within each sector (utility, manufacturing, service, IT, finance, and professional service) based on the share of job postings requiring a four-year college degree measured in 2010, a partially treated year. Given that job posting outcomes are missing for a subset of our firms in 2007, we use 2010 data to calculate baseline education requirements for consistency. The average share of job postings requiring a four-year college degree in 2010 is 14.5% in the low-requirement group and 59.6% in the high-requirement group, respectively, after weighting observations by the number of job postings by the firm in 2010. Standard errors are clustered at the firm level.

Figure 12: Effects of Reformulation on Skill Requirements: Heterogeneity by Minimum Wage Levels



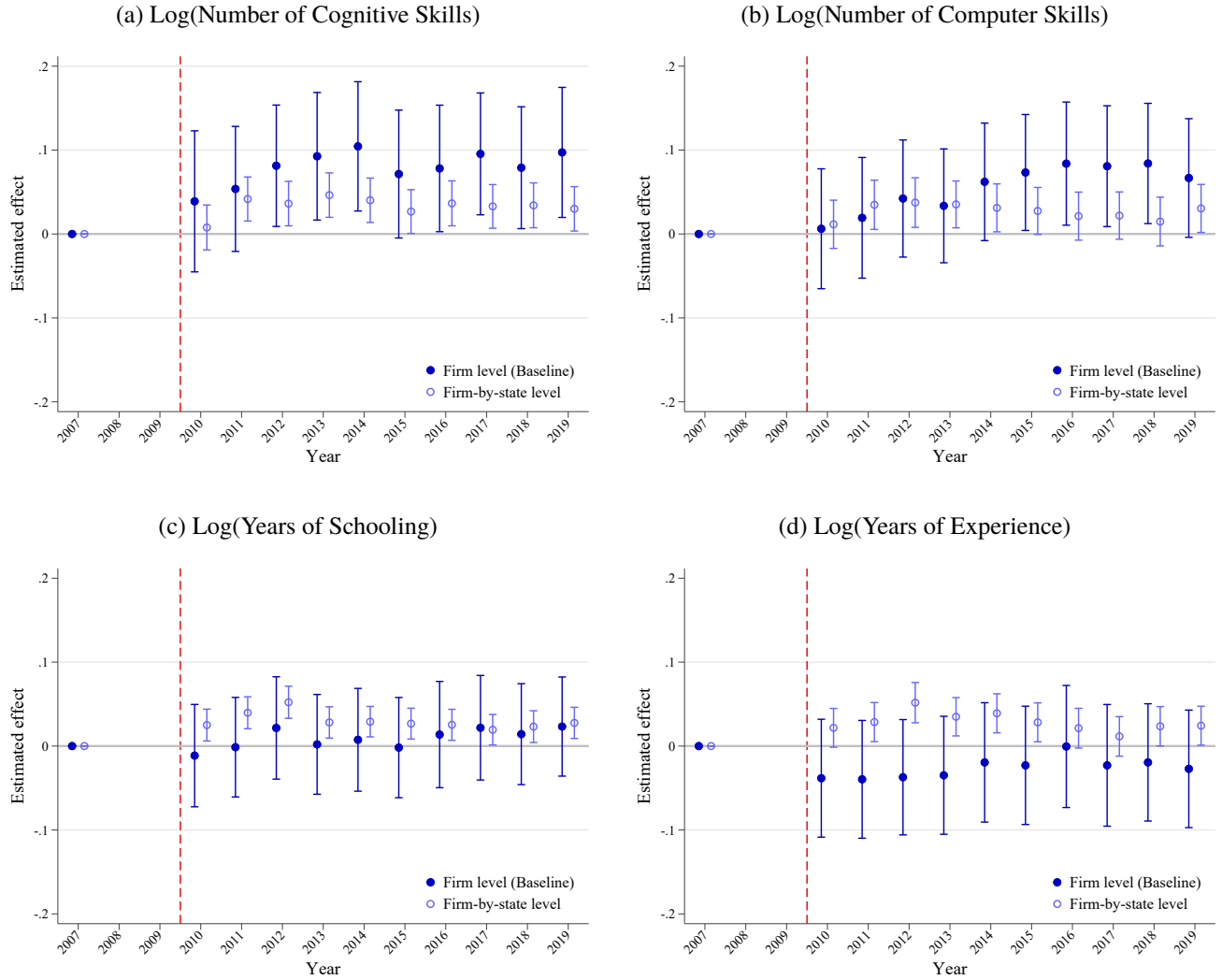
Notes: The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (14) separately for the sub-group denoted on the y-axis. Our baseline estimates are displayed at the top of each panel. Standard errors are clustered at the firm level.

Figure 13: Effects of Reformulation on Skill Requirements: Half-Yearly Data



Notes: The figure shows the impact of firm-level exposure to the OxyContin reformulation on job skill requirements using half-yearly data. The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (13) for the following outcomes: average number of cognitive skills (panel (a)), average number of computer skills (panel (b)), average years of schooling (panel (c)), and average years of experience (panel (d)) required in a job posting. All outcomes are log-transformed. The first half of 2010, which is one period prior to the OxyContin reformulation, is set as the reference point and normalized to zero. Standard errors are clustered at the firm level.

Figure 14: Effects of Reformulation on Skill Requirements: Firm-by-State Level Analysis



Notes: The figure presents output from estimation of equation (13) using our baseline sample (circles in dark blue) and our alternative sample based on state-by-year observations (hollow circles in light blue). For both samples, we plot the coefficients and 95% confidence intervals on the interactions between the exposure to reformulation and the full set of year dummies. The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level.

Table 1: Summary Statistics for Firm-Level Outcomes

	Mean	SD	25th percentile	75th percentile	Firm-year Observations
A. Pre-Intervention Rate of Prescription Opioid Use, 2006–2009					
Per capita opioid prescriptions	0.703	0.216	0.554	0.816	19,870
B. Firm Outcomes, 2005–2019					
Employment	16,894	46,320	442	11,400	19,870
C. Skill Requirements in a Job Posting, 2007, 2010–2019					
Number of job postings	1,634	6096	15	747	19,870
Number of cognitive skills	0.705	0.515	0.321	1.000	18,546
Number of computer skills	1.855	1.582	0.770	2.505	18,546
Years of schooling	9.509	4.236	6.894	12.553	18,546
Years of experience	2.394	1.540	1.282	3.326	18,546

Notes: This table presents summary statistics for the firm-level exposure measure and the main outcome variables for the 2,104 firms in our analysis. Panel A presents statistics for firm-level exposure to per capita opioid prescriptions from 2006 to 2009. Panel B reports statistics for firm-level employment over the years 2005–2019. Panel C presents statistics for the number of job postings per firm and for the average number of specific skills required in a firm’s job postings over the years 2007 and 2010–2019.

Table 2: Effects of the OxyContin Reformulation on Employment and Skill Requirements

	Employment (1)	Cognitive Skill (2)	Computer Skill (3)	Schooling Years (4)	Experience Years (5)
Panel A: Average Effect					
Post-reformulation (2010–2019)	-0.051*** (0.017) [0.002]	0.079** (0.036) [0.027]	0.054* (0.032) [0.095]	0.009 (0.029) [0.758]	-0.026 (0.033) [0.419]
Panel B: Short- and Medium-Run Effects					
Short-run post-reformulation (2010–2014)	-0.042*** (0.014) [0.004]	0.075** (0.036) [0.040]	0.032 (0.033) [0.322]	0.003 (0.029) [0.915]	-0.034 (0.033) [0.301]
Medium-run post-reformulation (2015–2019)	-0.062*** (0.021) [0.003]	0.083** (0.036) [0.021]	0.078** (0.034) [0.021]	0.015 (0.029) [0.609]	-0.018 (0.034) [0.591]
Sample unit	Firm	Firm	Firm	Firm	Firm
Data	Compustat	Lightcast	Lightcast	Lightcast	Lightcast
Outcome	Log	Log	Log	Log	Log
Untransformed outcome mean	16,370.58	0.765	1.963	9.963	2.516
Firm-year observations	26,302	16,954	17,425	17,583	17,551

Notes: This table presents coefficients, standard errors (in parentheses), and p-values [in brackets] from estimation of equation (14). The regressions include firm and industry-by-year fixed effects, where industry is defined by the 3-digit NAICS code. Standard errors are clustered at the firm level. Untransformed outcome means are calculated based on the pre-reformulation period. *p<0.1; **p<0.05; ***p<0.01.

Table 3: Robustness of Baseline Estimates to Controlling for Weighted Local Labor Force Participation by Worker Subgroups

	(1) Baseline	(2) Add Gender	(3) Add Education	(4) Add Race
Panel A: Average Effect				
Panel A.1: Employment				
Post-reformulation (2010–2019)	-0.051*** (0.017) [0.002]	-0.051*** (0.016) [0.002]	-0.052*** (0.016) [0.002]	-0.051*** (0.016) [0.002]
Panel A.2: Cognitive Skill Requirements				
Post-reformulation (2010–2019)	0.079** (0.036) [0.027]	0.075** (0.037) [0.042]	0.069* (0.037) [0.060]	0.069* (0.037) [0.062]
Panel A.3: Computer Skill Requirements				
Post-reformulation (2010–2019)	0.054* (0.032) [0.095]	0.052 (0.034) [0.123]	0.044 (0.033) [0.188]	0.044 (0.033) [0.185]
Panel B: Medium-Run Effect				
Panel B.1: Employment				
Medium-run post-reformulation (2015–2019)	-0.062*** (0.021) [0.003]	-0.062*** (0.021) [0.003]	-0.066*** (0.022) [0.003]	-0.065*** (0.022) [0.003]
Panel B.2: Cognitive Skill Requirements				
Medium-run post-reformulation (2015–2019)	0.083** (0.036) [0.021]	0.080** (0.037) [0.032]	0.072* (0.038) [0.056]	0.071* (0.038) [0.058]
Panel B.3: Computer Skill Requirements				
Medium-run post-reformulation (2015–2019)	0.078** (0.034) [0.021]	0.080** (0.035) [0.023]	0.075** (0.035) [0.032]	0.077** (0.035) [0.029]
LFP by Gender	No	Yes	Yes	Yes
LFP by Education	No	No	Yes	Yes
LFP by Race/Ethnicity	No	No	No	Yes

Notes: This table reports the sensitivity of our results when controlling for labor force participation rates for the subgroup of workers. In column 1, we reproduce our baseline estimates from the estimation of equation (14) reported in Table 2, separately for the average post-reformulation effect (Panel A) and the medium-run effect (Panel B). Each panel reports results for employment (sub-panel 1), cognitive skill requirements (sub-panel 2), and computer skill requirements (sub-panel 3). In column 2, we add female and male workers' labor force participation rates. In column 3, we add measures for education subgroups—college graduates and non-college graduates. Finally, column 4 adds measures for race/ethnicity subgroups—Non-Hispanic White, Non-Hispanic Black, and Hispanic. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

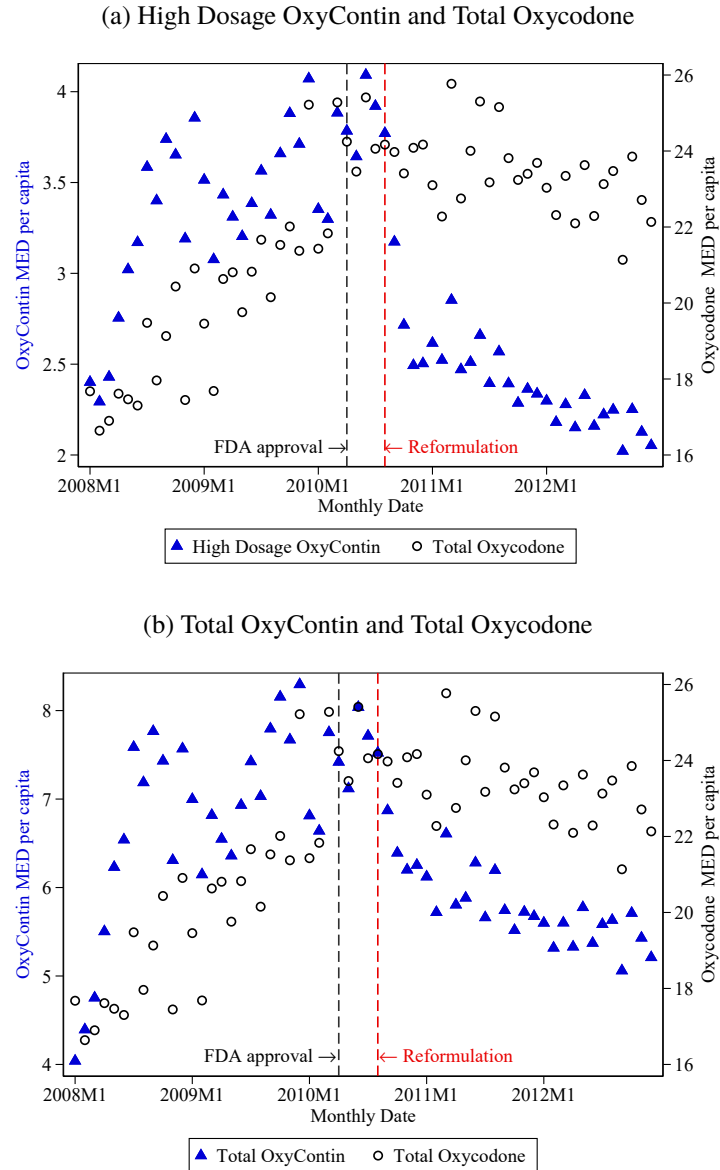
For Online Publication

“Who Gets Screened Out? The Opioid Crisis and
Employer Skill Requirements”

Kim, Kim, and Park (2026)

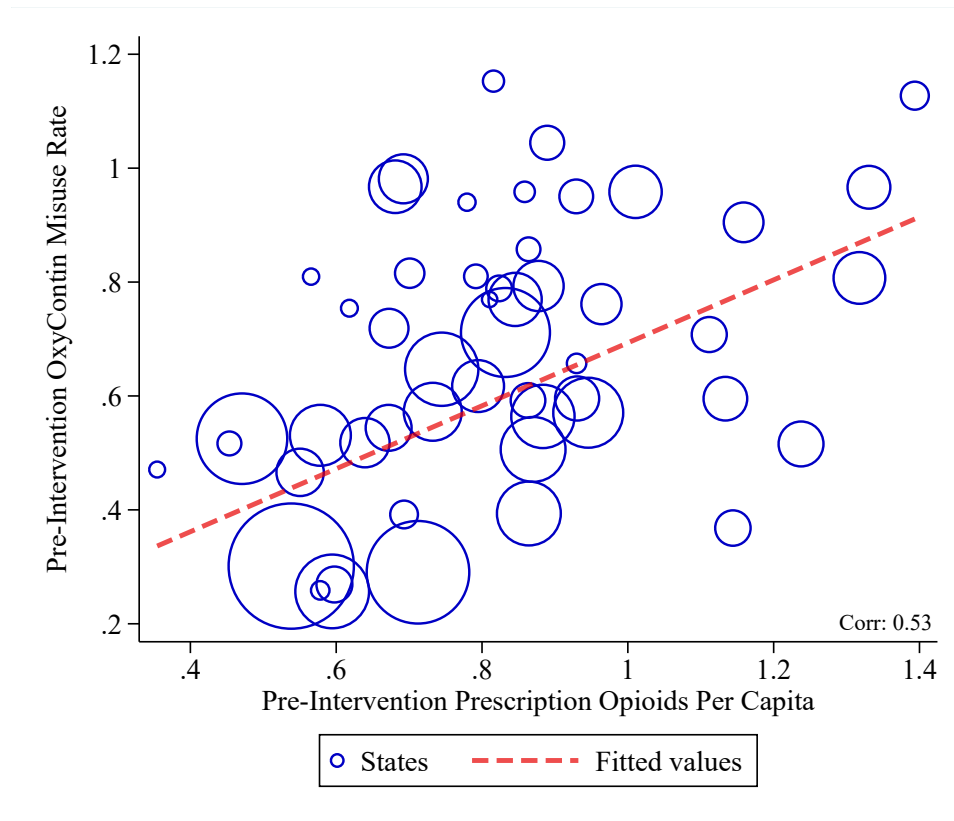
A Supplemental Figures and Tables

Figure A1: National Trends in Legal Distribution of OxyContin and Oxycodone



Notes: The figure presents national trends in the legal distribution of OxyContin and oxycodone from 2008m1 to 2012m12, based on the DEA's ARCOS data. Blue triangles represent the per capita Morphine Equivalent Dose (MED) of OxyContin, and black hollow circles indicate the per capita MED of oxycodone. In panel (a), we focus specifically on high-dosage OxyContin (80 mg), which is more susceptible to misuse. In panel (b), we present the trends in the total supply of OxyContin. The black dashed vertical line indicates April 2010, when the FDA approved the new OxyContin formula. The red dashed vertical line indicates August 2010, when the new formula was released and replaced the original version of OxyContin.

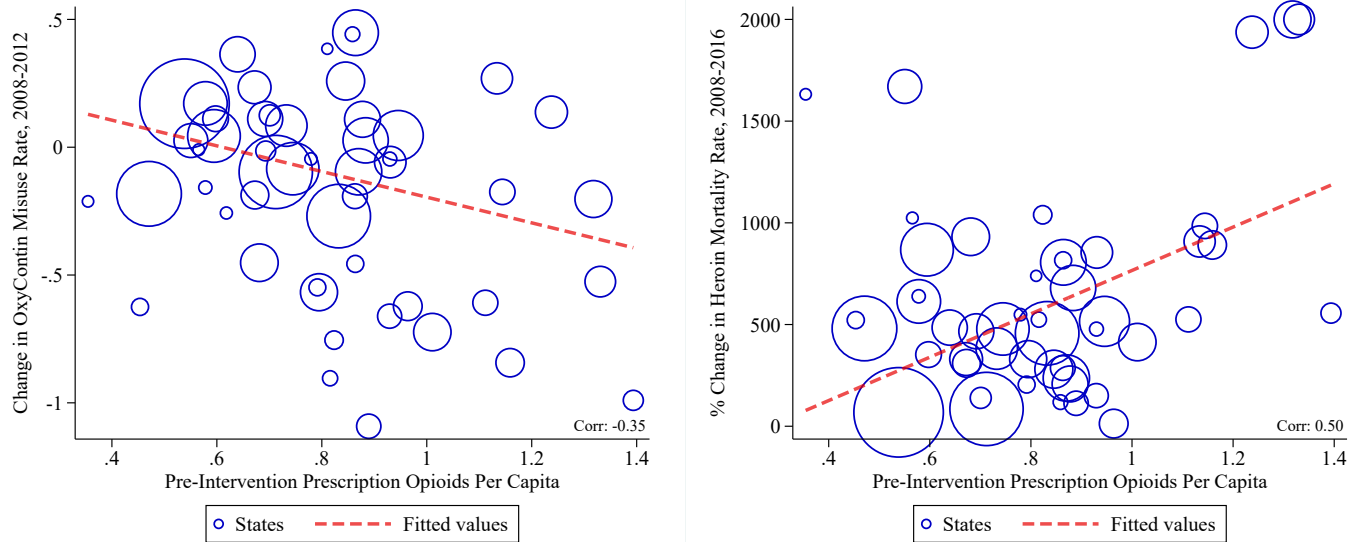
Figure A2: Relationship Between 2006–2009 Prescription Opioid Use Measure (CDC) and 2004–2009 OxyContin Misuse Measure (NSDUH)



Notes: This figure presents the relationship between the state-level average of opioid prescriptions per capita from 2006 to 2009, obtained using CDC data, and the state-level average of OxyContin misuse rates from 2004 to 2008, calculated using the National Survey on Drug Use and Health (NSDUH) data ([Alpert et al.](#)'s measure). Data on the NSDUH measure of OxyContin misuse are obtained from [Alpert et al. \(2018\)](#). To construct the state-level CDC measure of opioid prescription use, we calculate the population-weighted average of our county-level opioid exposure measure. The size of the markers indicates the population size of each state as of 2009.

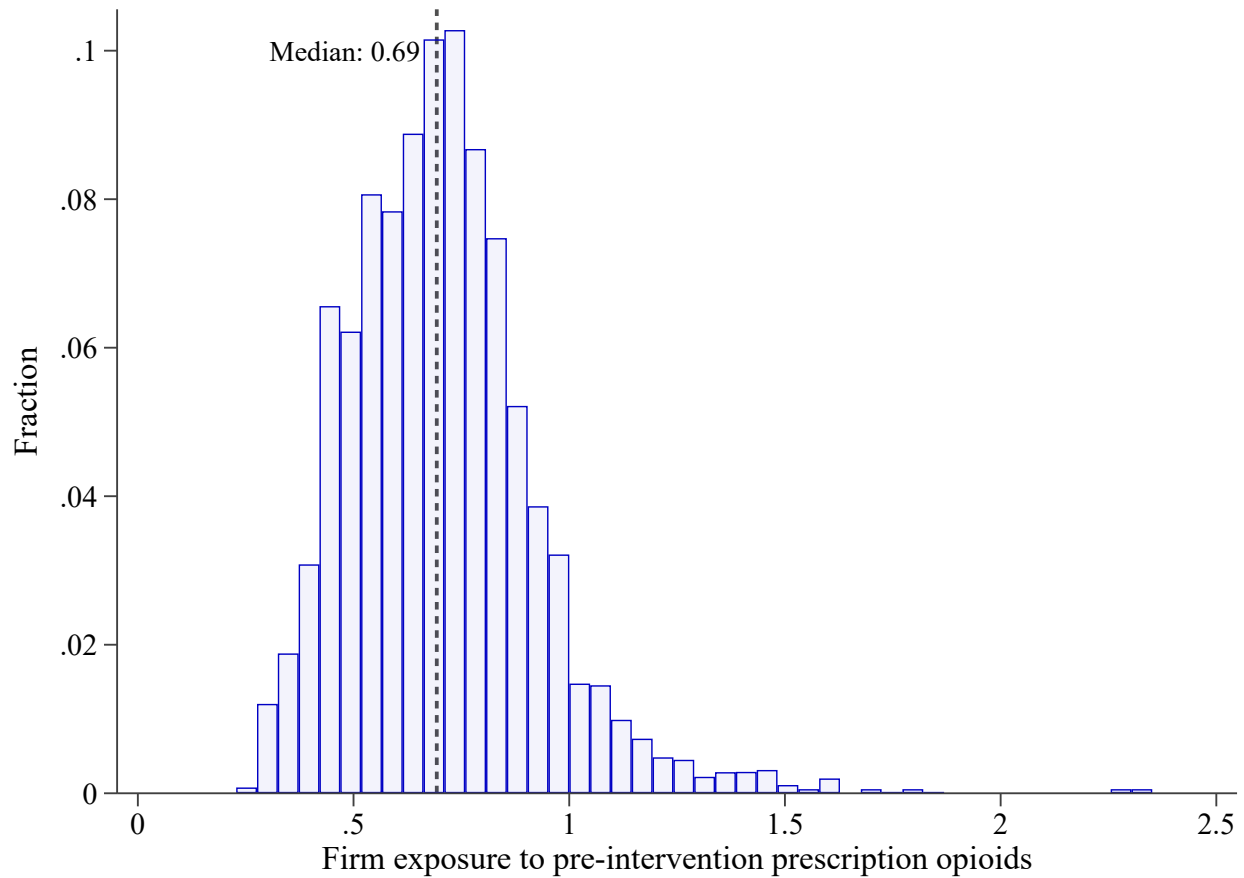
Figure A3: Relationship Between 2006–2009 Prescription Opioid Use Measure (CDC) and Changes in OxyContin Misuse and Changes in Heroin Mortality

(a) Pre-Intervention Exposure and Changes in OxyContin Misuse (b) Pre-Intervention Exposure and Changes in Heroin Mortality



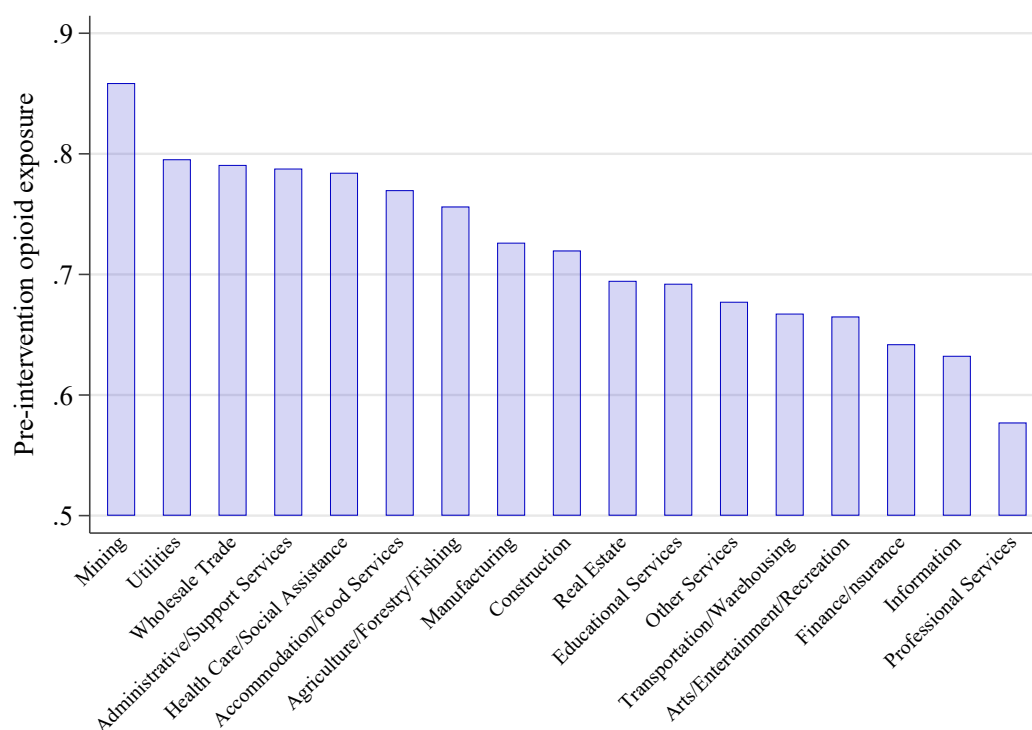
Notes: The figure presents the state-level averages of opioid prescriptions per capita from 2006 to 2009 (CDC data) and subsequent changes. Panel (a) shows the relationship between the CDC measure and level changes in OxyContin misuse rates between 2008 and 2012, and panel (b) presents the relationship between the same CDC measure and percentage changes in heroin death rates per 100,000 from 2008 to 2016. The NSDUH data on OxyContin misuse is obtained from [Alpert et al. \(2018\)](#), and the data on heroin mortality are obtained from the National Vital Statistics System (NVSS). The percentage change in mortality is winsorized at 2000 percent. The size of the markers indicates the population size of each state as of 2009.

Figure A4: Distribution of Firm Exposure to Pre-Intervention Prescription Opioids



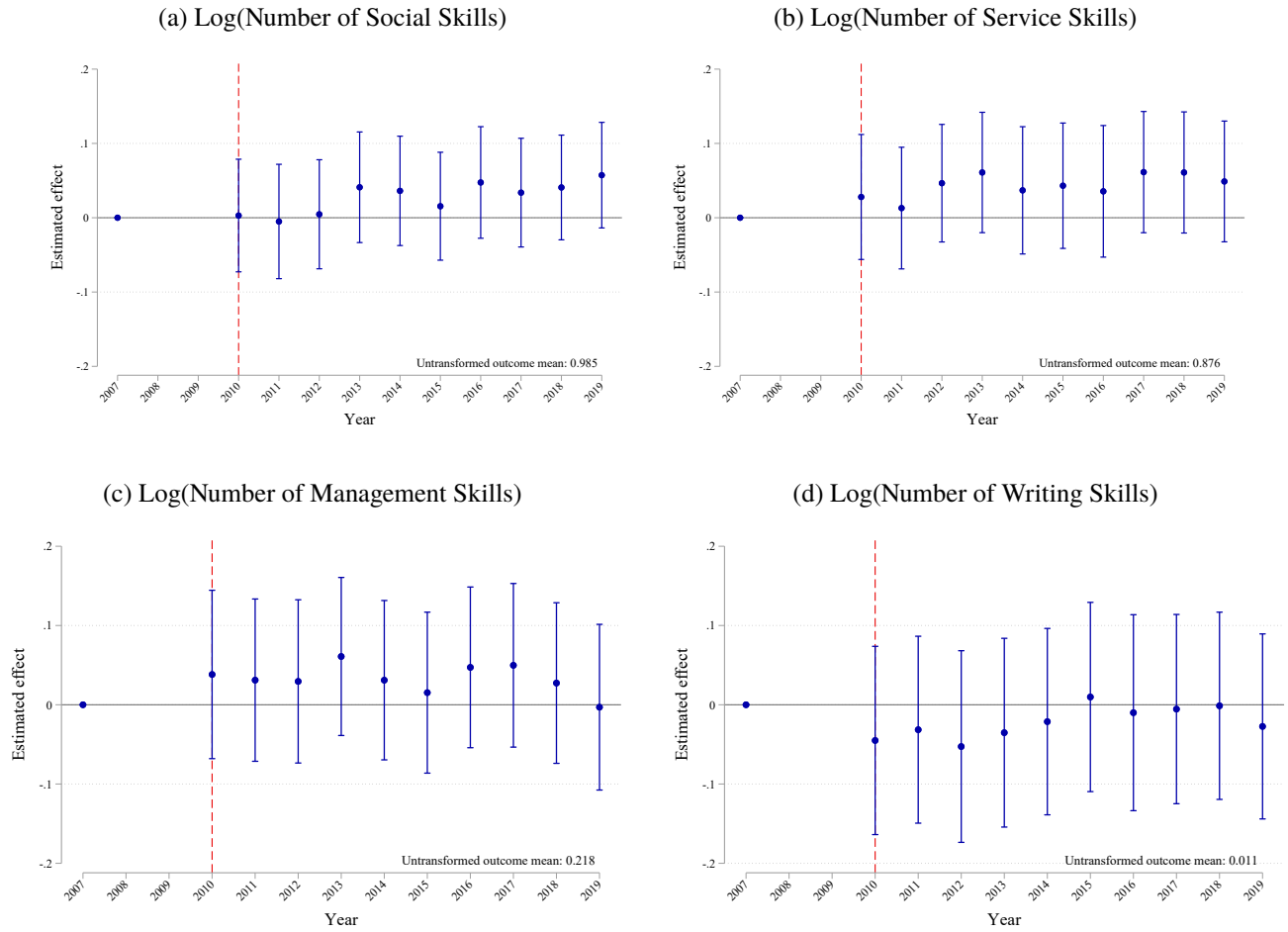
Notes: This figure shows a histogram of the firm-level pre-intervention exposure measure used in the analysis. A firm's exposure is constructed as a Bartik-style weighted average of county-level per-capita Schedule II prescription opioid prescriptions over the pre-intervention period (2006–2009) across the firm's establishments, where weights are based on each establishment's pre-intervention number of job ads (used to proxy establishment size). The vertical marker reports the sample median (0.69).

Figure A5: Firm Exposure to Pre-Intervention Prescription Opioids by Industry



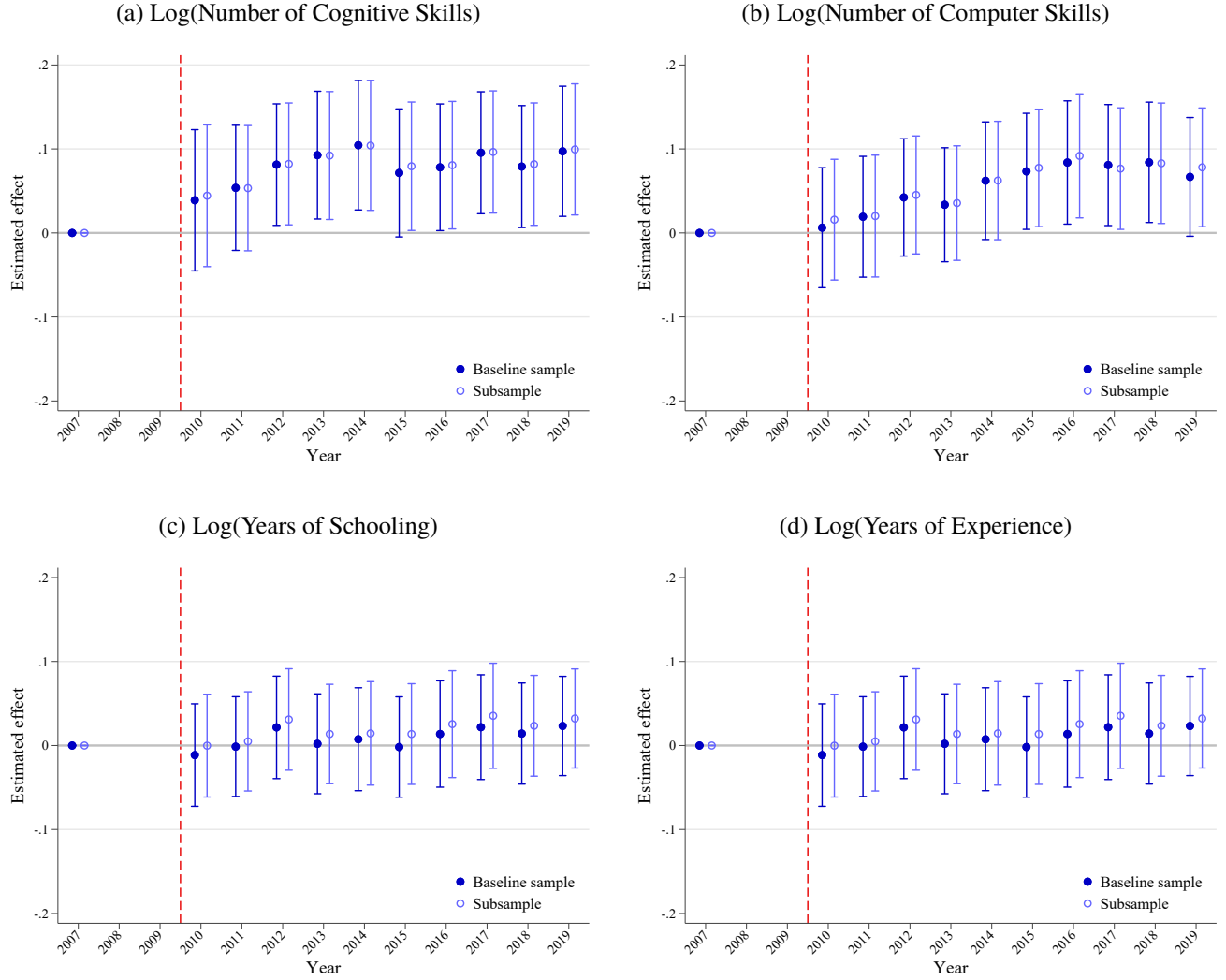
Notes: This figure summarizes firm exposure to pre-intervention prescription opioids by industry group. Firm exposure is the Bartik-style weighted average of county-level per-capita Schedule II prescription opioid prescriptions in 2006–2009 across the firm’s establishments, weighted by pre-intervention establishment-level job-ad counts.

Figure A6: Effects of Reformulation on Skill Requirements: Other Skills



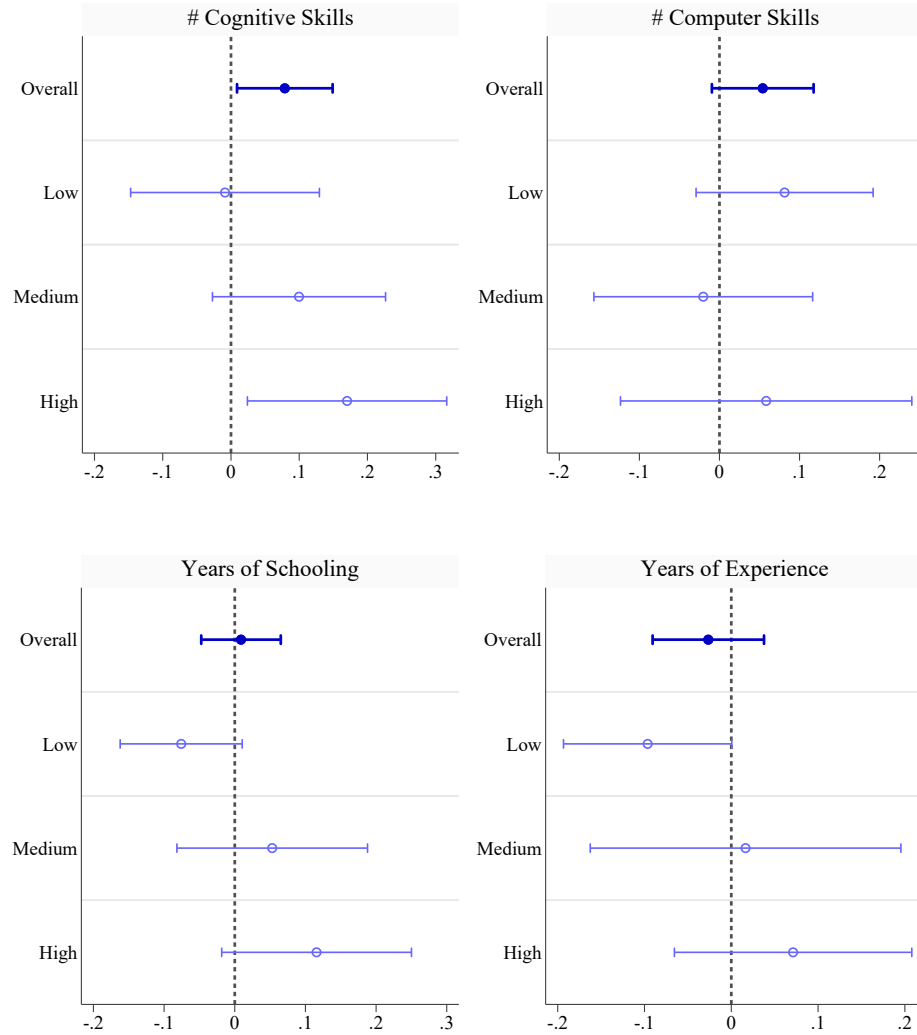
Notes: The figure shows the impact of firm-level exposure to the OxyContin reformulation on the following outcomes: average number of social skills (panel (a)), average number of service skills (panel (b)), average number of management skills (panel (c)), and average number of writing skills (panel (d)). All outcomes are log-transformed. The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (13). The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level. Untransformed outcome means are calculated based on the pre-reformulation period.

Figure A7: Robustness of Baseline Estimates to Using the Firm-Occupation Analysis Sample



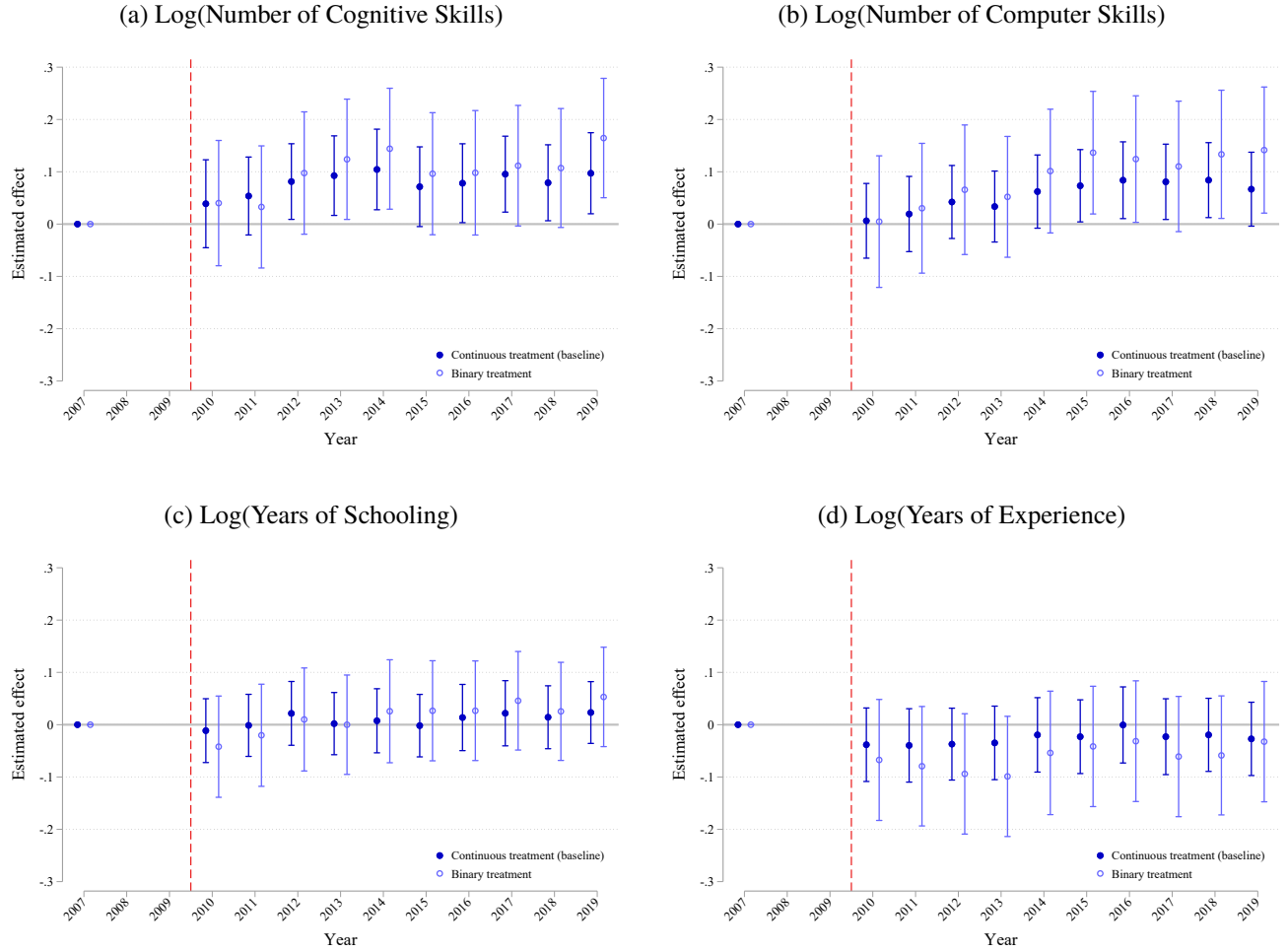
Notes: The figure presents output from the baseline regression model (13) estimated on two samples: the baseline sample (dark blue circles) and a subsample consisting of firms included in the firm-occupation analysis (light blue hollow circles). Both samples are based on firm-level panel data. For each sample, we plot the coefficients and 95% confidence intervals on the interactions between the exposure to reformulation and the full set of year dummies. The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level.

Figure A8: Effects of Reformulation on Skill Requirements: Heterogeneity by Employment Protection Levels



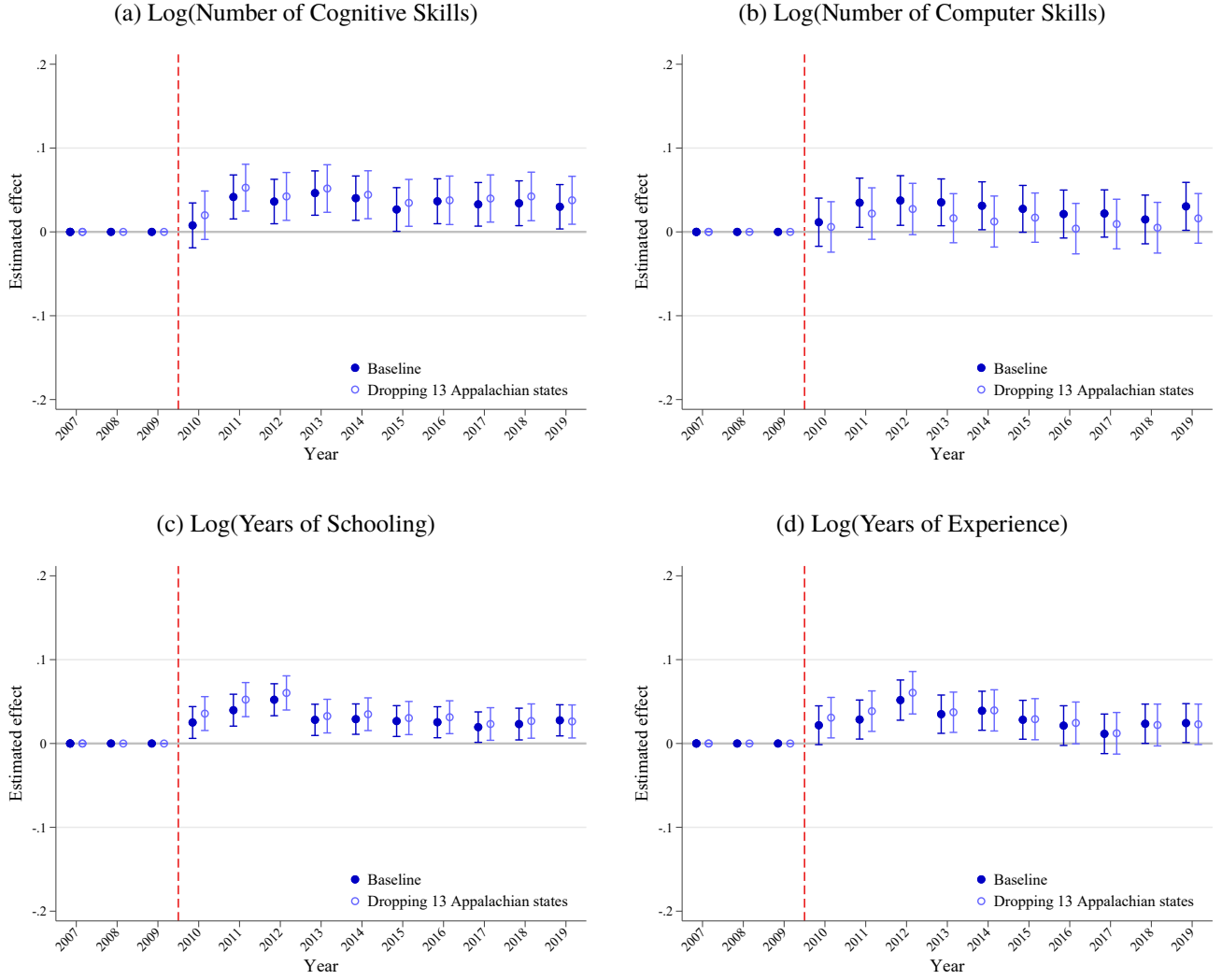
Notes: The figure displays the coefficients and their corresponding 95% confidence intervals on the interaction terms from equation (14) separately for the sub-group denoted on the y-axis. Our baseline estimates are displayed at the top of each panel. Standard errors are clustered at the firm level.

Figure A9: Robustness of Baseline Estimates to a Binary Treatment Measure of Pre-Reformulation Exposure



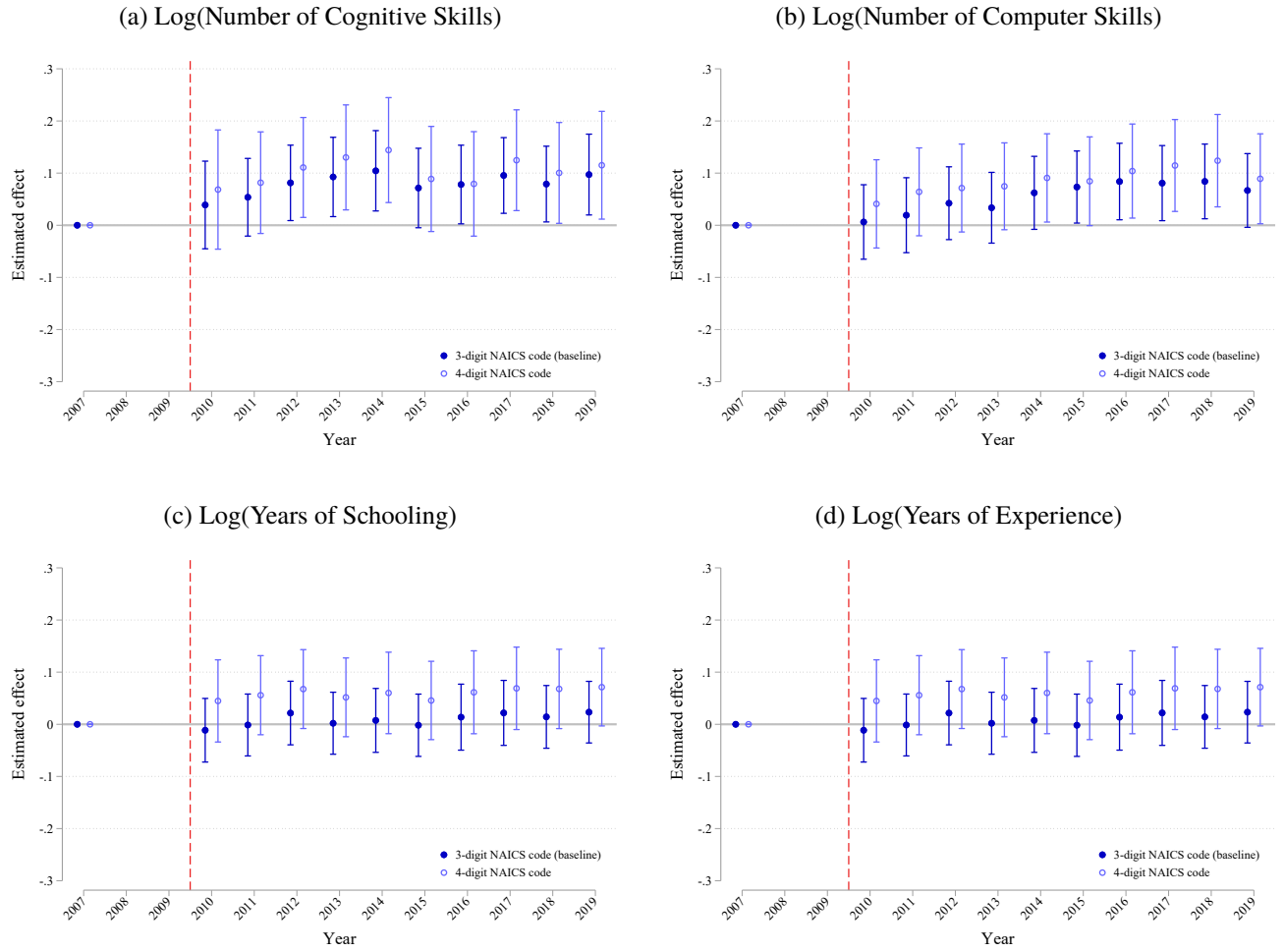
Notes: The figure presents estimates from two specifications of equation (13). The baseline specification uses the continuous measure of pre-reformulation exposure to local prescription opioid use (solid circles). The alternative specification replaces the continuous exposure measure with a binary treatment indicator (hollow circles), defined as an indicator for firms with above-median pre-reformulation exposure. The figure plots the coefficients and 95% confidence intervals on the interactions between treatment and year indicators for each specification. Outcomes are log-transformed measures of skill requirements in job postings: average number of cognitive skills (panel (a)), average number of computer skills (panel (b)), average years of schooling (panel (c)), and average years of experience (panel (d)). The reference year is 2007, normalized to zero. Standard errors are clustered at the firm level.

Figure A10: Robustness of Firm-by-State Analysis Estimates to Dropping Appalachian States



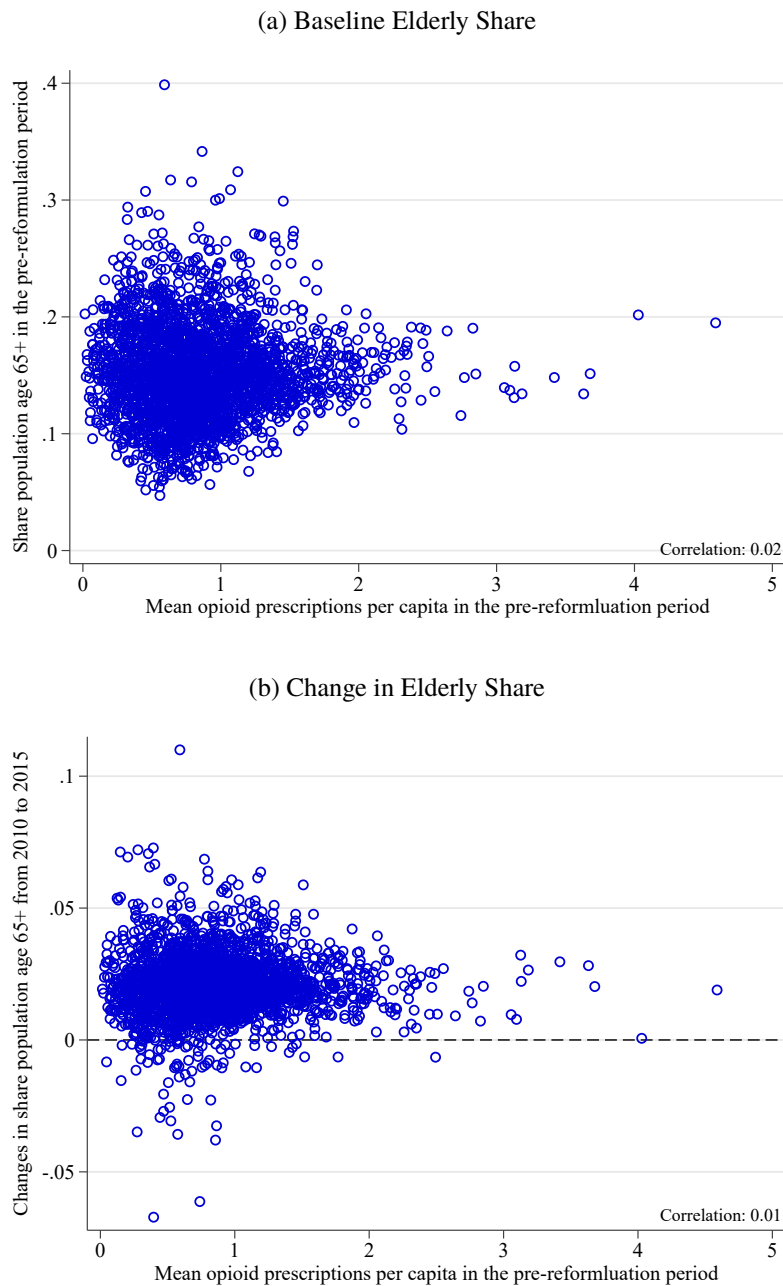
Notes: The figure presents output from estimation of equation (13) using our state-by-year sample (circles in dark blue) and the same sample with the exclusion of 13 Appalachian states (hollow circles in light blue). For both samples, we plot the coefficients and 95% confidence intervals on the interactions between the exposure to reformulation and the full set of year dummies. The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level.

Figure A11: Robustness of Baseline Estimates to Using a 4-Digit Industry Code



Notes: The figure presents results from equation (13) using two specifications: our baseline specification with industry-by-year fixed effects based on 3-digit industry code (represented by dark blue circles), and an alternative specification with industry-by-year fixed effects based on 4-digit industry code (represented by hollow circles in light blue). For both specifications, we plot the coefficients and 95% confidence intervals on the interactions between the exposure to reformulation and the full set of year dummies. The year 2007 is set as the reference point and normalized to zero. Standard errors are clustered at the firm level.

Figure A12: Correlation Between Local Prescription Opioid Use and Elderly Population Share



Notes: This figure presents the correlation between opioid exposure and elderly share. Panel (a) presents a scatter plot in which each observation represents the county-level population-weighted average opioid prescriptions per capita between 2006 and 2009—a measure for county-level exposure to the reformulation—on the x-axis, and the population-weighted average share of the population over age 65 in that county during the same period on the y-axis. In panel (b), we further examine the correlation between exposure to the OxyContin reformulation and the elderly share growth rate from 2010 to 2015.

Table A1: Employer Response to Substance Misuse by Substance Type

Response	Prescrip. Opioids	Prescrip. Stimulants	Benzodiazepines	Alcohol	Legal Marijuana	Illicit Marijuana	Illicit Opioids	Heroin, Fentanyl	Other Illicit Drugs
Ignore the problem	10%	9%	5%	4%	14%	4%	3%	2%	1%
Return them to position after treatment	41%	39%	39%	44%	27%	31%	25%	25%	22%
Ensure they are carefully monitored for rest of employment	31%	31%	26%	24%	27%	16%	17%	15%	17%
Relocate to position of lesser responsibility	9%	8%	8%	5%	7%	6%	6%	4%	5%
Dismiss them	9%	13%	21%	23%	24%	43%	49%	54%	55%

Notes: This table summarizes how employers respond to different types of substance misuse in the workplace, based on a survey conducted by the National Safety Council (NSC). The data reflect responses from 526 employer decision makers representing U.S. organizations with 50 or more employees. Respondents were asked: “Which of the following would you say best reflects your organization’s approach to an employee who is found to be misusing...?” Employers selected one of the following options: ignore the problem, return the employee to their position after treatment, ensure the employee is carefully monitored for the remainder of their employment, relocate the employee to a position of lesser responsibility, or dismiss them. Responses are shown by substance type, including prescription opioids, prescription stimulants, benzodiazepines, alcohol, legal marijuana, illicit marijuana, illicit opioids, heroin/fentanyl, and other illicit drugs.

Table A2: Skill Categorization

Category	Key words and phrases
Cognitive	Problem Solving, Research, Analytical, Critical Thinking, Math, Statistics
Computer	Computer, Spreadsheets, Common Software (e.g. Microsoft Excel, Powerpoint), Programming language or specialized software (e.g. Java, SQL, Python, etc.)
Social	Communication, Teamwork, Collaboration, Negotiation, Presentation
Customer Service	Customer, Sales, Client, Patient
Management	Project Management, People Management (Supervisory, Leadership, Management, Mentoring, Staff)
Writing	Writing

Notes: Categorization of skill requirements in Lightcast from [Deming and Kahn \(2018\)](#). Our computer skill category combines general computer skills and specific software skills, categorized separately in [Deming and Kahn \(2018\)](#). Computer skills are basic software, such as Microsoft Excel and PowerPoint, while specific software skills include names of specialized software.

Table A3: Robustness of Baseline Estimates to Controlling for Weighted Local Wage Level by Worker Subgroups

	(1) Baseline	(2) Add Gender	(3) Add Education	(4) Add Race
Panel A: Average Effect				
Panel A.1: Employment				
Post-reformulation (2010–2019)	-0.051*** (0.017) [0.002]	-0.050*** (0.017) [0.003]	-0.050*** (0.017) [0.004]	-0.044** (0.017) [0.011]
Panel A.2: Cognitive Skill Requirements				
Post-reformulation (2010–2019)	0.079** (0.036) [0.027]	0.084** (0.036) [0.019]	0.084** (0.036) [0.021]	0.085** (0.037) [0.021]
Panel A.3: Computer Skill Requirements				
Post-reformulation (2010–2019)	0.054* (0.032) [0.095]	0.065** (0.033) [0.048]	0.061* (0.033) [0.062]	0.062* (0.033) [0.062]
Panel B: Medium-Run Effect				
Panel B.1: Employment				
Medium-run post-reformulation (2015–2019)	-0.062*** (0.021) [0.003]	-0.063*** (0.021) [0.003]	-0.064*** (0.022) [0.004]	-0.057*** (0.022) [0.010]
Panel B.2: Cognitive Skill Requirements				
Medium-run post-reformulation (2015–2019)	0.083** (0.036) [0.021]	0.087** (0.036) [0.017]	0.086** (0.037) [0.019]	0.088** (0.037) [0.019]
Panel B.3: Computer Skill Requirements				
Medium-run post-reformulation (2015–2019)	0.078** (0.034) [0.021]	0.085** (0.034) [0.012]	0.084** (0.034) [0.014]	0.086** (0.034) [0.012]
Wage by Gender	No	Yes	Yes	Yes
Wage by Education	No	No	Yes	Yes
Wage by Race/Ethnicity	No	No	No	Yes

Notes: This table reports the sensitivity of our results when controlling for wage levels for the subgroup of workers. In column 1, we reproduce our baseline estimates from the estimation of equation (14) reported in Table 2, separately for the average post-reformulation effect (Panel A) and the medium-run effect (Panel B). Each panel reports results for employment (sub-panel 1), cognitive skill requirements (sub-panel 2), and computer skill requirements (sub-panel 3). In column 2, we add female and male workers' wage levels. In column 3, we add measures for education subgroups—college graduates and non-college graduates. Finally, column 4 adds measures for race/ethnicity subgroups—Non-Hispanic White, Non-Hispanic Black, and Hispanic. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

B Proof of Proposition and Policy Implications

In this section, we provide proofs of the proposition and policy implications presented in Section 3.

PROPOSITION 1: *The optimal hiring strategy, $\underline{q}^*$, can be implicitly defined as*

$$\begin{aligned}\underline{q}^* = & \frac{c}{c + (1 - \delta)v + k} - \frac{c\delta}{c + (1 - \delta)v + k} (1 - m_{HPH} - m_{LP_L}) \\ & + \frac{c\delta}{c + (1 - \delta)v + k} \int_0^{\underline{q}^*} G(q) dq.\end{aligned}$$

Proof of Proposition 1. We begin by proving equation (8) from Section 3.

$$\begin{aligned}V(\underline{q}) = & \delta \int_0^1 \max\{H(q, V(\underline{q})), V(\underline{q})\} dG(q) - k \\ = & \delta \int_0^1 \max\{qv + (1 - q)(V(\underline{q}) - c), V(\underline{q})\} dG(q) - k \\ = & \delta \int_0^{\underline{q}} V(\underline{q}) dG(q) + \delta \int_{\underline{q}}^1 qv + (1 - q)(V(\underline{q}) - c) dG(q) - k \\ = & \delta V(\underline{q}) [G(q)]_{q=0}^{\underline{q}} + \delta \int_{\underline{q}}^1 qv + (1 - q)[V(\underline{q}) - c] dG(q) - k \\ = & \delta V(\underline{q}) G(\underline{q}) + \delta \int_{\underline{q}}^1 vq dG(q) + \delta V(\underline{q}) \int_{\underline{q}}^1 (1 - q) dG(q) - c\delta \int_{\underline{q}}^1 (1 - q) dG(q) - k.\end{aligned}$$

So we have:

$$\begin{aligned}V(\underline{q}) \left[1 - \delta G(\underline{q}) - \delta \int_{\underline{q}}^1 (1 - q) dG(q) \right] &= \delta \left[\int_{\underline{q}}^1 vq dG(q) - c \int_{\underline{q}}^1 (1 - q) dG(q) \right] - k \\ &= \delta \int_{\underline{q}}^1 vq - c(1 - q) dG(q) - k.\end{aligned}$$

This gives:

$$V(\underline{q}) = \frac{\delta \int_{\underline{q}}^1 (vq + (1 - q)(-c)) dG(q) - k}{1 - \delta G(\underline{q}) - \delta \int_{\underline{q}}^1 (1 - q) dG(q)}.$$

In addition, the denominator is:

$$1 - \delta G(\underline{q}) - \delta \int_{\underline{q}}^1 (1 - q) dG(q) = 1 - \delta G(\underline{q}) - \delta \int_{\underline{q}}^1 1 dG(q) + \delta \int_{\underline{q}}^1 q dG(q)$$

$$\begin{aligned}
&= 1 - \delta G(\underline{q}) - \delta [G(q)]_{q=\underline{q}}^{q=1} + \delta \int_{\underline{q}}^1 q dG(q) \\
&= 1 - \delta G(\underline{q}) - \delta + \delta G(\underline{q}) + \delta \int_{\underline{q}}^1 q dG(q) = 1 - \delta + \delta \int_{\underline{q}}^1 q dG(q),
\end{aligned}$$

which completes the proof of equation (8). Also note that:

$$\begin{aligned}
\int_{\underline{q}}^1 (1 - q) dG(q) &= \int_{\underline{q}}^1 1 dG(q) - \int_{\underline{q}}^1 q dG(q) \\
&= [G(q)]_{q=\underline{q}}^{q=1} - \int_{\underline{q}}^1 q dG(q) \\
&= 1 - G(\underline{q}) - \int_{\underline{q}}^1 q dG(q).
\end{aligned}$$

Plug this into the value function of equation (8), we have:

$$\begin{aligned}
V(\underline{q}) &= \frac{\delta \nu \int_{\underline{q}}^1 q dG(q) - \delta c \left[1 - G(\underline{q}) - \int_{\underline{q}}^1 q dG(q) \right] - k}{1 - \delta \left(1 - \int_{\underline{q}}^1 q dG(q) \right)} \\
&= \frac{\delta \nu \int_{\underline{q}}^1 q dG(q) - \delta c(1 - G(\underline{q})) + \delta c \int_{\underline{q}}^1 q dG(q) - k}{1 - \delta \left(1 - \int_{\underline{q}}^1 q dG(q) \right)}.
\end{aligned}$$

Now let's define $N \equiv \delta \int_{\underline{q}}^1 [q\nu + (1 - q)(-c)] dG(q) - k$ and $D \equiv 1 - \delta \left(1 - \int_{\underline{q}}^1 q dG(q) \right)$. Then we have $V(\underline{q}) \equiv \frac{N}{D}$. Now the F.O.C condition gives:

$$\frac{\partial V(\underline{q})}{\partial \underline{q}} = \frac{N'D - D'N}{D^2} = 0.$$

From the Leibniz Integral Rule, we have:

$$\begin{aligned}
N' &\equiv \frac{\partial N}{\partial \underline{q}} = -\delta(\underline{q}\nu + (1 - \underline{q})(-c))g(\underline{q}) \cdot 1 \\
&= -\delta g(\underline{q})[\underline{q}\nu - c + c\underline{q}] = -\delta g(\underline{q})[\underline{q}(\nu + c) - c] \\
D' &\equiv \frac{\partial D}{\partial \underline{q}} = \delta[-\underline{q}g(\underline{q}) \cdot 1] = -\delta \underline{q}g(\underline{q}).
\end{aligned}$$

From this we have:

$$\begin{aligned}
\frac{\partial V(\underline{q})}{\partial \underline{q}} &= \frac{-\delta g(\underline{q})[\underline{q}(\nu + c) - c]D + \delta \underline{q}g(\underline{q})N}{D^2} \\
&= \delta g(\underline{q}) \cdot \frac{-D[\underline{q}(\nu + c) - c] + \underline{q}N}{D^2} \\
&= \delta g(\underline{q}) \cdot \frac{-D\underline{q}(\nu + c) + Dc + \underline{q}N}{D^2} = 0.
\end{aligned}$$

Hence, $-D(\nu + c)q + DC + Nq = 0$, so we have

$$\underline{q}^* = \frac{Dc}{D(\nu + c) - N}.$$

We now substitute D & N to get:

$$\begin{aligned}
\underline{q}^* &= \frac{c \left[1 - \delta \left(1 - \int_{\underline{q}^*}^1 q dG(q) \right) \right]}{\left[1 - \delta \left(1 - \int_{\underline{q}^*}^1 q dG(q) \right) \right] (\nu + c) - \delta \int_{\underline{q}^*}^1 [\nu q + (1 - q)(-c)] dG(q) + k} \\
&= \frac{\left(1 - \delta \left(1 - \int_{\underline{q}^*}^1 q dG(q) \right) \right) c}{\left(1 - \delta G(\underline{q}^*) \right) c + (1 - \delta)\nu + k}.
\end{aligned}$$

We further rewrite this condition:

$$\begin{aligned}
\underline{q}^* \left[c - \delta c G(\underline{q}^*) + (1 - \delta)\nu + k \right] &= c - c\delta \left[1 - \int_{\underline{q}^*}^1 q dG(q) \right] \\
\underline{q}^* (c + (1 - \delta)\nu + k) - \delta c \underline{q}^* G(\underline{q}^*) &= c - c\delta \left[1 - \int_{\underline{q}^*}^1 q dG(q) \right] \\
\underline{q}^* (c + (1 - \delta)\nu + k) &= c - c\delta \left[1 - \int_{\underline{q}^*}^1 q dG(q) \right] + \delta c \underline{q}^* G(\underline{q}^*).
\end{aligned}$$

Applying integration by parts yields:

$$\begin{aligned}
\int_{\underline{q}^*}^1 q dG(q) &= [qG(q)]_{\underline{q}^*}^1 - \int_{\underline{q}^*}^1 G(q) dq \\
&= G(1) - \underline{q}^* G(\underline{q}^*) - \int_{\underline{q}^*}^1 G(q) dq \\
&= 1 - \underline{q}^* G(\underline{q}^*) - \int_{\underline{q}^*}^1 G(q) dq
\end{aligned}$$

$$1 - \int_{\underline{q}^*}^1 q dG(q) = \underline{q}^* G(\underline{q}^*) + \int_{\underline{q}^*}^1 G(q) dq.$$

Plug this into the equation above to get:

$$\begin{aligned} \underline{q}^*(c + (1 - \delta)\nu + k) &= c - c\delta \left[\underline{q}^* G(\underline{q}^*) + \int_{\underline{q}^*}^1 G(q) dq \right] + \delta c \underline{q}^* G(\underline{q}^*) \\ &= c - c\delta \int_{\underline{q}^*}^1 G(q) dq. \end{aligned}$$

Now adding and subtracting $c\delta \int_0^{\underline{q}^*} G(q) dq$ to RHS to get:

$$\begin{aligned} \underline{q}^*(c + (1 - \delta)\nu + k) &= c - c\delta \int_0^{\underline{q}^*} G(q) dq - c\delta \int_{\underline{q}^*}^1 G(q) dq + c\delta \int_0^{\underline{q}^*} G(q) dq \\ &= c - c\delta \int_0^1 G(q) dq + c\delta \int_0^{\underline{q}^*} G(q) dq. \end{aligned}$$

Then we get the expression:

$$\begin{aligned} \underline{q}^* &= \frac{c}{c + (1 - \delta)\nu + k} - \frac{c\delta}{c + (1 - \delta)\nu + k} \int_0^1 G(q) dq \\ &\quad + \frac{c\delta}{c + (1 - \delta)\nu + k} \int_0^{\underline{q}^*} G(q) dq \end{aligned}$$

$$\text{where } G(q) = (1 - m_L) G_H(q) + m_L G_L(q)$$

$$G_H(q) = P_H \cdot \Phi\left(\frac{S_H(q) - 1}{\sigma}\right) + (1 - P_H) \Phi\left(\frac{S_H(q)}{\sigma}\right)$$

$$G_L(q) = P_L \cdot \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) + (1 - P_L) \Phi\left(\frac{S_L(q)}{\sigma}\right).$$

Plug in the equation $\int_0^1 q(q) dq = 1 - E(q) = 1 - m_H P_H - m_L P_L$, we have:

$$\begin{aligned} \underline{q}^* &= \frac{c}{c + (1 - \delta)\nu + k} - \frac{c\delta}{c + (1 - \delta)\nu + k} (1 - m_H P_H - m_L P_L) \\ &\quad + \frac{c\delta}{c + (1 - \delta)\nu + k} \int_0^{\underline{q}^*} G(q) dq. \end{aligned}$$

POLICY IMPLICATION 1: *When the proportion of low-productivity workers (drug users) increases in the population of less-skilled applicants for this occupation (i.e. p_L decreases), the optimal hiring threshold, $\underline{q}^*$, increases.*

Proof of Policy Implication 1. First, when the solution y is given implicitly, we have following relationship from the differentiation of implicit functions:

$$\begin{aligned} \text{If } F(y(\theta), \theta) = 0 \text{ then } \frac{d}{d\theta} F(y(\theta), \theta) &= 0 \\ \Rightarrow \frac{\partial F}{\partial y} \frac{\partial y}{\partial \theta} + \frac{\partial F}{\partial \theta} &= 0 \Rightarrow \frac{\partial y}{\partial \theta} = -\frac{\partial F / \partial \theta}{\partial F / \partial y}. \end{aligned}$$

Let's define:

$$\begin{aligned} F(\underline{q}^*, \vec{\theta}) &= \underline{q}^* - \frac{c}{c + (1 - \delta)\nu + k} + \frac{c\delta}{c + (1 - \delta)\nu + k} (1 - m_H P_H - m_L P_L) \\ &\quad - \frac{c\delta}{c + (1 - \delta)\nu + k} \int_0^{\underline{q}^*} G(q) dq. \end{aligned}$$

Then differentiate F with respect to $\underline{q}^*$ gives:

$$\frac{\partial F}{\partial \underline{q}^*} = -\frac{c\delta}{c + (1 - \delta)\nu + k} \frac{d}{d\underline{q}^*} \left[\int_0^{\underline{q}^*} G(q) dq \right] = -\frac{c\delta}{c + (1 - \delta)\nu + k} \cdot G(\underline{q}^*) < 0.$$

We want to check the sign of $\frac{\partial \underline{q}^*}{\partial P_L} = -\frac{\partial F / \partial P_L}{\partial F / \partial \underline{q}^*}$, and since we know that $\frac{\partial F}{\partial \underline{q}^*} < 0$, we only need to check the sign of $\frac{\partial F}{\partial P_L}$.

$$\begin{aligned} \frac{\partial F}{\partial P_L} &= \frac{-c\delta m_L}{c + (1 - \delta)\nu + k} - \frac{c\delta}{c + (1 - \delta)\nu + k} \int_0^{\underline{q}^*} \frac{\partial G(q)}{\partial P_L} dq \\ &= \frac{-c\delta}{c + (1 - \delta)\nu + k} \left[m_L + \int_0^{\underline{q}^*} \frac{\partial G(q)}{\partial P_L} dq \right] \\ &= \frac{-c\delta}{c + (1 - \delta)\nu + k} \left[m_L + m_L \int_0^{\underline{q}^*} \frac{\partial G_L(q)}{\partial P_L} dq \right] \\ &= \frac{-c\delta m_L}{c + (1 - \delta)\nu + k} \left[1 + \int_0^{\underline{q}^*} \frac{\partial G_L(q)}{\partial P_L} dq \right]. \end{aligned}$$

In the meantime, we have:

$$\begin{aligned}
\frac{\partial G_L(q)}{\partial P_L} &= \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) + P_L \cdot \phi\left(\frac{S_L(q) - 1}{\sigma}\right) \cdot \frac{1}{\sigma} \cdot \frac{\partial S_L(q)}{\partial P_L} \\
&\quad - \Phi\left(\frac{S_L(q)}{\sigma}\right) + (1 - P_L) \cdot \phi\left(\frac{S_L(q)}{\sigma}\right) \cdot \frac{1}{\sigma} \cdot \frac{\partial S_L(q)}{\partial P_L} \\
&= \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) + \frac{1}{\sigma} \cdot \frac{\partial S_L(q)}{\partial P_L} \left[P_L \cdot \phi\left(\frac{S_L(q) - 1}{\sigma}\right) + (1 - P_L) \phi\left(\frac{S_L(q)}{\sigma}\right) \right].
\end{aligned}$$

Note that since $S_\gamma(q) = \frac{1}{2} - \sigma_\gamma^2 * \log\left[\frac{1-q}{q} \cdot \frac{P_r}{1-P_\gamma}\right]$, we have $\frac{\partial S_L(q)}{\partial P_L} = \frac{-\sigma^2}{P_L(1-P_L)}$ and this gives:

$$\begin{aligned}
\frac{\partial G_L(q)}{\partial P_L} &= \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) - \frac{\sigma}{P_L(1 - P_L)} \left[P_L \phi\left(\frac{S_L(q) - 1}{\sigma}\right) + (1 - P_L) \phi\left(\frac{S_L(q)}{\sigma}\right) \right] \\
&= \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) - \frac{\sigma}{1 - P_L} \phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \frac{\sigma}{P_L} \phi\left(\frac{S_L(q)}{\sigma}\right).
\end{aligned}$$

Since $\Phi\left(\frac{S_L(q)-1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) < 0$, we have $\frac{\partial G_L(q)}{\partial P_L} < 0$ and this gives the inequality below:

$$\begin{aligned}
1 + \int_0^{q^*} \frac{\partial G_L(q)}{\partial P_L} dq &> 1 + \int_0^1 \frac{\partial G_L(q)}{\partial P_L} dq \\
&= 1 + \int_0^1 \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) dq \\
&\quad - \frac{\sigma}{1 - P_L} \int_0^1 \phi\left(\frac{S_L(q) - 1}{\sigma}\right) dq - \frac{\sigma}{P_L} \int_0^1 \phi\left(\frac{S_L(q)}{\sigma}\right) dq \\
&= 1 + \int_0^1 \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) dq - \frac{\sigma}{1 - P_L} - \frac{\sigma}{P_L} \\
&= 1 + \int_0^1 \Phi\left(\frac{S_L(q) - 1}{\sigma}\right) - \Phi\left(\frac{S_L(q)}{\sigma}\right) dq - \frac{\sigma}{(1 - P_L)P_L} \\
&= 1 + \left(\frac{\sigma}{(1 + P_L)P_L} - 1 \right) - \frac{\sigma}{(1 - P_L)P_L} = 0.
\end{aligned}$$

This proves that $\frac{\partial F}{\partial P_L} < 0$, and hence finalizes the proof of $\frac{\partial q^*}{\partial P_L} = -\frac{\partial F / \partial P_L}{\partial F / \partial q^*} < 0$.

POLICY IMPLICATION 2: *As the proportion of drug users among less-skilled workers increases (i.e., as P_L decreases), the permanent composition of less-skilled workers declines, given a recruiting threshold $\underline{q}$ (i.e. $\frac{\partial \gamma_L}{\partial P_L} > 0$).*

Proof of Policy Implication 2. When we have $m_A = m_B = \frac{1}{2}$:

$$r_\gamma = \frac{P_\gamma (1 - G_{\gamma 1})}{P_\gamma (1 - G_{\gamma 1}) + P_{-\gamma} (1 - G_{-\gamma 1})}.$$

So we have:

$$r_L = \frac{P_L (1 - G_{L1})}{P_L (1 - G_{L1}) + P_H (1 - G_{H1})}$$

where $G_{H1} = \Phi \left(\frac{S_H(\underline{q}) - 1}{\sigma} \right) = \Phi \left(-\frac{1}{2\sigma} - \sigma * \log \left[\frac{P_H}{1 - P_H} \cdot \frac{1 - \underline{q}}{\underline{q}} \right] \right)$

$$G_{L1} = \Phi \left(\frac{S_L(\underline{q}) - 1}{\sigma} \right) = \Phi \left(-\frac{1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1 - P_L} \cdot \frac{1 - \underline{q}}{\underline{q}} \right] \right).$$

By differentiating r_L with respect to P_L we get:

$$\begin{aligned} \frac{\partial r_L}{\partial P_L} &= \frac{\frac{\partial \{P_L(1-G_{L1})\}}{\partial P_L} \cdot [P_L (1 - G_{L1}) + P_H (1 - G_{H1})] - \frac{\partial \{P_L(1-G_{L1})+P_H(1-G_{H1})\}}{\partial P_L} \cdot P_L (1 - G_{L1})}{[P_L (1 - G_{L1}) + P_H (1 - G_{H1})]^2} \\ &= \frac{\frac{\partial \{P_L(1-G_{L1})\}}{\partial P_L} [P_L (1 - G_{L1}) + P_H (1 - G_{H1}) - P_L (1 - G_{L1})]}{[P_L (1 - G_{L1}) + P_H (1 - G_{H1})]^2} \\ &= \frac{\frac{\partial \{P_L(1-G_{L1})\}}{\partial P_L} [P_H (1 - G_{H1})]}{[P_L (1 - G_{L1}) + P_H (1 - G_{H1})]^2}. \end{aligned}$$

Note that:

$$\begin{aligned} \frac{\partial \{P_L (1 - G_{L1})\}}{\partial P_L} &= 1 - G_{L1} + P_L \cdot \frac{\partial \{(1 - G_{L1})\}}{\partial P_L} \\ &= 1 - G_{L1} + P_L \cdot \left(-\frac{\partial G_{L1}}{\partial P_L} \right) \\ &= 1 - G_{L1} - P_L \cdot \frac{\partial G_{L1}}{\partial P_L}, \end{aligned}$$

in which

$$\begin{aligned}
\frac{\partial G_{L1}}{\partial P_L} &= \frac{\partial \left\{ \Phi \left(\frac{S_L(q)-1}{\sigma} \right) \right\}}{\partial P_L} = \partial \left\{ \Phi \left(\frac{\frac{1}{2} - \sigma^2 * \log \left[\frac{P_L}{1-P_L} \cdot \frac{1-q}{q} \right] - 1}{\sigma} \right) \right\} / \partial P_L \\
&= \partial \left\{ \Phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} \cdot \frac{1-q}{q} \right] \right) \right\} / \partial P_L \\
&= \phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} \cdot \frac{1-q}{q} \right] \right) * \frac{\partial \left\{ -\sigma * \log \left[\frac{P_L}{1-P_L} \cdot \frac{1-q}{q} \right] \right\}}{\partial P_L} \\
&= \phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} \cdot \frac{1-q}{q} \right] \right) * \left[-\sigma * \frac{\frac{1}{(1-P_L)^2} \frac{1-q}{q}}{\frac{P_L}{1-P_L} \cdot \frac{1-q}{q}} \right] \\
&= \phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} * \frac{1-q}{q} \right] \right) * \left[-\sigma * \frac{1}{P_L (1-P_L)} \right].
\end{aligned}$$

Combining the equations above, we have:

$$\begin{aligned}
\frac{\partial \{P_L (1 - G_{L1})\}}{\partial P_L} &= 1 - G_{L1} - P_L \cdot \phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} * \frac{1-q}{q} \right] \right) * \left[-\sigma * \frac{1}{P_L (1-P_L)} \right] \\
&= 1 - G_{L1} + \frac{\sigma}{1-P_L} \phi \left(\frac{-1}{2\sigma} - \sigma * \log \left[\frac{P_L}{1-P_L} * \frac{1-q}{q} \right] \right) > 0.
\end{aligned}$$

And this proves that $\frac{\partial r_L}{\partial P_L} > 0$.

POLICY IMPLICATION 3: *As the recruiting threshold $\underline{q}$ increases, the permanent composition of less-skilled workers declines.*

Proof of Policy Implication 3. Note that:

$$r_\gamma = \frac{m_\gamma P_\gamma (1 - G_{\gamma 1})}{m_\gamma P_\gamma (1 - G_{\gamma 1}) + (1 - m_\gamma) P_{-\gamma} (1 - G_{-\gamma 1})}.$$

Let the denominator:

$$D \equiv m_L P_L (1 - G_{L1}) + (1 - m_L) P_H (1 - G_{H1}).$$

Then we have:

$$\frac{\partial r_L}{\partial \underline{q}} = \frac{-m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} \cdot N - \frac{\partial N}{\partial \underline{q}} m_L P_L (1 - G_{L1})}{D^2}.$$

Note that:

$$\begin{aligned}\frac{\partial N}{\partial \underline{q}} &= -m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} - (1 - m_L) P_H \frac{\partial G_{H1}}{\partial \underline{q}} \\ &= -m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} - m_H P_H \frac{\partial G_{H1}}{\partial \underline{q}}.\end{aligned}$$

If we denote the nominator of $\frac{\partial r_L}{\partial \underline{q}}$ as N' , then we have:

$$\begin{aligned}N' &= -m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} (m_L P_L (1 - G_{L1}) + m_H P_H (1 - G_{H1})) \\ &\quad - \left(-m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} - m_H P_H \frac{\partial G_{H1}}{\partial \underline{q}} \right) m_L P_L (1 - G_{L1}) \\ &= -m_L P_L \frac{\partial G_{L1}}{\partial \underline{q}} (m_L P_L (1 - G_{L1}) + m_H P_H (1 - G_{H1})) \\ &\quad + m_L^2 P_L^2 \frac{\partial G_{L1}}{\partial \underline{q}} (1 - G_{L1}) + m_H P_H m_L P_L (1 - G_{L1}) \frac{\partial G_{H1}}{\partial \underline{q}} \\ &= -m_L^2 P_L^2 (1 - G_{L1}) \frac{\partial G_{L1}}{\partial \underline{q}} - m_H m_L P_H P_L (1 - G_{H1}) \frac{\partial G_{L1}}{\partial \underline{q}} \\ &\quad + m_L^2 P_L^2 (1 - G_{L1}) \frac{\partial G_{L1}}{\partial \underline{q}} + m_H m_L P_H P_L (1 - G_{L1}) \frac{\partial G_{H1}}{\partial \underline{q}} \\ &= m_H m_L P_H P_L \left[(1 - G_{L1}) \frac{\partial G_{H1}}{\partial \underline{q}} - (1 - G_{H1}) \frac{\partial G_{L1}}{\partial \underline{q}} \right].\end{aligned}$$

Since we know that:

$$\begin{aligned}\frac{\partial G_{H1}}{\partial \underline{q}} &= \frac{\sigma}{\underline{q}(1 - \underline{q})} \phi \left(\frac{S_H(\underline{q}) - 1}{\sigma} \right) \\ \frac{\partial G_{L1}}{\partial \underline{q}} &= \frac{\sigma}{\underline{q}(1 - \underline{q})} \phi \left(\frac{S_L(\underline{q}) - 1}{\sigma} \right).\end{aligned}$$

We can plug in these equations above to get:

$$\begin{aligned}N' &= m_H m_L P_H P_L \left[(1 - G_{L1}) \frac{\sigma}{\underline{q}(1 - \underline{q})} \phi \left(\frac{S_H(\underline{q}) - 1}{\sigma} \right) - (1 - G_{H1}) \frac{\sigma}{\underline{q}(1 - \underline{q})} \phi \left(\frac{S_L(\underline{q}) - 1}{\sigma} \right) \right] \\ &= m_H m_L P_H P_L \frac{\sigma}{\underline{q}(1 - \underline{q})} \left[(1 - G_{L1}) \phi \left(\frac{S_H(\underline{q}) - 1}{\sigma} \right) - (1 - G_{H1}) \phi \left(\frac{S_L(\underline{q}) - 1}{\sigma} \right) \right].\end{aligned}$$

To prove $\frac{\partial r_L}{\partial \underline{q}} < 0$, it is equivalent to prove $N' < 0$. Hence, we need to prove the distribution G_{H1} dominates

G_{L1} in terms of hazard rate, which is equivalent to prove:

$$\begin{aligned} \frac{(1 - G_{L1}) \cdot \phi\left(\frac{S_H(q)-1}{\sigma}\right) \frac{\sigma}{q(1-q)}}{(1 - G_{H1}) \cdot \phi\left(\frac{S_L(q)-1}{\sigma}\right) \frac{\sigma}{q(1-q)}} &= \frac{\frac{1}{1-G_{H1}} \cdot \phi\left(\frac{S_H(q)-1}{\sigma}\right) \frac{\sigma}{q(1-q)}}{\frac{1}{1-G_{L1}} \cdot \phi\left(\frac{S_L(q)-1}{\sigma}\right) \frac{\sigma}{q(1-q)}} = \frac{\frac{1}{1-G_{H1}(q)} g_{H1}(q)}{\frac{1}{1-G_{L1}(q)} g_{L1}(q)} \\ &= \frac{\text{Hazard Rate of Distribution } G_{H1}}{\text{Hazard Rate of Distribution } G_{L1}} < 1. \end{aligned}$$

Before proving the statement above, we first prove that the distribution G_{H1} dominates G_{L1} in terms of likelihood ratio, which is equivalent to proving $\frac{\partial^2 \ln g}{\partial \sigma \partial q} > 0$. This inequality can be proved by:

$$\begin{aligned} \ln g(q; P) &= \ln \phi\left(\frac{s(q)-1}{\sigma}\right) \frac{\sigma}{q(1-q)} \\ &= \ln \left[\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{s(q)-1}{\sigma}\right)^2} \frac{\sigma}{q(1-q)} \right] = \ln \frac{1}{\sqrt{2\pi}} - \frac{1}{2} \left(\frac{s(q)-1}{\sigma} \right)^2 + \ln \frac{\sigma}{q(1-q)} \\ \frac{\partial \ln g(q; P)}{\partial p} &= -\frac{1}{2p(1-p)} - \frac{\sigma^2}{p(1-p)} \ln \left[\frac{p}{1-p} \frac{1-q}{q} \right] \\ \frac{\partial \ln g(q; P)}{\partial p \partial q} &= \frac{\sigma^2}{p(1-q)} \cdot \frac{1}{(1-p)} \cdot \frac{1}{q} > 0. \end{aligned}$$

For all $q < q'$, the fact that the distribution G_{H1} dominates G_{L1} in terms of likelihood ratio gives:

$$\begin{aligned} \frac{g_{L1}(q')}{g_{L1}(q)} < \frac{g_{H1}(q')}{g_{H1}(q)} &\Rightarrow \int_q^1 \frac{g_{L1}(t)}{g_{L1}(q)} < \int_q^1 \frac{g_{H1}(t)}{g_{H1}(q)} \Rightarrow \frac{1 - G_{L1}(q)}{g_{L1}(q)} < \frac{1 - G_{H1}(q)}{g_{H1}(q)} \\ &\Rightarrow \frac{g_{H1}(q)}{1 - G_{H1}(q)} < \frac{g_{L1}(q)}{1 - G_{HL}(q)}. \end{aligned}$$

This finalizes the proof that $N' < 0$ and $\frac{\partial r_L}{\partial q} < 0$.